Voting Rules and the Price of Peace*

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Abstract

In a conflicted controlling-shareholder transaction, corporate law often relies on a vote of the minority to screen out value-destroying deals, and the conventional view holds that a tougher vote screens better. In a one-stage signaling model the conventional view fails: raising the required margin relocates the controllers who game the threshold, as raising an exam's passing mark relocates the candidates who cram to clear it. I add the enforcement stage. A controller who over-sells a deal to win approval may later pay to settle the litigation that over-selling invites, and when the threshold that approves a deal also governs its ratification, a stricter rule raises the price of persuasion and the price of peace together. The second price can dominate, so a tougher vote can reduce misrepresentation. Welfare effects turn on enforcement waste and the quality of deterred deals, clarifying when either/or safe harbors leave minority investors underprotected.

Keywords: controlling shareholders, majority-of-minority approval, ratification, shareholder litigation, related-party transactions, signaling.

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1 Introduction

Controlling-shareholder transactions pose a recurring problem in corporate law: the party who proposes the transaction often stands on both sides of it. A freeze-out merger, a parent–subsidiary restructuring, or a related-party sale may create value for the firm, yet it may also transfer value from the minority investors to the controller (Johnson et al., 2000; Bebchuk and Kahan, 2002). Since ordinary market discipline operates weakly when a single shareholder controls the outcome, corporate law assigns the screening task to some combination of minority votes, independent committees, and judicial review. A controller may be required to condition the transaction on approval by an independent committee, on an uncoerced majority of the disinterested minority, or on both, with courts reviewing whether those procedures were genuinely satisfied. Concentrated ownership makes this a global screening problem rather than a feature peculiar to any single jurisdiction (Faccio and Lang, 2002; Kraakman and Armour, 2017).

A basic design question is whether the disinterested-stockholder standard that authorizes such a transaction ex ante should also govern its ratification ex post, when the deal is later challenged, or whether the two should be set independently. As a matter of general ratification law the two are coupled: an act that required a heightened approval threshold can be confirmed only at that same threshold. In the United Kingdom this is the rule in *Foss v. Harbottle*, as restated in *Edwards v. Halliwell* and *Prudential Assurance v. Newman Industries (No. 2)*—ordinary-majority ratification is available only where a simple majority could validly have authorized the act, so that "a simple majority cannot confirm a transaction which requires the concurrence of a greater majority."[1] Delaware reaches the same result by statute for defective corporate acts: under DGCL §204 the quorum and vote required to ratify are those the act originally required, with judicial validation under §205 as the separate alternative route, so that a supermajority or class-vote defect cannot be cured by an ordinary-majority ratification.[2] On this dimension both systems *couple* the standards: authorization law supplies the coupled pole of the design space studied below.

Conflicted-controller transactions are subject to a second, analytically distinct inquiry. Beyond the authorization question—whether a vote met the threshold the act required—fiduciary doctrine asks whether a disinterested vote *cleanses* the controller's conflict and thereby fixes the applicable standard of review.[3] The model holds these two questions apart. The coupling baseline above is

[1] *Foss v. Harbottle* (1843) 2 Hare 461, 67 E.R. 189; *Edwards v. Halliwell* [1950] 2 All E.R. 1064 (C.A.) (Jenkins L.J., restating the rule in *Foss* and its exceptions); *Prudential Assurance Co. Ltd v. Newman Industries Ltd (No. 2)* [1982] Ch. 204 (C.A.). The special-resolution case is Companies Act 2006, s. 283: the 75% threshold governs both an original special resolution and one ratifying an act for which that threshold is required. Ratifying a director's conflicted conduct falls under a distinct regime, s. 239, which instead disregards the votes cast in favour by the interested director and any member connected with him. The statutory derivative-claim regime in Part 11 of the Companies Act 2006 recast the procedure for derivative litigation, yet it left this supermajority principle in place and does not permit a simple majority to ratify an act for which the constitution or statute requires a higher threshold.

[2] 8 Del. C. §§204–205. Section 204 ties the ratifying stockholder vote to the authorization the act originally required, carrying forward any larger or class vote that the defective act required when it was taken; §205, as the separate alternative route, vests the Court of Chancery with jurisdiction to validate a defective act.

[3] A court credits a stockholder vote as cleansing only when it is fully informed and uncoerced. The conflicted-controller cleansing authorities, which bear on the standard of review rather than on corporate authorization, are collected in Section 6.

a proposition of authorization law, under which the ratifying threshold carries forward mechanically, whereas the cleansing question enters only through a court's residual discretion to deny credit to a vote that was not fully informed and uncoerced, a residual that the model summarizes in the single parameter σ. On the ratification standard r itself the analysis studies two polar designs: a *coupled* regime, in which r tracks the approval threshold π along the authorization default (the case $r = \pi$), and a *decoupled* regime, in which r is set independently of π; the decoupled pole is likewise instantiated by statute: under German group law the settlement of a dependent company's claims against its controller requires a special resolution of the outside shareholders after a three-year wait, a standard fixed without reference to the approval threshold.[4] A higher coupled threshold raises two prices together, the price of persuasion before approval and the price of peace once litigation begins, and decoupling severs the second from the first. This paper asks how that choice—whether the ex ante and ex post standards move together (coupled) or independently (decoupled)—shapes a controller's incentive to misrepresent deal quality, and when each design serves the minority investors.

The conventional intuition is that a tougher vote screens out more bad deals. If conflicted controllers can exaggerate quality, raising the majority-of-the-minority or supermajority requirement forces a controller to persuade more disinterested shareholders, which should deter low-quality transactions. This intuition is incomplete even before litigation enters, for much the same reason that raising the passing mark on an examination does not by itself improve it as a screen. The candidates with the strongest incentive to inflate their answers are the ones who sit just below the mark; if the mark moves from 50 to 70, that incentive is relocated rather than removed, because a fresh group of candidates now sits just below 70 with the most to gain from exaggeration. A higher cutoff can shift the distortion to a new margin rather than eliminate it, and the screen then continues to admit some candidates who fall short of the required standard while inducing costly efforts to manipulate the signal.

Controlling-shareholder approval behaves the same way. A higher approval bar removes some low-quality controllers, for whom the exaggeration needed to clear the bar becomes too costly, while it simultaneously creates a new population of near-threshold controllers who fall just short and need only a small exaggeration to pass. As the first group exits, the second enters. In a one-stage signaling model with costly misrepresentation (Kartik, 2009), the second effect can dominate, so that a stricter vote widens pooling at the threshold rather than narrowing it—a comparative static consistent with the bunching observed around discrete voting and disclosure cutoffs (Saez, 2010; Kleven, 2016; Burgstahler and Dichev, 1997; Degeorge et al., 1999).

Approval, however, is not the terminal step in a controller transaction. Minority shareholders who suspect that they were misled can sue, and the resulting litigation is typically resolved through settlement, ratification, or a judicial determination about whether the disinterested vote cleanses the conflict (Bebchuk, 1984; Restrepo, 2021).[5] A favorable vote therefore does double

[4] Aktiengesetz §309(3), applied to claims against a controlling enterprise in a de facto group by §317(4): the company may waive or settle its compensation claims only three years after the claim arises, only with a special resolution (*Sonderbeschluss*) of the outside shareholders, and only if no minority holding one-tenth of the share capital represented at the vote objects on the record.

[5] A court treats a disinterested vote as cleansing only when it is fully informed and uncoerced. Outside the

duty: it secures approval at the first stage, and it later serves as evidence of cleansing, as a precondition for settlement, or as the basis for a court-creditable release. A controller who exaggerated quality to win approval may consequently have to pay again, after the fact, to extinguish the very claims that the exaggeration invites. The empirical record is consistent with such reshuffling: Restrepo (2021) finds that after Delaware moved to encourage dual protections, majority-of-the-minority conditions rose sharply while deal premiums did not change significantly.

Modeling this second stage is what changes the analysis. Let π denote the approval threshold, g the gap between a controller's disclosure and the true quality of the deal, and r the ratification standard that governs settlement. Throughout, the minority shareholders are modeled as atomistic voters who vote on the deal terms disclosed in the proxy materials through a primitive perception of the disclosure, while the legal system prices the gap between disclosed and true value ex post—the empirically natural case for the dispersed, inattentive minorities of controlled firms (Section 5.1.2). The device is a benchmark, and it is conservative for the central result: the reversal survives fully Bayesian voters, where the equilibrium disclosure gap is strictly larger and the ratification channel's grip weakly stronger (Proposition 3). The cost of buying peace is endogenous to the legal architecture: it is increasing in the size of the disclosure gap, in the toughness of the ratification standard, and in the intensity of the judicial scrutiny that decides whether a release will later be credited. When approval and ratification are coupled—so that the threshold π that approves the deal also fixes the disinterested-support standard r that governs its ratification ex post (with r rising in π)—a stricter vote raises both the persuasion margin before approval and the settlement margin after litigation begins. The feedback that this second margin exerts on the controller's first-stage disclosure is what I call the enforcement-feedback channel.

The central result is that the enforcement-feedback channel can reverse the one-stage prediction. Raising the approval threshold moves the lower boundary of types still willing to misrepresent through two routes: directly, because a higher bar demands a larger exaggeration, and (indirectly) because the coupled ratification standard makes that exaggeration more expensive to defend after the fact. The indirect route acts purely on the cost side: a tougher ratification standard raises the price the boundary controller pays for the marginal lie without raising the prize that lie secures or altering the quality the pivotal voter demands, so it supplies a second lever on the boundary that the one-stage model, which moves only the approval bar, does not possess. In the one-stage benchmark only the direct route operates, so a higher bar draws new misrepresenting types into the top of the pool faster than the rising boundary deters them at the bottom, and the pool grows. Once the indirect route is present and strong enough—formally, when the ratification channel's contribution exceeds the density-weighted shortfall that the bunching force would otherwise create—the boundary deters types at the bottom faster than the bar draws them in at the top, and the pool contracts. A stricter vote then reduces misrepresentation, the opposite of the one-stage benchmark. The reversal requires neither sluggish beliefs nor thin distributional tails, only that the settlement technology respond sufficiently to the ratification standard.

A contraction of the pooling region is not automatically desirable. Its welfare consequences de-

controller context such a vote restores business-judgment review (*Corwin v. KKR Fin. Holdings LLC*, 125 A.3d 304 (Del. 2015)). The residual risk that a court withholds that credit is the parameter σ, and the conflicted-controller cleansing authorities are taken up in Section 6.

pend on what the deterred transactions look like and on how costly the enforcement process is. I decompose the shortfall of any governance regime, relative to a first-best, into false positives, false negatives, and enforcement costs, and I show that the sign of a reform's welfare effect is governed by four institutionally interpretable parameters: the deadweight share of enforcement costs, the convexity of those costs in the disclosure gap, the sensitivity of enforcement to the ratification standard, and the quality of the marginal deterred deal. Coupling improves welfare when the transactions it deters destroy value and when enforcement waste is substantial; it reduces welfare when the binding problem is over-deterrence of efficient restructuring.

A final implication concerns the relationship between the two screening instruments. Independent committees and minority votes operate on different margins: a committee changes the composition of the deals that reach a vote, whereas a tougher ratification standard changes the discipline imposed on the controllers who survive the vote. Since the cost of committee independence is largely invariant to the toughness of enforcement, while the enforcement-cost savings from screening are increasing in the difficulty of ratification, the marginal value of committee quality is increasing in the ratification standard. In this sense, committees and votes are procedural complements rather than interchangeable substitutes. This conditional complementarity bears directly on the design of safe harbors that allow a controller to satisfy one procedure in place of the other.

The paper connects three economic literatures usually studied apart. The signaling tradition models costly misrepresentation but stops at the receiver's decision (Crawford and Sobel, 1982; Kartik, 2009); the work on voting thresholds and freeze-outs studies the approval margin without a subsequent enforcement stage (Dalkır et al., 2019; Subramanian, 2005); and the litigation-and-settlement literature studies the resolution of claims without endogenizing the disclosure that generates them (Bebchuk, 1984). The contribution here is to couple the ex ante screening problem to the ex post settlement technology, so that a single procedural parameter operates in both stages. The remainder of the paper proceeds as follows. Section 2 reviews the related literature; Section 3 presents the two-stage model; Section 4 develops the Stage-2 litigation and ratification technology; Section 5 characterizes the Stage-1 equilibrium and the litigation-adjusted comparative static under coupled and decoupled rules; Section 6 analyzes welfare and the design implications, including the design of either/or safe harbors that permit one procedure to substitute for the other; and Section 7 concludes. The Appendix collects the deferred proofs; the Online Appendix contains the microfoundations, the benchmark model, the extension proofs, and robustness material.

2 Literature Review

The transactions we study sit at the center of the agency conflict between a controlling shareholder and the minority, the *principal–principal* conflict that a substantial literature treats as the dominant governance problem outside the dispersed-ownership firm (Goshen and Squire, 2017; Kraakman and Armour, 2017). Concentrated ownership is the global norm. Once one looks beyond the United States and the United Kingdom, controlling blockholders are present in most

listed firms (La Porta et al., 1999; Claessens et al., 2000), and the private benefits they can extract are large and sensitive to legal protection, as the wide dispersion of estimated block and control premia across countries attests (Barclay and Holderness, 1989; Albuquerque and Schroth, 2010; Atanasov et al., 2014). Whether procedural law can constrain that extraction is contested, since a controller's long horizon and reputational stake may (partly) align its interests with the minority (Choi, 2018), even as minority approval and judicial review remain the principal external checks on controller self-dealing (Gilson, 2006). The 2025 Delaware reform sharpens the question, and a growing commentary debates whether a move toward an either/or safe harbor weakens minority protection (Bainbridge, 2025; Whitehead, 2025). We speak to that debate by tracing how a single procedural threshold operates across both the approval and the enforcement stages of such a transaction, drawing on four bodies of work that have largely been studied in isolation: a doctrinal literature on controller transactions and cleansing, together with three economic literatures on the approval problem, the signaling problem, and the litigation-and-settlement problem.

A first body of work, on controller transactions and cleansing doctrine, examines how procedural protections discipline self-dealing by controllers and how courts should review the resulting transactions (Bebchuk and Kahan, 2002; Subramanian, 2005; Enriques, 2014; Bebchuk and Kahan, 2000). Goshen (2003) develops the theoretical case for regulating controller self-dealing through procedural cleansing rather than outright prohibition, treating an approval requirement as a 'screen' that blocks value-reducing self-dealing while permitting the efficient related-party transactions that a flat ban would also deter; the same concern with making a controller's commitments to the minority credible animates the broader account of investor protection under concentrated ownership (Gilson and Gordon, 2003), and Goshen and Hamdani (2019) and Bebchuk and Kastiel (2024) carry the design question further, asking which controller conflicts a cleansing vote actually reaches and how the vote should be structured. A largely empirical and doctrinal literature evaluates these protections in operation: studies of freeze-outs and of the cleansing effect of a minority vote assess majority-of-the-minority approval and special-committee review as screening devices and trace how Delaware's standard of review has shaped freeze-out structure, pricing, and litigation outcomes (Subramanian, 2007; Restrepo and Subramanian, 2015; Restrepo, 2021; Rock, 2019; Booth, 2013), while a comparative strand maps the institutional design of related-party-transaction regulation across jurisdictions (Enriques and Tröger, 2019; Enriques, 2014; Ventoruzzo, 2015). This work characterizes the protections largely as ex ante screens whose value is assessed at the moment of approval. We treat the minority vote instead as an instrument that also operates ex post, so that the same threshold which screens a deal before approval reshapes the cleansing-and-settlement technology that governs it after a challenge; the screening and enforcement roles of the vote are thus analyzed together.

A second body of work studies how voting rules and reporting thresholds affect behavior. The classical analysis of corporate voting asks how the allocation of votes and the choice of majority or supermajority rules aggregate dispersed information and screen proposals (Easterbrook and Fischel, 1983; Harris and Raviv, 1988; Grossman and Hart, 1988), weighing the protection of minorities against the risk that a determined minority blocks value-creating action (Maug and Rydqvist, 2009; Miller, 2021); a more recent strand characterizes optimal and strategically robust voting rules and models when shareholders acquire information before voting and how it is impounded into the outcome (Gershkov et al., 2017; Harris and Raviv, 2010; Meirowitz and

Pi, 2023; Levit et al., 2024, 2026). A parallel applied literature evaluates majority-of-the-minority voting directly, framing the choice between rules as a trade-off between the Type I error of approving value-reducing deals and the Type II error of blocking value-creating ones (Thompson, 1995) and reporting decidedly mixed evidence: Rock (2019) finds little average effect on premia or completion, studies of institutional minority voting document support for insider-sponsored deals even where minority power is formally enhanced (Hamdani and Yafeh, 2013), evidence from related-party transactions in China links minority disapproval to efficiency gains while also raising concerns about distorted investment around the vote (Cao et al., 2024; Lin et al., 2025; Gözlügöl, 2021), and a complementary line traces this heterogeneity to frictions within the minority electorate itself, such as conflicts of interest and empty voting, and proposes refinements of the aggregation mechanism (Cunningham, 2021; Hu and Hamermesh, 2025; Booth, 2013). Across these treatments the vote is modeled as an aggregator of pre-existing private information, and the controller's disclosure is taken as given rather than chosen in anticipation of the rule.

Closest to the comparative static we study, the one-stage lying-cost model of strategic communication shows how a higher approval requirement shifts the marginal misrepresenting type and generates pooling near the threshold, since types just below the bar inflate their disclosure to clear it (Kartik, 2009); the freeze-out literature separately studies how a higher approval requirement shifts the population of deals that clear it, though through a tendering and free-rider channel rather than disclosure bunching (Dalkır et al., 2019). The bunching force is thus derived within a single approval stage. We carry the threshold into a second, post-approval stage, where a coupled ratification standard can move the marginal pooling type faster than the approval bar itself does, so that the sign of the one-stage bunching comparative static reverses once the enforcement channel is active.

A third body of work models communication between an informed sender and an uninformed receiver, and two traditions within it bear on the controller's disclosure problem: costly signaling, in which an informed party separates from worse types once distorting its message is sufficiently expensive (Spence, 1973; Milgrom and Roberts, 1986), and verifiable disclosure, in which the receiver reads an adverse inference into silence and unraveling forces full disclosure except where it stalls (Grossman, 1981; Milgrom, 1981; Verrecchia, 1983; Dye, 1985; Okuno-Fujiwara et al., 1990; Shin, 1994). The controller's problem combines features of both, because disclosure is costly to falsify yet the minority's approval rule fixes how much exaggeration is required and hence how much concealment survives in equilibrium.

The strategic-information-transmission and cheap-talk frameworks characterize equilibrium communication when sender and receiver interests are imperfectly aligned (Crawford and Sobel, 1982; Farrell and Rabin, 1996), and the lying-cost models that build on them describe how a sender's report responds to the marginal cost of distortion, supplying the bar-boundary logic that underlies the controller's decision at the approval stage (Kartik, 2009). A related information-design literature instead lets the sender commit in advance to how informative the receiver's signal will be (Kamenica and Gentzkow, 2011), and the monotone, single-crossing structure on which our equilibrium partition rests reflects the log-concavity conditions that are standard across these threshold signaling models (Bagnoli and Bergstrom, 2005). Throughout this work the cost of misrepresentation is a primitive of the sender's technology, fixed independently of the institution

that the receiver applies. We endogenize that cost through the second stage, so that the price of exaggeration is the anticipated cost of buying peace, $L(g, r, \sigma)$, generated by the litigation-and-ratification subgame and therefore responsive to the very procedural parameter, the threshold r coupled to π, that governs approval.

A fourth body of work studies how legal claims are resolved under asymmetric information and why the resolution dissipates value rather than effecting a pure transfer. The canonical models of suit, settlement, and trial show how the prospect of trial, the allocation of legal costs, and the parties' private information about the merits jointly determine whether a dispute is filed and on what terms it settles (Shavell, 1982; P'ng, 1983). A screening-and-signaling literature then locates the deadweight loss in the information asymmetry itself, since private knowledge about the strength of a claim drives the probability of suit and the breakdown of settlement, so that bargaining cannot avoid some litigation cost (Bebchuk, 1984; Reinganum and Wilde, 1986; Daughety and Reinganum, 2005).

This work takes the dispute as given and analyzes its resolution—with empirical work on shareholder and derivative litigation documenting how much of that resolution is absorbed as fees and process cost rather than recovered by the plaintiff class (Thompson and Thomas, 2004)—without endogenizing the ex ante disclosure that generates the claim in the first place. A recent exception, Choi and Spier (2026), endogenizes an issuer's non-disclosure against an ex post liability stage and characterizes the damages rule that deters inefficient concealment, though it treats the liability standard as a fixed policy instrument set independently of the approval decision. We instead close this loop through a procedural threshold coupled across the two stages, making the disclosure that produces the dispute respond to the anticipated cost of settling it, so that the deadweight component of enforcement, scaled in the welfare analysis by the parameter ω, feeds back into the controller's first-stage incentive to misrepresent. The litigation stage thus supplies the disclosure cost whose comparative statics drive our main result.

Taken together, the existing work studies the approval problem, the signaling problem, or the litigation-and-settlement problem separately. Our contribution is to couple them within a single model, so that one procedural rule both screens conflicted transactions ex ante and reshapes the settlement and ratification technology ex post. That coupling is what makes the comparative static on the approval threshold depend on the second-stage enforcement channel, and it is also what allows the framework to ask, of revised §144 and of comparable regimes elsewhere, when treating committee approval and minority approval as substitutes rather than complements serves minority investors. Throughout, we retain the one-stage threshold-bunching model as a benchmark 'foil' against which the two-stage results are read.

3 Model

This section presents the core model. Subsequent results cite only the numbered statements below.

3.1 Timeline (Stages 0–2)

The game proceeds in three stages:

Stage 0. Nature. Project quality θ is drawn from F on $[\underline{\theta}_0, \overline{\theta}_0]$ and observed privately by the controlling shareholder (CS).

Stage 1. Disclosure & Vote. The CS sends a public message $m \in \mathbb{R}$. Minority shareholders update beliefs and vote under the prevailing approval rule (majority-of-the-minority, MoM, or supermajority with quota π). If the proposal fails, the game ends with payoff zero.

Stage 2. Extraction, Litigation & Ratification. Conditional on approval:

(a) The CS extracts private benefits $B(\theta)$ (or, in the continuous-extraction extension, chooses $x \in [0, B(\theta)]$).

(b) Minority shareholders may file suit. The probability and expected cost of litigation are increasing in the *misrepresentation gap* $g := m - \theta$ (or $g := m^{\text{req}}(u) - \theta$).

(c) The CS may attempt to extinguish claims by buying support for a *ratification/settlement vote* requiring threshold r. Courts may decline to treat a stockholder vote as cleansing if it is not fully informed and uncoerced; in the model, this is captured by probability $\sigma \in [0, 1)$ that the court refuses to credit the ratification vote.

(d) The litigation/ratification subgame resolves, producing a reduced-form expected enforcement cost $L(g, r, \sigma)$.

We solve backward from Stage 2. The enforcement cost $L(g, r, \sigma)$ enters the CS's Stage-1 problem exactly as a signaling cost would, with the additional feature that L depends on the ratification threshold r and the court-independence parameter σ, which, in the carry-forward baseline, includes $r = \pi$ (the same quota as the approval rule).

Because the timeline is solved in reverse, the controller, contemplating Stage 1 disclosure, anticipates the Stage 2 enforcement exposure that follows. This forward-looking calculation is what converts the institutional parameters—r, σ, and shareholder resistance—into a disciplinary force on Stage 1 behavior.

The model treats the map $r(\pi)$ with $r'(\pi) \geq 0$ as the design space. At the *coupled* pole, the carry-forward baseline $r(\pi) = \pi$, the same disinterested-stockholder standard governs both the ex ante approval and the ex post ratification of a transaction; at the *decoupled* pole, r is set without reference to the approval threshold. Authorization law supplies the coupled pole, through the general rule that an act requiring a heightened threshold can be confirmed only at that threshold: in the United Kingdom under *Foss v. Harbottle*, as restated in *Edwards v. Halliwell* and *Prudential* (with Companies Act 2006, s. 283, as the special-resolution instance), and in Delaware by statute for defective corporate acts under DGCL §204, with judicial validation under §205 as the alternative route. The coupling is doctrinally exact when the transaction's validity itself turns on a heightened vote—a merger or charter amendment requiring a class or supermajority vote, or, in the regimes that mandate majority-of-the-minority approval of material related-party transactions (India's SEBI regime; Hong Kong's Chapter 14A on connected transactions; Section 6), a disinterested-minority threshold that operates as an authorization requirement rather than an optional cleansing device. The decoupled pole has statutory instances of its own: in German group

law, a dependent company may waive or settle its compensation claims against the controlling enterprise only three years after the claim arises, only by special resolution of the outside shareholders, and only if no minority holding one-tenth of the capital represented at the vote objects on the record (Aktiengesetz §309(3), applied to the de facto group by §317(4)), so the settlement standard is fixed by statute whatever vote the underlying transaction required. Fiduciary-cleansing regimes, such as Delaware's controller doctrine, set no ratification threshold on this axis at all: where the minority vote serves only to cleanse a fiduciary conflict, an ex post vote at most shifts the burden of review rather than confirming the act, and its force runs through judicial credit rather than through r.[6] The residual court-credit risk, namely a court declining to credit a purchased or insufficiently informed ratification vote, is captured by σ. When r moves with π, this feedback creates a disciplinary channel that is absent from one-stage models.

Unless otherwise stated, the model uses

$$\alpha \in (0,1), \quad \pi \in [1/2,1), \quad r \in (0,1], \quad \sigma \in [0,1), \quad \lambda = 1/N \in (0,1),$$

with $\theta \in [\underline{\theta}_0, \overline{\theta}_0]$ and message $m \in \mathbb{R}$. The misrepresentation gap is

$$g := m - \theta,$$

and at the boundary we write $g^* := m^{\text{req}}(u) - \theta^*$. In the ratification-vote microfoundation, recovery satisfies $R \in [0,1]$ and cleansing is feasible only if

$$0 < r \leq 1 - \sigma.$$

For $r > 1 - \sigma$, the microfounded cleansing cost is unbounded, so those points are excluded from the comparative-statics corridors. Accordingly, whenever the paper specializes to that microfoundation and differentiates with respect to r or σ, the analysis is restricted to the feasible interior region.

3.2 Controlling Shareholders

3.2.1 Project Quality

Suppose that there is a single *controlling shareholder* (CS) who permanently owns a strictly positive but non-unanimous voting stake $\alpha \in (0,1)$, while a large but finite number N of symmetric minority shareholders—discussed below—hold the residual $1-\alpha$.

We denote the quality of a project by the variable θ. It is realised once, observed privately by the CS, and thereafter remains fixed throughout the game. The type θ is distributed $\theta \sim F$, where F

[6] This authorization baseline is analytically separate from the fiduciary-cleansing doctrine that governs the standard of review for conflicted-controller transactions, collected with its authorities in Section 6. The model does not tie r to that doctrine; a court's residual discretion to deny a vote cleansing credit enters only through σ.

is continuous on the compact interval $\left[\underline{\theta}_0, \overline{\theta}_0\right] \subset \mathbb{R}$ and the support endpoints satisfy

$$[\,\underline{\theta}_0,\ \overline{\theta}_0], \qquad \underline{\theta}_0 < 0 < \overline{\theta}_0. \tag{P1}$$

We assume the following.

Assumption 1 (Log-concavity and monotone shoulders (P2))**.** *The type distribution F has a density $f := F'$ that is strictly positive on $[\underline{\theta}_0, \overline{\theta}_0]$ and* log-concave*:*

$$f(x) > 0 \quad \textit{and} \quad \log f(\cdot) \textit{ is concave on } [\underline{\theta}_0, \overline{\theta}_0].$$

Let $\theta_{\text{mode}} \in (\underline{\theta}_0, \overline{\theta}_0)$ denote a mode of f (unique when f is strictly log-concave; the flat uniform density used in several specializations is the degenerate log-concave case with zero-slope shoulders, where the mode is not unique). Then f has monotone shoulders*: it is* nondecreasing *on $[\underline{\theta}_0,\ \theta_{\text{mode}}]$ (left shoulder) and* nonincreasing *on $[\,\theta_{\text{mode}},\ \overline{\theta}_0]$ (right shoulder). Under the affine baseline introduced below, the mode is normalized to the majority-of-the-minority bar, $\theta_{\text{mode}} = u_{\text{MoM}} = 0$.*

3.2.2 Private Benefits (pure perk + diversion)

Implementing the project gives the controller private benefits $B(\theta)$ on top of the pro-rata payoff $\alpha\theta$. We explicitly decompose

$$B(\theta) \;=\; b_0 \;+\; D(\theta),$$

where $b_0 \in \mathbb{R}$ is a *pure perk* (does not reduce minorities' payoff) and $D(\theta)$ is the diversionary component that does. We assume

$$D \;:\; [\,\underline{\theta}_0, \overline{\theta}_0] \;\longrightarrow\; \mathbb{R}, \qquad D \in C^1, \qquad 0 < B'(\theta) < 1 - \alpha \quad \big(\forall\, \theta\big), \tag{P3}$$

and fix the additive constant of D by normalization. (The affine baseline of §5.2.6 sets $D(\theta) = b_1\theta$, i.e. $D(0) = 0$, which yields $u_{\text{MoM}} = 0$; the alternative normalization $D(\underline{\theta}_0) = 0$ used in some Online Appendix limit arguments differs only by the constant $b_1\underline{\theta}_0$, absorbed into the perk b_0, and changes no comparative static.) Since b_0 is constant, we have, without loss of generality, $B'(\theta) = D'(\theta)$ on $[\underline{\theta}_0, \overline{\theta}_0]$. The controller's prize uses the full $B(\theta) = b_0 + D(\theta)$, while minorities subtract only $D(\theta)$. Hence the net minority payoff

$$(1 - \alpha)\,\theta \;-\; D(\theta)$$

is strictly increasing on $[\underline{\theta}_0, \overline{\theta}_0]$ by (P3), a property used in the single-crossing arguments that follow. Condition (P3) has a marginal reading. A one-unit rise in true quality increases the controller's private benefit by $B'(\theta)$ and the minority's pro-rata stake by $1-\alpha$; because $B'(\theta) < 1-\alpha$, the diversion the controller skims grows more slowly than the value the minority gains, so the minority's net payoff $(1 - \alpha)\theta - D(\theta)$ still rises in θ and both sides prefer a better project. The controller's incentive at the disclosure stage is therefore to talk the project up rather than down. If $B'(\theta) > 1 - \alpha$ (the diversion–dominant case, deferred from the main text), the acceptance criterion reverses.

After observing θ, the controller sends a publicly observable message $m \in \mathbb{R}$; disclosure itself is costless ex ante, but any misrepresentation exposes the controller to expected Stage-2 enforcement costs (litigation, settlement, or failed ratification). Section 4 presents that enforcement mechanism first and derives from it the reduced-form enforcement cost $C(m, \theta) \equiv L(g, r, \sigma)$, so the signaling cost the controller faces is an equilibrium object of the enforcement subgame rather than a primitive; the induced specification (S1)–(S3) carried through the signaling analysis is stated in §4.1.

4 Stage 2: Litigation, settlement, and ratification technology

We now describe the enforcement mechanism that operates after Stage-1 approval and derive from it the reduced-form enforcement cost $L(g, r, \sigma)$ that the Stage-1 disclosure problem inherits; the induced cost specification is stated in §4.1 at the end of the section.

Define the *misrepresentation gap* as $g := m - \theta$ (equivalently, $g := m^{\text{req}}(u) - \theta$ for the minimal persuasive lie). Let $r \in (0, 1]$ denote the *ratification/settlement vote threshold* (the fraction of non-conflicted shareholders whose approval is needed to extinguish claims). In the carry-forward baseline, $r = \pi$, the same quota as the Stage-1 approval rule; more generally, $r = r(\pi)$ with $r'(\pi) \geq 0$.

After approval and extraction, minority shareholders may file suit. Consider how the mechanism works. Foremost, the *probability of suit* $p(g, r)$ is increasing in $|g|$: a larger gap between the controller's disclosure and the true project quality is stronger prima facie evidence of misrepresentation, which encourages plaintiff attorneys and reduces the controller's likelihood of early dismissal. Next, the controller can attempt to extinguish claims by buying shareholder support for a *ratification vote* at threshold r (analogous to familiar cleansing logic for informed, disinterested stockholder votes). Because more shareholders must be persuaded, ratification requires the controller to compensate enough minority shareholders to assemble an r-supermajority—effectively a settlement-support cost that is increasing in r. Finally, courts may decline to credit a ratification vote as cleansing when the vote is not fully informed and uncoerced; in the model, a deterministic credit discount $\sigma \in [0, 1)$ captures this, so that only fraction $(1 - \sigma)$ of purchased votes is credited. A larger σ therefore raises the expected enforcement burden and can make cleansing infeasible when $r > 1 - \sigma$.

The distinctive feature of Stage 2 is that the controller actively buys peace through ratification rather than passively waiting to be sued. Three primitives interact multiplicatively. First, a larger gap g raises the probability of suit and makes ratification harder. Second, a tougher threshold r forces the controller deeper into the shareholder resistance distribution. Third, stricter scrutiny σ means purchased votes are more likely discounted. Since all three reinforce each other—large gaps are costliest when the threshold is high and scrutiny is strict—the cost surface exhibits the convexity and supermodularity formalized below.

Solving the Stage-2 subgame backward yields a reduced-form expected enforcement cost $L(g, r, \sigma)$. Its properties follow from the mechanism above and inherit the supermodularity structure standard in complementarity economics (see Amir, 2005, for a survey):

Assumption 2 (Properties of the enforcement cost $L(g, r, \sigma)$)**.** *The enforcement cost function* $L : \mathbb{R} \times (0, 1] \times [0, 1) \to \mathbb{R}_+$ *satisfies:*

- (L1) $L(0, r, \sigma) = 0$, $L_g(g, r, \sigma) > 0$ *for* $g > 0$, *and* $L_{gg}(g, r, \sigma) \geq 0$ *(truthful disclosure is free, and exaggeration has an accelerating effect: suit probability, expected damages, and settlement difficulty all rise with the gap).*
- (L2) $L_r(g, r, \sigma) \geq 0$ *for* $g > 0$, *and* $L_{gr}(g, r, \sigma) \geq 0$ *(a tougher ratification threshold forces the controller to buy more votes at increasing marginal cost, a burden amplified when the gap is larger because more shareholders must be compensated at a higher per-vote price).*
- (L3) $L_\sigma(g, r, \sigma) \geq 0$ *for* $g > 0$, *and* $L_{g\sigma}(g, r, \sigma) \geq 0$ *(stricter judicial scrutiny discounts purchased votes and so raises the enforcement burden, the more so the larger the misrepresentation).*

As a robustness extension, court-credit risk can be made endogenous in the disclosure gap: $\sigma = \sigma(g)$ with $\sigma'(g) \geq 0$, which captures the intuition that courts are less likely to credit ratification votes when the disclosure gap is large. This extension amplifies the litigation channel and relaxes the conditions needed for ratification-cost dominance; details are in the Online Appendix. For tractability, all equilibrium characterizations and proofs in the main text keep σ exogenous.

When extraction is endogenous ($x \in [0, B(\theta)]$), the reduced-form net cost is isomorphic to the fixed-extraction specification; the continuous formulation is in the Online Appendix.

The controller extinguishes claims by compensating the lowest-resistance minority voters until a credited r-supermajority is assembled. Each voter supports settlement once the transfer covers her reservation price $R + v_i$—a common litigation value R plus idiosyncratic resistance $v_i \sim G$—so the controller climbs an upward-sloping supply curve; with court-credit discount σ, cleansing is feasible only on $0 < r \leq 1 - \sigma$. The resulting minimum settlement-support cost $C(R, r, \sigma)$ and its distribution-free sign properties are stated in Lemma 1 and derived in the Online Appendix, §S9.1.

Lemma 1 (Ratification cost – general resistance distribution)**.** *Under the buying-peace model above with* $R \in [0, 1]$, $\sigma \in [0, 1)$, *threshold* $r \in (0, 1]$, *and resistance distribution* G *with continuous positive density* g_v *on* $(\underline{v}, \bar{v})$. *Define the settlement-support cutoff, for* $0 < r \leq 1 - \sigma$,

$$v^*(r, \sigma) \;:=\; G^{-1}\left(\frac{r}{1 - \sigma}\right),$$

and the kink threshold

$$r_0(R, \sigma) \;:=\; (1 - \sigma)\, G(-R),$$

with the convention that $r_0 = 0$ *if* $-R \leq \underline{v}$.

The controller's minimum ratification expenditure is:

$$C(R, r, \sigma) \;=\; \begin{cases} \infty, & r > 1 - \sigma, \\ 0, & 0 < r \le r_0(R, \sigma), \\ \displaystyle\int_{\max\{\underline{v}, -R\}}^{v^*(r,\sigma)} (R + v)\, g_v(v)\, dv, & r_0(R, \sigma) < r \le 1 - \sigma. \end{cases} \tag{1}$$

On the active interior region $r_0(R, \sigma) < r < 1 - \sigma$ where $R + v^* > 0$, the following sign conditions hold:

$$\text{(a)}\quad C_R \;=\; G(v^*) - G\big(\max\{\underline{v}, -R\}\big) \;>\; 0, \qquad \text{(b)}\quad C_{RR} \;=\; g_v(-R)\,\mathbf{1}\{-R > \underline{v}\} \;\ge\; 0,$$

$$\text{(c)}\quad C_r \;=\; \frac{R + v^*}{1 - \sigma} \;>\; 0, \qquad \text{(d)}\quad C_\sigma \;=\; \frac{r\,(R + v^*)}{(1 - \sigma)^2} \;>\; 0,$$

$$\text{(e)}\quad C_{Rr} \;=\; \frac{1}{1 - \sigma} \;>\; 0, \qquad \text{(f)}\quad C_{R\sigma} \;=\; \frac{r}{(1 - \sigma)^2} \;>\; 0.$$

In particular, $C_{RR} > 0$ whenever $-R \in (\underline{v}, \bar{v})$, i.e. whenever some voters have negative reservation values. At the kink $r = r_0(R, \sigma)$, right-derivatives in r are zero and all inequalities hold weakly.

Proof. Proof in the Online Appendix, §S9.1. □

The controller faces an upward-sloping supply curve of ratification votes: more votes (higher r) or greater judicial discounting (higher σ) force him deeper into the resistance distribution. The key takeaway is that the comparative-statics signs are determined at the pivotal voter rather than by the shape of G. Since the density $g_v(v^*)$ cancels in every Leibniz evaluation at the pivotal voter, the marginal cost of a tougher threshold is governed by the pivotal voter's reservation price $R + v^*$ and not by the density of voters near the margin (with the single kink exception noted in Remark 4).

Below the kink ($r \le r_0$), the controller extinguishes claims at zero cost, so that enforcement is toothless. The disciplinary force of Stage 2 activates only when r pushes past the kink into the active region. This is the core argument for coupling: under $r(\pi) = \pi$, raising the approval bar simultaneously pushes enforcement from the zero-cost into the active region, so the two margins move together.

Remark (Distribution-free signs). The sign conditions $C_r > 0$, $C_\sigma > 0$, $C_{Rr} > 0$, and $C_{R\sigma} > 0$ hold for any continuous resistance distribution G with positive density. The density $g_v(v^*)$ cancels in every Leibniz evaluation because the marginal cost of a tougher threshold is determined by the pivotal voter's reservation price $R + v^*$, not the density of voters near the margin. The one exception is $C_{RR} = g_v(-R)\,\mathbf{1}\{-R > \underline{v}\}$, which depends on the density at the kink rather than at the pivotal voter. The cancellation reflects two modeling choices—the additive reservation price $R + v_i$ and the deterministic court-credit factor $1 - \sigma$—rather than a distribution-free property of vote-buying as such; under a stochastic credit or an interaction between R and v_i the Leibniz

cancellation would fail and the signs would become density-weighted. See the Online Appendix (§S9.4) for the full economic argument.

Under uniform resistance $v_i \sim \text{Unif}[-1, 1]$, the ratification cost admits a closed-form quadratic expression with all derivative signs confirmed; see the Online Appendix (§S9.3) for the statement and proof.

Define the *recovery at stake* as $R(g, x) = \ell(g) \cdot x$, where $\ell : \mathbb{R}_+ \to \mathbb{R}_+$ is increasing and convex in $|g|$ (litigation exposure per unit extracted). The reduced-form enforcement cost is then

$$L(g, r, \sigma) \;:=\; p(g, r) \cdot C\big(R(g, x^*),\; r,\; \sigma\big),$$

where $p(g, r)$ is the probability of suit and x^* is the optimal extraction. On the feasible cleansing region $r \le 1-\sigma$, C is increasing-convex in R and increasing in r, σ on the active branch. Lemma 6 in the Online Appendix gives explicit sufficient conditions under which the composite $L(g, r, \sigma)$ inherits (L1)–(L3) on the up-wedge $g \ge 0$. For $r > 1 - \sigma$, cleansing is infeasible and the effective enforcement burden is unbounded in this microfoundation.

The reduced-form $L(g, r, \sigma)$ treats the gap as the sufficient statistic for enforcement costs, integrating over litigation paths, which is a standard approach (Bebchuk, 1984).

An alternative settlement-bargaining microfoundation (cf. Bebchuk, 1984) confirms (L1)–(L3) under stated monotonicity, zero-at-zero, and convexity conditions; see the Online Appendix.

4.1 The induced signaling cost

We can now state the disclosure technology that Stage 2 induces. After observing θ, the controller sends a publicly observable message $m \in \mathbb{R}$. Disclosure itself is *costless ex ante*; any misrepresentation, however, exposes the controller to the expected enforcement costs just derived, so the controller's Stage-1 problem carries a reduced-form *enforcement cost* $C(m, \theta) \equiv L(g, r, \sigma)$ in the misrepresentation gap $g := m - \theta$, the ratification threshold r, and the court-independence parameter σ. For notation, we occasionally use $\Delta := m - \theta$ inside wedge-by-wedge derivative formulas; g and Δ denote the same object.

Throughout the signaling analysis we work with the piecewise-smooth, convex, gap-based parametric form (kinked at the truth) (cf. the lying-cost formulation in Kartik, 2009), whose parameters $\eta > 0$ and $\kappa \ge 0$ are enforcement-cost primitives increasing in the ratification threshold r and the court-independence parameter σ (Assumption 2). Specifically, for all $(m, \theta) \in \mathbb{R}^2$:

$$C(m, \theta) \;\equiv\; L(g, r, \sigma) \;=\; \eta(r, \sigma)\,|m - \theta| \;+\; \frac{\kappa(r, \sigma)}{2}\,(m - \theta)^2, \qquad \text{with } C(\theta, \theta) = 0. \tag{S1}$$

Equivalently, writing $\Delta := m - \theta$,

$$C(m,\theta) = \begin{cases} \eta(r,\sigma)\,(\theta - m) \;+\; \dfrac{\kappa(r,\sigma)}{2}\,(m-\theta)^2, & m < \theta, \\ 0, & m = \theta, \\ \eta(r,\sigma)\,(m - \theta) \;+\; \dfrac{\kappa(r,\sigma)}{2}\,(m-\theta)^2, & m > \theta. \end{cases} \tag{S2}$$

This form satisfies (L1)–(L3). The parameter $\eta(r,\sigma)$ captures the *fixed marginal exposure* from any non-zero misrepresentation (the per-unit cost of litigation risk at the origin of the gap), and it is increasing in both r and σ because a higher ratification threshold or more stringent independence scrutiny makes it costlier to buy peace even for a small misrepresentation. The parameter $\kappa(r,\sigma)$ governs the *convexity* of the cost in the gap, reflecting the accelerating difficulty of defending or settling larger misrepresentations; we have $\kappa_r \geq 0$ and $\kappa_\sigma \geq 0$ because harder ratification or stricter scrutiny makes the settlement probability fall faster as the gap grows. When r and σ are held fixed, all subsequent results reduce to the original signaling-cost model. The novel channel arises when $r = r(\pi)$ (carry-forward baseline), so that tightening π simultaneously raises $\eta(r,\sigma)$ and $\kappa(r,\sigma)$.

Remark (Nesting). The parametric form $\eta(r,\sigma)|g| + \frac{\kappa(r,\sigma)}{2} g^2$ is a tractable kinked local approximation to the enforcement cost on the up-wedge $g \geq 0$, not a smooth Taylor expansion: the $\eta|g|$ term carries a positive one-sided marginal exposure at the truth, which the composite $L(g,r,\sigma)$ inherits when the suit probability turns on linearly at $g = 0$. All propositions stated under Assumption 2 remain valid under the ratification-vote microfoundation on the active branch $r > r_0(R(g),\sigma)$, where the sufficient conditions of Lemma 6 deliver (L1)–(L3) with $L_g > 0$ for $g > 0$; at and below the kink the inequalities hold weakly and the ratification channel is inactive. The microfoundation provides additional closed-form structure but does not alter the general results.

4.1.0.1 Derived properties of the cost specification.

Everything the proofs use about C follows from (S1)–(S2) by direct computation. The function is convex in the gap Δ (the sum of the convex map $\Delta \mapsto |\Delta|$ and the quadratic $\Delta \mapsto \frac{\kappa}{2}\Delta^2$), symmetric in the sense $C(\theta + \Delta, \theta) = C(\theta - \Delta, \theta)$, and C^∞ on each wedge $\{\Delta \neq 0\}$ with a single kink at $\Delta = 0$. Because C depends on (m,θ) only through $\Delta = m - \theta$, the type derivative is $C_\theta = -C_m$ off the kink; no sign restriction on C_θ is imposed, and the comparative statics that follow refer directly to C_m on the relevant wedge. The wedge derivatives, cited throughout as (S3), are

$$C_m(m,\theta) = \begin{cases} -\eta + \kappa\Delta, & \Delta < 0, \\ \eta + \kappa\Delta, & \Delta > 0, \end{cases} \qquad C_\theta = -C_m, \qquad C_m(0^-) = -\eta, \quad C_m(0^+) = +\eta, \tag{S3}$$

with subgradient $C_m(0) \in [-\eta, \eta]$ at the truth; hence $|C_m(\Delta)| \geq \eta$ for all $\Delta \neq 0$, and $|C_m(\Delta)|$ is weakly increasing in $|\Delta|$.

Three features of (S3) do the economic work in what follows. The strictly positive *linear lip*

of size η at the truth is the marginal enforcement cost of the *first* unit of misrepresentation: because even an infinitesimal exaggeration already exposes the controller to a discrete jump in litigation risk, the marginal cost of lying is bounded away from zero at the truth, so the first increment of exaggeration is never free. This lip sharpens the separation at the margin; it is the convex cost of the full gap, together with the endpoint and single-crossing structure of the lying-profit function, that makes the lowest types—those facing the largest gaps—separate truthfully into a rejected region of positive measure. The *tail growth* of the cost, $C(\Delta) \to \infty$ as $|\Delta| \to \infty$ whenever the feasible message set allows gaps of unbounded size, makes sufficiently large lies eventually prohibitive; when the feasible message set implies a bounded gap envelope, tail growth need not trigger separation and a finite-bar dominance test is required instead (see the Online Appendix). Finally, the *down-wedge* cost is never incurred on the equilibrium path: the litigation microfoundation prices over-optimistic disclosure, and equilibrium poolers over-report, so the symmetric under-reporting cost serves only as an off-path signaling discipline securing the uniqueness arguments (Lemma 4 in the Online Appendix). A one-sided cost that is zero on the down-wedge would leave the up-talk allocation unchanged but weaken those uniqueness results to allocation uniqueness under a boundary tie-break.

The specification (S1)–(S2) thus says that exaggeration is costly, with a discrete jump at truth-telling and a cost that grows convexly in the size of the lie. Crucially, the wedge properties (S3) are derived rather than assumed as primitives: they come from the Stage 2 enforcement technology just described, where η and κ inherit their values from the ratification threshold r and judicial scrutiny σ.

This is what distinguishes the model from standard cheap-talk or costly-signaling setups (cf. Kartik, 2009): the cost function has institutional content. Its parameters encode the difficulty of buying peace through ratification (r-sensitive) and the probability that courts refuse to credit the outcome (σ-sensitive). When the carry-forward baseline links r to π, tightening the vote simultaneously steepens the cost surface, an interaction that no one-stage model can generate. Two further consequences follow. Foremost, enforcement costs are only partially deadweight: some fraction represents transfers, so the welfare cost of pooling is governed by the deadweight parameter ω. Next, changing the screening rule simultaneously changes the cost surface, which is impossible when signaling costs are exogenous.

4.2 Controlling Shareholder's Utility Function

Given the primitives in Sections 3.2.1 and 4.1, and the voting cut-off in (V2), define the rule-specific approval indicator

$$\mathbf{1}\{\text{approve}(m)\} := \mathbf{1}\Big\{(1-\alpha)\,\mu(m) \;-\; \beta^{\text{Div}}(m) \;\geq\; \kappa^{\text{rule}}(\lambda)\Big\} \;=\; \mathbf{1}\{\mu(m) \;\geq\; u\}.$$

where u is the belief cut-off from (V3). The controlling shareholder chooses a message anticipating minority voting and the expected Stage-2 enforcement cost:

$$\max_{m} \; U_{CS}(m,\theta) \;=\; \mathbf{1}\{\text{approve}(m)\} \cdot \Big(\alpha\,\theta + B(\theta) - C(m,\theta)\Big), \tag{C1}$$

where $C(m, \theta) \equiv L(g, r, \sigma)$ is the reduced-form enforcement burden from Stage-2 litigation (Section 4), with $g = m - \theta$ the misrepresentation gap, r the ratification threshold, and σ the court-independence parameter. When r and σ are fixed, the Stage-1 problem reduces to a signaling model with a lying cost borne on approval; since truthful disclosure carries no enforcement cost ($C(\theta, \theta) = 0$), charging C only upon approval leaves the pooling boundary unchanged from the unconditional formulation, and the novel interaction arises under the carry-forward baseline $r = \pi$.

4.3 Controlling Shareholder's Participation Constraint

Absent additional structure, Assumption P3 together with the payoff criterion in (C1) implies that a controller has little incentive to initiate projects with $\theta \leq 0$. To ensure well-posed participation at the voting cut-off, we adopt the affine specification $B(\theta) = b_0 + b_1\theta$ with $b_1 \in (0, 1 - \alpha)$ and choose b_0 large enough that the controller finds it (weakly) profitable to proceed at the bar.[7] This is encoded in the following assumption.

Assumption 3 (Controller participation at the rule cut-off (CPC) and interior scope)**.** *Maintain Assumptions P1–P3, S1–S2, and M1. In the baseline, take $B'(\theta) \in (0, 1 - \alpha)$ and $B(\theta) = b_0 + b_1\theta$ with $b_1 \in (0, 1-\alpha)$. For any voting environment that induces a belief cut-off u via (V2)-(V3), require the controller's participation constraint at the cut-off u to hold:*

$$\alpha\, u + B(u) \;\geq\; 0. \tag{CPC}$$

Under $B(\theta) = b_0 + b_1\theta$ this is equivalent to

$$(\alpha + b_1)\, u + b_0 \;\geq\; 0 \qquad \Longleftrightarrow \qquad u \;\geq\; u_{\text{CPC}} := -\frac{b_0}{\alpha + b_1}.$$

Since $\alpha + b_1 > 0$, (CPC) imposes a lower bound u_{CPC} on admissible cut-offs. In the baseline, b_0 is chosen so that $u \geq u_{\text{CPC}}$ along every interior corridor of cut-offs considered, which ensures the controller participates at the bar. Notably, under $u_{\text{MoM}} = 0$, CPC reduces to $b_0 \geq 0$, so the pure perk can alone ensure participation at the MoM bar. Two features of this specification are modeling assumptions rather than derived properties. First, the controller's willingness to pursue some value-reducing deals ($\theta < 0$) is secured by taking b_0 large enough to satisfy (CPC). Second, b_0 is taken to leave the minority's payoff unchanged, so that it shifts the controller's participation margin without entering the minority approval test. Were part of b_0 instead borne by minorities, the MoM bar and the pooling region would shift accordingly.

[7] Henceforth, we refer to the "bar" as the belief cut-off, u defined in V3.

5 Stage 1 equilibrium given Stage 2 enforcement costs

5.1 Minority Shareholders

5.1.1 Minority Shareholder Voting and Posterior Beliefs

As established in Section 3.2.1, a large but finite number N of symmetric minority shareholders collectively hold the residual $1-\alpha$ of the project's valuation, θ. To economize on notation, we define the variables s and λ:

$$s \;=\; \frac{1-\alpha}{N}, \qquad \lambda \;:=\; \frac{s}{1-\alpha} \;=\; \frac{1}{N} \;\in\; (0,1).$$

Each minority shareholder therefore owns the fraction λ of the residual stake and, correspondingly, is exposed to a λ-proportion of any private benefits extracted by the controlling shareholder. We assume $\sum_{n=1}^{N} \lambda\, D(\theta) = D(\theta)$; that is, only the *diversionary* (zero-sum) component is borne pro-rata by minorities, while the pure perk b_0 does not reduce minorities' payoff. Except where noted, we work going forward with a single representative minority shareholder, without loss of generality.

5.1.2 Beliefs and Updating

The information structure mirrors the institutional structure of proxy voting. The controller (sender) is fully strategic, and its message m is the disclosed deal value in the proxy materials; minority shareholders (receivers) are atomistic and vote on the disclosed terms, reading the disclosure through a *primitive*, monotone perception $\mu(m)$—under the identity calibration below, at face value—rather than through the equilibrium Bayesian posterior that a fully strategic receiver would compute. The truth of the disclosure is policed by the legal system rather than by the electorate: courts price the objective gap $g = m - \theta$ ex post through the enforcement cost $L(g, r, \sigma)$, independent of voter sophistication. The device is in the spirit of Levit et al. (2024) and of the rational-inattention approach to voting (Sims, 2003; Matějka and Tabellini, 2021).[8]

[8] Were minority voters fully sophisticated, inverting the controller's equilibrium strategy rather than reading the message through the perception μ, a Bayesian receiver observing the pooled message $m^{\text{req}}(u)$ would infer $\mathrm{E}[\theta \mid \theta \in [\theta^*(u), u)] < u$ and reject; the atomistic-voter assumption is therefore load-bearing for the pooling equilibrium. The reading we maintain here is positive, since the results concern dispersed, inattentive minority electorates, for which $\mu(m)$ is naturally the coarse posterior of a rationally inattentive voter (Sims, 2003; Matějka and Tabellini, 2021) who optimally does not acquire the information needed to undo the controller's inflation. Under fully strategic voters the model becomes a costly-signaling problem of the Kartik (2009) type, with boundary pooling at the lowest approvable message and the approval test applied to the equilibrium posterior; there a higher ratification standard still raises the cost of the inflation that a stricter bar demands and pushes toward separation. Proposition 3 establishes that the reversal survives fully Bayesian voters as a perfect-Bayesian-equilibrium result in the general log-concave environment, in the threshold-pooling equilibrium it characterizes (proof in the Online Appendix, §S9.6): Bayesian consistency forces the pooled message above the bar, enlarging the equilibrium misrepresentation gap, so the ratification channel binds weakly harder there; in the parametric example the Bayesian equilibrium reverses at primitives

In the core model we adopt the *identity calibration* $m_{\text{req}}(u) = \mu^{-1}(u) = u$ (the marginal voter reads disclosure at face value over the type support), so that the belief cut-off u, the minimum approvable message, and the marginal truthful type coincide; §5.1.3 and the Online Appendix relax this to a general monotone μ satisfying (M1)—including the attention-weighted effective posterior—carrying the message cut-off $m_{\text{req}}(u) = \mu^{-1}(u)$ separately. Each shareholder observes the disclosed message m and votes according to the aggregation rule (V2). Shareholders do not reverse-engineer the controller's equilibrium strategy to update further. This is a maintained behavioral assumption about a dispersed, largely inattentive electorate, since empirically most minority shares are not actively voted. The partial-attention microfoundation developed below shows that the assumption can be reconciled with Bayesian updating, but only on a restricted range: when attention is partial the effective belief is pulled toward the prior mean $\mathrm{E}[\theta]$, which sustains approval under Bayesian updating only where the prior mean lies strictly above the bar ($\mathrm{E}[\theta] > u$) and attention is low enough (the exact threshold on τ is given in the Online Appendix, §S9.5). At the stringent bars $u \geq \mathrm{E}[\theta]$ that drive the paper's reversal the rescue fails, so there approval rests on the atomistic-voter assumption itself rather than on the microfoundation. The Online Appendix (§S9.5) provides a detailed discussion.

For each minority shareholder, a *belief system* specifies, for each publicly observed message $m \in \mathbb{R}$, a posterior distribution $\rho(\cdot \mid m) \in \Delta([\underline{\theta}_0, \overline{\theta}_0])$ over types. On the equilibrium path, the benchmark attentive Bayesian posterior is obtained by Bayes' rule from the prior F and the sender's message strategy $m(\cdot)$;[9] off the equilibrium path we adopt the Grossman–Perry *credible-beliefs* refinement (Online Appendix, §S1): upon observing an unexpected message, minorities place probability 1 on the set of types that (weakly) benefit most from that deviation (Grossman and Perry, 1986); when a specific supporting belief is exhibited, we use the canonical conditional-prior selection, the prior restricted to that set.

We summarize the fully attentive Bayesian posteriors by their expectations:

$$\bar{\mu}(m) := \mathrm{E}[\theta \mid m], \qquad \beta(m) := \mathrm{E}[B(\theta) \mid m], \qquad \beta^{\mathrm{Div}}(m) := \mathrm{E}[D(\theta) \mid m].$$

The controller's payoff uses $\beta(m)$ (full B), while the voting test for minorities uses $\beta^{\mathrm{Div}}(m)$ (diversion only).

Not all shareholders are attentive. Let $\tau \in (0, 1]$ denote the fraction of minority shares (equivalently, voters) that process the controller's message m.[10] The remaining $1 - \tau$ are inattentive, abstain, or vote mechanically using the prior mean $\mu_0 := \mathrm{E}[\theta]$. The *effective posterior mean* that enters the voting aggregator is

$$\mu(m) := \tau\, \bar{\mu}(m) + (1 - \tau)\, \mu_0.$$

In the general specification, $\mu(m)$ denotes this effective voting statistic, not the fully attentive

where the identity calibration does not.

[9] See Section 5.2.5 for strategy notation.

[10] Broadridge ProxyPulse reports that retail investors vote only a minority of the shares they own (roughly 28–30%, including 28% in the 2025 update), a rate far below institutional voting; see https://www.broadridge.com/_assets/pdf/2025proxypulse-updated.pdf. See also Brav et al. (2022) and Lauterbach and Mugerman (2020).

Bayesian posterior $\bar{\mu}(m)$; the core model adopts the identity calibration $m_{\text{req}}(u) = u$. The mixture implies $\mu'(m) = \tau\,\bar{\mu}'(m)$, so responsiveness is scaled down when attention is low. The benchmark is nested at $\tau = 1$ (so $\mu = \bar{\mu}$).

We assume that:

$$\begin{aligned} &\mu(m) \text{ is } C^1 \text{ with } \mu'(m) > 0 \text{ for all } m; \text{ accordingly,} \\ &\mu^{-1} \text{ exists and is } C^1 \text{ with } (\mu^{-1})'(u) = 1/\mu'(\mu^{-1}(u)) > 0. \end{aligned} \tag{M1}$$

We further impose tail-limit regularity on the attentive benchmark: as messages become arbitrarily small or large, the attentive posterior mean converges to the endpoints of the valuation support, i.e.,

$$\lim_{m\to-\infty} \bar{\mu}(m) = \underline{\theta}_0, \qquad \lim_{m\to+\infty} \bar{\mu}(m) = \overline{\theta}_0. \tag{M1$'$}$$

Under the constant-attention mixture the effective statistic μ then has range $\big(\tau\underline{\theta}_0+(1-\tau)\mu_0,\ \tau\overline{\theta}_0+(1-\tau)\mu_0\big)$, which collapses to the support endpoints at $\tau = 1$; interior bars are taken inside the range of μ.

5.1.3 Endogenous participation and turnout (robustness summary)

The baseline keeps belief responsiveness in reduced form. A turnout microfoundation can be added without changing any core results: let attentive participation be $\tau(m) \in (0, 1]$, with attentive voters using $\bar{\mu}(m) = \mathrm{E}[\theta \mid m]$ and inattentive voters using the prior $\mu_0 = \mathrm{E}[\theta]$. The effective voting belief is

$$\mu(m) = \tau(m)\,\bar{\mu}(m) + \big(1 - \tau(m)\big)\mu_0.$$

When votes are not expected to be close, participation is low and beliefs are sluggish; near the bar, pivotality raises participation and hence responsiveness. This interpretation supports the same comparative statics and maps the local responsiveness parameter in §S4 of the Online Appendix to attention-weighted slope near the boundary (cf. Meirowitz and Pi, 2022, for a related voting-trading dilemma that generates similar sluggishness). The mapping from attention-weighted responsiveness to the local belief slope near the boundary is developed in the Online Appendix, §S4.

5.1.4 Expected Utility from Approving (Interim Stage)

Following Levit et al. (2024) and consistent with evidence on institutional investors' limited engagement in concentrated-ownership settings (Hamdani and Yafeh, 2013), we endow each minority investor i with a privately known bias b_i that perturbs her cut-off for approving the proposal. The bias is normalised as

$$b_i \ \sim\ \text{Uniform}\,[-1,\ 1], \qquad \mathbb{E}[b_i] = 0, \tag{M2}$$

independent of θ and distributed across investors. Using the posterior summaries from Section 5.1.2, investor i's interim utility upon approving is

$$\mathbb{E}[u_i \mid m] = \lambda\big[(1-\alpha)\,\mu(m) - \beta^{\mathrm{Div}}(m)\big] + b_i, \tag{M3}$$

where $G(b) = (b+1)/2$ for $b \sim U[-1,1]$.

5.1.5 Yes/No rule and Pivotal bias

We define

$$t_\lambda(m) \;:=\; -\,\lambda\big[(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m)\big].$$

A minority shareholder i therefore votes "Yes" if and only if $b_i \geq t_\lambda(m)$. Accordingly, the fraction of "Yes" votes is

$$1 - G\big(t_\lambda(m)\big) = \frac{1 - t_\lambda(m)}{2}, \tag{M4}$$

given the distribution of $b_i \sim U[-1,1]$, for $t_\lambda(m) \in [-1,1]$ ($t_\lambda(m)$ saturates at 0 or 1 outside). We treat the minority as a continuum of shares indexed by b_i, so vote shares are deterministic; for finite N, (M4) is the expected-share (law-of-large-numbers) approximation. The cutoff $t_\lambda(m)$ locates the *pivotal* minority shareholder—the voter whose private bias just offsets the perceived net benefit of the proposal and who is therefore (just) indifferent between approval and rejection. A higher quota π demands a larger favorable share and so shifts the pivotal vote to a more demanding shareholder, one whose bias is less favorable to the controller; the controller can restore approval only by raising perceived value $\mu(m)$, which under costly disclosure means sending a larger gap. The threshold thus disciplines the controller by relocating the pivotal voter, and the exaggeration needed to clear the vote is pinned to that voter's reservation.

5.2 Voting Rules

5.2.1 Definition

We consider two types of Voting Rules.

1. Majority of the Minority (MoM). A proposal is implemented if and only if at least one-half of the $(1-\alpha)$ minority shares vote "Yes" (ties are approved, matching the weak inequality in the aggregator (V2) below).

2. Super-majority (SM). Fix a quota $S \in (0.5, 1)$. Implementation requires at least a fraction S of *all* shares in favour. Equivalently, the fraction of minority votes needed is

$$\pi \;=\; \frac{S-\alpha}{1-\alpha}. \tag{V1}$$

An SM rule is defined to be *lenient* when $\pi < 0.5$ and *stringent* when $\pi > 0.5$.[11]

With these primitives fixed, all subsequent analysis derives equilibrium signalling and voting outcomes and compares welfare across the two rule families.

5.2.2 Cut-offs under MoM and super-majority S

For the MoM rule, the cutoff is pinned down by the minority shareholder who is (just) indifferent between voting "Yes" and "No" for a given proposal, conditional on the message m sent by the CS. Using expressions M4 and V1, then, under the MoM rule shareholders vote "Yes" if and only if

$$\frac{1 - t_\lambda(m)}{2} \;\geq\; \tfrac{1}{2} \;\Longrightarrow\; t_\lambda(m) \leq 0 \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta^{\text{Div}}(m) \;\geq\; 0.$$

Similarly, under the SM rule, let $\pi = (S-\alpha)/(1-\alpha)$. Requiring $(1 - t_\lambda(m))/2 \geq \pi$ as per expressions M4 and V1 delivers the condition under which shareholders vote "Yes" if and only if

$$t_\lambda(m) \leq 1 - 2\pi \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta^{\text{Div}}(m) \;\geq\; -\frac{1-2\pi}{\lambda}.$$

We can then define the rule-specific cutoff

$$\underline{\kappa}^{(\lambda)}_{\text{rule}} := -\frac{k_{\text{rule}}}{\lambda}, \quad k_{\text{MoM}} = 0, \quad k_{\text{SM}} = 1 - 2\pi,$$

so that in both cases minorities approve if and only if

$$(1-\alpha)\,\mu(m) - \beta^{\text{Div}}(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\text{rule}}. \tag{V2}$$

5.2.3 Belief Cut-offs u

Fix a voting rule and exposure λ. Let $\kappa^{(\lambda)}_{\text{rule}}$ be the constant in (V2). Define the minimal persuasive message which would pass the proposal given a voting rule:

$$m_{\text{req}} := \inf\left\{ m \in \mathbb{R} \;:\; (1-\alpha)\mu(m) - \beta^{\text{Div}}(m) \;\geq\; \kappa^{(\lambda)}_{\text{rule}} \right\}, \qquad u := \mu(m_{\text{req}}). \tag{V3}$$

[11] Equation (V1) is where nominal and minority-effective thresholds diverge: whenever the controller's own shares count toward the denominator, a formally demanding quota may ask little of the minority. A 60%-controller facing the UK's 75% special-resolution requirement (Companies Act 2006, s. 283) needs only about fifteen percentage points of independent support—$\pi = 0.375$, a lenient rule in the model's classification—whereas a majority-of-the-minority rule excludes the controller's block entirely; at the 90% squeeze-out thresholds (Companies Act 2006, s. 979; Hong Kong Companies Ordinance, Part 13) the required independent support is at its maximum. The comparison of MoM and supermajority rules below accordingly tracks the minority-effective stringency π: the ratification channel operates with force only when the threshold is demanding from the minority's perspective.

We assume throughout that the voting aggregator $\Phi(m) := (1-\alpha)\mu(m) - \beta^{\text{Div}}(m)$ is continuous and strictly increasing and crosses $\kappa_{\text{rule}}^{(\lambda)}$; together with (M1) this makes m_{req} exist and be unique with $m_{\text{req}} = \mu^{-1}(u)$. ($\mu$ increasing alone does not order Φ: under the affine baseline $D(\theta) = b_1\theta$ the diversion perception is $\beta^{\text{Div}} = b_1\bar{\mu}$, so Φ is strictly increasing exactly when the attentive share satisfies $(1-\alpha)\tau > b_1$—automatic at full attention by (P3)—and immediate under the identity calibration of the core model.) Since $\mu(m) = \mathbb{E}[\theta \mid m]$, u is the conditional expected valuation of the pivotal shareholder who induces the proposal to pass. We refer to u as the belief cut-off (threshold) for passage.

Voting is governed by the primitive perception μ satisfying (M1) (§5.1.2); the threshold algebra is in the Online Appendix (§S9.5).

The mechanism is one of persuasion rather than payment. The pivotal voter approves once perceived value $\mu(m)$ clears her reservation u, and under the identity calibration she reads the disclosure at face value, so the controller wins her vote by sending a message above the cutoff. The gap $g = m - \theta$ that this requires is the cost of persuasion at the disclosure stage—a misrepresentation that courts can detect and price ex post through the enforcement cost $L(g, r, \sigma)$—and is a separate object from the transfer the controller later pays to assemble a ratification supermajority at Stage 2.

5.2.4 Regime Orientation

When diversion is modest, $0 < B'(\theta) < 1-\alpha$ (*payoff-dominant, talk-up*), minorities' net payoff $(1-\alpha)\theta - D(\theta)$ is increasing in true quality, so they welcome higher perceived value. The approval test is $\mu(m) \geq u$, any pooling (if present) lies *below* u, and persuasion is *upward*. Equivalently, with $\Phi(m) := (1-\alpha)\mu(m) - \beta^{\text{Div}}(m)$, Φ is strictly increasing (the aggregator condition maintained at (V3)), yielding a unique interior u.

When diversion is steep, $B'(\theta) > 1-\alpha$ (*diversion–dominant*), the approval test flips to $\mu(m) \leq u$ and persuasion is downward. This extension is deferred from the main text.

5.2.5 Interior cut-offs and strategy notation

Standing assumption (interior cut-off): Unless stated otherwise, $u \in (\underline{\theta}_0, \overline{\theta}_0)$.[12] The degenerate cases $u \notin (\underline{\theta}_0, \overline{\theta}_0)$ (all-Yes/all-No) are collected in Corollary 5 (Online Appendix).

Strategy notation: The controller's type-contingent message strategy is a function $m_u : [\underline{\theta}_0, \overline{\theta}_0] \to \mathbb{R}$, written $m_u(\theta)$; when the u-dependence is not important, we write $m(\theta)$.

[12] The standing interior-cutoff assumption keeps the analysis away from the degenerate atomistic limits. As $N \to \infty$, each minority investor's exposure becomes negligible and screening is not worth its cost—the classic free-rider logic in corporate control (Shleifer and Vishny, 1986)—so lenient supermajority rules approach universal approval while stringent rules approach universal rejection, and MoM is neutral; these are boundary artifacts of the $\lambda = 1/N$ scaling that do not affect the comparative statics on the interior of the policy domain. We therefore state the paper's results for fixed finite N with interior u.

5.2.6 Closed-Form Cut-Offs with Affine Private Benefits

Assume $B(\theta) = b_0 + b_1\theta$ with $b_1 \in (0,\ 1-\alpha)$. Write the diversionary part as $D(\theta) = b_1\theta$ (so b_0 is a pure perk). Given any message m,

$$\beta^{\mathrm{Div}}(m) \;=\; \mathrm{E}\big[D(\theta) \mid m\big] \;=\; b_1\,\mu(m).$$

Substituting into the common voting inequality (V2),

$$(1-\alpha)\,\mu(m) \;-\; \beta^{\mathrm{Div}}(m) \;=\; (1-\alpha-b_1)\,\mu(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\mathrm{rule}}.$$

Because $1-\alpha-b_1 > 0$, this is equivalent to the belief threshold

$$\mu(m) \;\geq\; u_{\mathrm{rule}}(\lambda) \;:=\; \frac{\underline{\kappa}^{(\lambda)}_{\mathrm{rule}}}{1-\alpha-b_1} \tag{V4}$$

for each voting rule.

Recalling $\underline{\kappa}^{(\lambda)}_{\mathrm{MoM}} = 0$ and $\underline{\kappa}^{(\lambda)}_{\mathrm{SM}} = -(1-2\pi)/\lambda$, (V4) gives

$$u_{\mathrm{MoM}} = 0, \qquad u_{\mathrm{SM}}(\pi,\lambda) \;=\; -\frac{1-2\pi}{\lambda(1-\alpha-b_1)}.$$

Thus, the pure perk b_0 does not enter the minorities' bar: it affects only the controller's prize and the CPC constraint below.

Keeping (α, b_1) fixed,

$$\frac{\partial u_{\mathrm{SM}}}{\partial \pi} = \frac{2/\lambda}{1-\alpha-b_1} > 0.$$

Hence stricter quotas (higher π) tighten the belief cut-offs u required to pass a given proposal in proportion to $(1-\alpha-b_1)^{-1}$. In a separating equilibrium (so that $\mu(m) = \theta$), the belief thresholds above translate directly into type thresholds, $\theta \geq u_{\mathrm{rule}}(\lambda)$. In a pooling equilibrium, minorities compare the primitive voting statistic $\mu(m)$ at the pooled message with $u_{\mathrm{rule}}(\lambda)$ from V4.

5.3 Equilibrium structure at a fixed bar

We now characterize Stage-1 behavior, taking the Stage-2 enforcement technology as given. The key object is the lying-profit function, which values approval net of expected enforcement costs. We begin with lemmas that apply to *any* voting rule in the baseline talk-up regime.

Maintained calibration (main text). Recall from §5.1.2 that the core model adopts the *identity calibration* $m_{\mathrm{req}}(u) = \mu^{-1}(u) = u$, so that the belief cut-off u, the minimal approvable message, and the marginal truthful type coincide and $(\mu^{-1})'(u) = 1$. We maintain it throughout the main-text equilibrium and welfare analysis (this section and Section 6): the three-region partition of Proposition 1, the pooling mass $P = \int_{\theta^*}^{u} f$, and the boundary-slope bound $\partial\theta^*/\partial u \in (0,1)$ are

stated under this calibration. The general monotone-belief case $m_{\text{req}}(u) = \mu^{-1}(u) \neq u$—including sluggish beliefs, where $(\mu^{-1})'(u) = 1/K > 1$—is developed as a robustness exercise in the Online Appendix (§S4) and summarized in Corollary 1.

5.3.1 Lying Profit Function

We reserve $u \in (\underline{\theta}_0, \overline{\theta}_0)$ for the rule-specific belief cut-off from (V3) and θ for types. Foremost, we redefine the "minimal persuasive message" in V3 as the "cheapest effective lie" $m_{\text{req}}(u)$—the lowest-cost message required to induce approvals by the pivotal minority shareholders given their belief thresholds.

Next, we define the CS's net gain from lying as a "lying-profit function", $\Lambda(\theta; u)$, which is a function both of the project's valuation θ and the cutoff u. A "lying-profit function" determines the payoff to the CS from misrepresenting the true valuation of the project, θ, such that minority shareholders will accept the proposal in equilibrium. We define $\Lambda(\theta; u)$ as:

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big) \;=\; \alpha\,\theta + B(\theta) - L\big(m_{\text{req}}(u) - \theta,\; r, \sigma\big),$$

where the second equality uses $C(m, \theta) \equiv L(g, r, \sigma)$ with $g = m_{\text{req}}(u) - \theta$. When $\Lambda(\theta; u) > 0$ for types just below u, at least some such types strictly prefer to shade up to $m_{\text{req}}(u)$ to secure approval. When r varies with π (carry-forward baseline), Λ acquires an additional dependence on r through the enforcement cost:

$$\Lambda(\theta; u, r) \;:=\; \alpha\,\theta + B(\theta) - L\big(m_{\text{req}}(u) - \theta,\; r, \sigma\big).$$

The lying-profit function asks whether a controller with true quality θ should exaggerate to the bar. High-quality types (close to u) face small gaps and low enforcement costs, so they pool. Low-quality types face large gaps and convex costs, so they separate truthfully and get rejected. The boundary type θ^* is exactly (just) indifferent. Everything flows from how θ^* responds to the policy parameters: when the ratification threshold is coupled to the approval threshold, the enforcement cost surface steepens, pushing θ^* upward and shrinking the pool. Conditional on exaggerating, every pooler sends exactly $m_{\text{req}}(u)$, since any higher message creates a strictly larger gap with zero additional benefit.

5.3.2 General Existence and Uniqueness

Our solution concept is an *atomistic-voter equilibrium* (AVE): the controller's message strategy is sequentially rational given the voting rule, while each atomistic minority voter approves a message according to the primitive monotone perception $\mu(m)$ and the aggregator (V2) (§5.1.2) rather than a Bayesian posterior reverse-engineered from the controller's equilibrium strategy; off-path messages are disciplined by Grossman–Perry credibility. The AVE is the proxy-voting implementation of the model—voting on disclosed terms as read, with the legal system pricing

the gap ex post—and it is conservative for the central result: under a fully Bayesian pivotal voter the reversal survives as a perfect-Bayesian-equilibrium result, with a strictly larger equilibrium gap and a weakly stronger ratification lever (Proposition 3). As noted by Burkart et al. (2024), signaling problems often induce multiple equilibria; we impose Grossman–Perry-credible off-path beliefs for uniqueness, with supporting lemmas in Appendix A. The result is stated in the payoff-dominant up-talk region $0 < B'(\theta) < 1-\alpha$ (P3). Fix any rule that induces the belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0]$. Under the maintained identity calibration $\mu(m) = m$, the cheapest message that clears the bar is $m_{\text{req}}(u) = \mu^{-1}(u) = u$, so that the belief cut-off, the minimal approvable message, and the marginal truthful type coincide; the three-region partition below is stated under this calibration, the general monotone-belief case being treated in the Online Appendix (§S4). Define the lying-profit function

$$\Lambda(\theta; u) \;:=\; \alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big),$$

and set $\theta^*(u) := \sup\{\theta \in [\underline{\theta}_0, u) \,:\, \Lambda(\theta; u) < 0\}$, with the convention $\sup\emptyset := \underline{\theta}_0$. Minorities implement if and only if $\mu(m) \geq u$ (equivalently, if and only if $(1-\alpha)\mu(m) - \beta^{\text{Div}}(m) \geq \underline{\kappa}^{(\lambda)}_{\text{rule}}$ by (V2)).

Proposition 1 (Fixed-bar equilibria have a threshold-pooling form)**.** *Maintain* (CPC) *at the cut-off. At any belief cut-off u, equilibrium disclosure has a three-region, threshold-pooling form, with Grossman–Perry-credible off-path beliefs.*

(i) Existence. There exists an AVE with sender message strategy

$$m(\theta) = \begin{cases} \theta, & \theta < \theta^*(u) \quad \textit{(truth; rejected)}, \\ m_{\text{req}}(u), & \theta \in [\theta^*(u), u) \quad \textit{(cheapest lie; approved)}, \\ \theta, & \theta \geq u \quad \textit{(truth; approved)}. \end{cases}$$

(ii) Allocation uniqueness for a given u. For the same cut-off u, every AVE yields the same implement/reject allocation: the project is implemented if and only if $\theta \geq \theta^*(u)$, up to the indifferent boundary type $\theta^*(u)$ itself (a measure-zero tie; see the proof).

(iii) Pooling vs. separation (endpoint test). Write

$$\Psi_\ell(u) \;:=\; \Lambda(\underline{\theta}_0; u), \qquad \Psi_h(u) \;:=\; \lim_{\theta \uparrow u} \Lambda(\theta; u) \;=\; \alpha u + B(u),$$

where the last equality uses the maintained identity calibration $m^{\text{req}}(u) = u$ and $C(u,u) = 0$. Since $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing on $[\underline{\theta}_0, u]$, exactly one of the following obtains:

(A) Partial pooling: $\Psi_\ell(u) < 0 < \Psi_h(u) \implies \exists!\ \theta^*(u) \in (\underline{\theta}_0, u),\ F(u) - F(\theta^*(u)) > 0.$

(B) Full separation: $\Psi_h(u) \leq 0 \implies \theta^*(u) = u$ *and the pooling set is empty.*

(C) Full pooling below u: $\Psi_\ell(u) \geq 0 \implies \theta^*(u) = \underline{\theta}_0$ *and all $\theta < u$ pool at $m_{\text{req}}(u)$.*

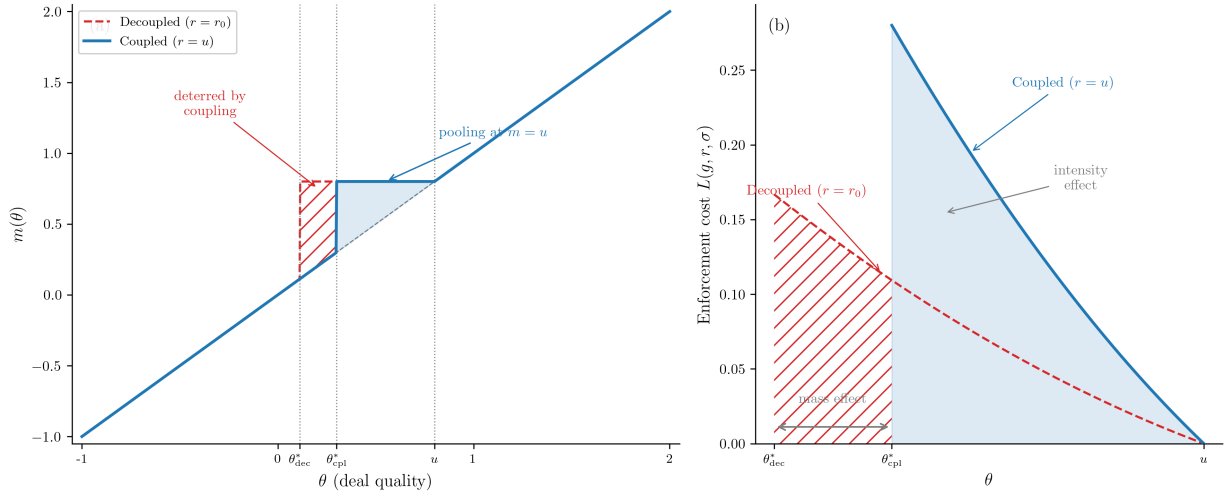
For general monotone beliefs, the upper endpoint is $\min\{m^{\text{req}}(u), \overline{\theta}_0\}$*; the corresponding endpoint test is stated in Lemma 3(iv).*

Proof sketch in main text; full proof in Appendix A.1. The fixed-u allocation follows from strict single-crossing of the lying-profit function, endpoint tests at $\underline{\theta}_0$ and u, and GP-credible beliefs. A complete type-by-type deviation argument is reported in Appendix A.1. □

The equilibrium has a three-region structure. Bottom types ($\theta < \theta^*$) tell the truth and get rejected. Middle types ($\theta \in [\theta^*, u)$) pool at $m_{\text{req}}(u)$—they exaggerate just enough to clear the bar. Top types ($\theta \geq u$) tell the truth and get approved. Single-crossing ensures that this partition is contiguous, since the lying-profit function is increasing in true quality, so the set of exaggerators is always a connected interval.

The pool has a skewed cost structure: every pooler sends $m_{\text{req}}(u)$, so the gap $g(\theta) = m_{\text{req}}(u) - \theta$ is largest at the boundary θ^*, and by convexity of L bottom-of-pool types are disproportionately expensive. When coupling raises θ^*, it trims the pool from the expensive end—the mass effect formalized in Section 6.

Figure 1: Equilibrium message policy and per-type enforcement cost



Panel (a): the equilibrium disclosure function $m(\theta)$ under coupled (solid blue, $r = u$) and decoupled (dashed red, $r = r_0$) regimes at bar $u = 0.8$. Types below θ^* separate truthfully and are rejected; types in $[\theta^*, u)$ pool at $m = u$; types above u separate truthfully and are approved. Coupling raises θ^* (from θ^*_{dec} to θ^*_{cpl}), deterring the hatched types who would have pooled under the lax regime. Panel (b): per-type enforcement cost $L(g, r, \sigma)$ within the pooling region, where $g = u - \theta$. The coupled curve (solid) is steeper—each unit of gap is costlier under tough ratification (intensity effect)—but starts further right because the most expensive types are deterred (mass effect). Parameters: $\alpha = 0.4$, $b_0 = 0.1$, $b_1 = 0.2$, $\sigma = 0.2$, $r_0 = 0.5$. The coupled bar $u = 0.8$ coincides with the cleansing-feasibility boundary $r = 1 - \sigma$, where the ratification cost is maximal but finite.

5.4 Main Result: Carry-Forward Ratification Reverses Pooling Comparative Statics

We now state the paper's central result. Under the carry-forward baseline $r(\pi) = \pi$, the pooling derivative $dP/d\pi$ acquires a *ratification-channel* term absent from any one-stage model; when this term is large enough, it reverses the benchmark prediction that tighter votes increase misrepresentation. The result is stated in the payoff-dominant up-talk regime $0 < B'(\theta) < 1 - \alpha$, maintaining (P1)–(P3), (S1)–(S2), Assumption 2, and (M1), under the maintained identity calibration $m^{\text{req}}(u) = u$ (Corollary 1 relaxes this to general μ, and Proposition 3 restates the reversal under a fully Bayesian pivotal voter); where the ratification-vote microfoundation of Lemma 1 is invoked, statements are evaluated on the feasible corridor $r(\pi) \le 1 - \sigma$. Let $P(\pi) := \int_{\theta^*(u(\pi),r(\pi))}^{u(\pi)} f(\theta)\, d\theta$ denote the pooling mass under approval rule π, with $u(\pi)$ the induced belief cut-off, $r(\pi)$ the ratification threshold ($r'(\pi) \ge 0$), and $g^*(\pi) := m^{\text{req}}\big(u(\pi)\big) - \theta^*$ the equilibrium misrepresentation gap at the boundary. We evaluate the comparative static on an open pooling corridor of rules on which $u(\pi)$ and $r(\pi)$ are C^1 with $u'(\pi) > 0$, the boundary is interior ($\underline{\theta}_0 < \theta^* < u < \overline{\theta}_0$) with $g^*(\pi) > 0$, and f is continuous and positive there; discrete rules such as majority-of-minority are compared by evaluating P at the corresponding cut-offs rather than by differentiating in π.

Proposition 2 (Coupled ratification can make stricter voting reduce pooling)**.** (i) *The pooling derivative acquires a* ratification channel *term:*

$$\frac{dP}{d\pi} \;=\; u'(\pi)\left[\, f(u) - f(\theta^*)\,\frac{\partial \theta^*}{\partial u}\,\right] \;-\; f(\theta^*)\,\frac{\partial \theta^*}{\partial r}\, r'(\pi),$$

where the first bracketed term is the standard bar–boundary effect—a density-weighted race between the cheap near-threshold types swept into the top of the pool at rate $f(u)$ and the expensive boundary types deterred at the bottom at rate $f(\theta^*)\,\partial\theta^*/\partial u$—and the second term is the extra bottom deterrence bought purely by the coupled rise in the ratification standard, which deletes boundary types at rate $f(\theta^*)\,(\partial\theta^*/\partial r)\,r'(\pi)$ while admitting no new poolers at the top.

(ii) *Under the* ratification-cost dominance *condition*

$$\frac{\partial \theta^*}{\partial u} \;+\; \frac{\partial \theta^*}{\partial r}\,\frac{r'(\pi)}{u'(\pi)} \;\ge\; \frac{f(u)}{f(\theta^*)}, \tag{2}$$

the pooling mass $P(\pi)$ is (weakly) decreasing in π along any pooling corridor. The ratification channel binds only on the active branch $r(\pi) > r_0$, where $\partial\theta^*/\partial r > 0$; below the kink $\partial\theta^*/\partial r = 0$ and (2) reduces to the ordinary bar–boundary condition $\partial\theta^*/\partial u \ge f(u)/f(\theta^*)$, which fails under a uniform prior but can still hold under a sufficiently thin right tail (Proposition 9); any reversal below the kink is therefore density-driven rather than enforcement-driven.

(iii) Uniform prior + identity beliefs corollary. *If f is uniform and $\mu(m) = m$ (so $f(u) = f(\theta^*)$ and*

$m^{\text{req}}(u) = u$), *condition* (2) *simplifies to:*

$$\frac{L_r(g^*, r, \sigma)\, r'(\pi)}{u'(\pi)} \;\geq\; \alpha + B'(\theta^*).$$

That is, the marginal tightening of Stage-2 enforcement from ratification must dominate the marginal prize slope $\alpha + B'(\theta^*)$.

Proof. The boundary θ^* solves $\Lambda(\theta^*; u, r) = 0$ for the lying-profit function $\Lambda(\theta; u, r) = \alpha\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big)$ with $C(m, \theta) \equiv L(g, r, \sigma)$; applying the implicit function theorem (using $\Lambda_\theta > 0$ from Lemma 3(i)) gives $\partial\theta^*/\partial u = C_m(\mu^{-1})'(u)/(\alpha + B' + C_m)$, equal to $C_m/(\alpha + B' + C_m) \in (0, 1)$ under the maintained identity calibration $(\mu^{-1})'(u) = 1$ (and in $(0, (\mu^{-1})'(u))$ in general; Lemma 3(iii)), and $\partial\theta^*/\partial r = L_r/(\alpha + B' + C_m) \geq 0$. Differentiating $P(\pi) = \int_{\theta^*}^{u} f\, d\theta$ by Leibniz's rule yields part (i); the sign condition $dP/d\pi \leq 0$ rearranges to (2), giving part (ii); and the uniform/identity specialization ($f(u) = f(\theta^*)$, $m^{\text{req}}(u) = u$) gives part (iii). Full derivation in Appendix A.3. □

The central tension in the result is that the approval threshold does two jobs simultaneously in the coupled regime: it sets the bar for persuasion (how good must the controller's disclosure look to win the vote?) and the bar for peace (how expensive is it to extinguish litigation after the vote?). In the one-stage model the threshold does only the first job, and the benchmark prediction—pooling increases with the threshold—follows from a dampening bound. When u rises by one unit, the bar moves up by one unit, but the boundary θ^* moves up by less than one unit ($\partial\theta^*/\partial u < 1$ under the maintained identity calibration): the prize slope $\alpha + B'(\theta^*)$ absorbs part of the bar increase without translating it into boundary movement, because a higher bar does not make approval more valuable to the marginal type. The bar outruns the boundary, widening the pooling interval.

The two-stage model adds the second job, which pushes in the opposite direction. The ratification channel supplies a second, independent lever on θ^*: the simultaneous increase in r pushes the boundary further upward through $\partial\theta^*/\partial r > 0$ (strict on the active ratification branch), and although this push carries the same $\alpha + B'(\theta^*) + C_m$ denominator as the bar channel, it is a pure cost-side lever—tougher ratification raises the *cost* of the marginal lie through L_r without raising its *prize* or shifting the bar u. The combined movement $\theta^*_u\, u' + \theta^*_r\, r'$ can then exceed u', and once each margin is weighted by the type density it carries, the boundary removes pooling mass at the bottom faster than the bar admits it at the top, so the pooling region contracts. This is the content of ratification-cost dominance: the ratification-channel push ($\theta^*_r\, r'/u'$) must be large enough to compensate for the shortfall ($f(u)/f(\theta^*) - \theta^*_u$) that the dampening force creates. Convexity of enforcement costs ($\kappa > 0$) amplifies the channel, because bottom-of-pool types—those with the largest gaps—bear disproportionately high costs, so the types removed by coupling are precisely those whose deterrence yields the largest per-type enforcement saving. The reversal requires neither sluggish beliefs nor thin tails, only that the litigation/ratification technology be sufficiently sensitive to the ratification threshold.

Remark. When $r'(\pi) = 0$ (ratification threshold decoupled from the approval rule), the second

term in the $dP/d\pi$ formula vanishes and the ratification-cost dominance condition reduces to the standard density-weighted boundary condition of the Online Appendix, §S4 (Proposition 9). The carry-forward baseline $r(\pi) = \pi$ with $r'(\pi) = 1$ represents the polar case in which the litigation channel is maximally active. As shown in Section S5, allowing $\sigma = \sigma(g)$ with $\sigma'(g) \geq 0$ adds the term $L_\sigma\, \sigma'(g)$ to the marginal enforcement cost of lying, which strengthens the litigation channel and relaxes the parameter conditions needed for ratification-cost dominance.

Corollary 1 (General monotone beliefs: the hybrid link)**.** *Outside the maintained identity calibration, under* (M1) *alone, two objects change. Foremost, the truthful-approval boundary is* $m^{\text{req}}(u) = \mu^{-1}(u)$*, which coincides with the belief cut-off* u *only when* μ *is the identity, and the pool's upper limit is the support-truncated image* $\theta^{\top}(u) := \min\{m^{\text{req}}(u), \overline{\theta}_0\}$*; the pooling mass is* $P_\mu(u) = \int_{\theta^*(u)}^{\theta^{\top}(u)} f$*. On the uncapped interior branch* $m^{\text{req}}(u) < \overline{\theta}_0$*,*

$$P'_\mu(u) = f\big(m^{\text{req}}(u)\big)(\mu^{-1})'(u) - f(\theta^*)\,\frac{\partial \theta^*}{\partial u},$$

while on the capped branch $m^{\text{req}}(u) \geq \overline{\theta}_0$ *the upper endpoint is fixed at* $\overline{\theta}_0$*, the* $f(m^{\text{req}}(u))(\mu^{-1})'(u)$ *term drops out, and whether pooling survives is governed by the endpoint test of Lemma 3(iv). Next, the boundary slope carries the belief-responsiveness factor,*

$$\frac{\partial \theta^*}{\partial u} = \frac{C_m\,(\mu^{-1})'(u)}{\alpha + B'(\theta^*) + C_m} \;\in\; \big(0, (\mu^{-1})'(u)\big) \qquad (\textit{Lemma 3(iii)}).$$

Proposition 2 is the specialization $(\mu^{-1})'(u) = 1$*,* $m^{\text{req}}(u) = u$*, in which* $P_\mu = \int_{\theta^*}^{u} f$ *and* $\partial\theta^*/\partial u \in (0, 1)$*. Because* $(\mu^{-1})'(u)$ *multiplies the movement of* both *endpoints and* $\partial\theta^*/\partial u < (\mu^{-1})'(u)$*, the general-*$\mu$ *bar–boundary comparison is inherently density-weighted—under a uniform prior* $P'_\mu(u) > 0$ *on the uncapped interior branch, so belief responsiveness does not by itself contract the pool. Its treatment away from identity, including the role of belief sluggishness, is carried out in the Online Appendix, §S4 (Remark S4.4), where belief sluggishness reproduces the interior-peak pattern of Proposition 9 rather than supplying a monotone reversal. The litigation/ratification channel of Proposition 2 contributes the separate term* $-f(\theta^*)(\partial\theta^*/\partial r)\, r'$*, which contracts the pool independently of that density comparison.*

The reversal also does not depend on the primitive-perception device itself. Suppose the pivotal minority voter is fully Bayesian, so that a message is approved only when the *equilibrium posterior mean* clears the bar, $\mathrm{E}[\theta \mid m] \geq u$, and the unraveling logic of §5.1.2 applies: no approved pool can sit entirely below the bar. The following proposition states the reversal in this setting as a perfect-Bayesian-equilibrium result—within the threshold-pooling class that is the Bayesian counterpart of the paper's boundary-pooling equilibria, under the sender-preferred selection maintained throughout—in the general environment of the paper: log-concave prior (P2), the affine baseline of Assumption 3, and a general enforcement cost satisfying (L1)–(L3).

Proposition 3 (The reversal survives fully Bayesian voters)**.** *Maintain* (P1)–(P3) *with the affine baseline* $B(\theta) = b_0 + b_1\theta$ *and Assumption 2, with* L *continuously differentiable in* (g, r) *on the up-wedge* $g > 0$ *and nonnegative on the down-wedge. Let the pivotal minority voter be fully Bayesian,*

and evaluate on an open corridor with $u(\pi), r(\pi) \in C^1$, $u'(\pi) > 0$, $r'(\pi) \geq 0$, and σ fixed, on which the Bayesian pool is interior and uncapped (condition (BI) in the Online Appendix, §S9.6).

(i) Existence, uniqueness, straddling. Among threshold-pooling profiles with the lowest approvable pooled message, there is exactly one perfect Bayesian equilibrium. Its boundary pair $(\theta_B^*, \hat{m})$ is the unique solution of

$$\mathrm{E}\left[\theta \mid \theta_B^* \leq \theta \leq \hat{m}\right] = u, \qquad \alpha\theta_B^* + B(\theta_B^*) = L\left(\hat{m} - \theta_B^*,\, r, \sigma\right),$$

and satisfies $\underline{\theta}_0 < \theta_B^* < u < \hat{m} < \overline{\theta}_0$: Bayesian consistency forces the pool to straddle the bar.

(ii) Reversal condition. The pooling mass $P(\pi) = F(\hat{m}) - F(\theta_B^*)$ is C^1 and satisfies $dP/d\pi \leq 0$ if and only if

$$\frac{L_r\left(g_B^*, r, \sigma\right) r'(\pi)}{u'(\pi)} \geq \bar{f}\left[\frac{\alpha + B'(\theta_B^*) + L_g}{f(\theta_B^*)} - \frac{L_g}{f(\hat{m})}\right], \qquad \bar{f} := \frac{P}{g_B^*}, \tag{3}$$

where $g_B^* := \hat{m} - \theta_B^*$ and L_g is evaluated at (g_B^*, r, σ). If the pool lies where f is nonincreasing (the right shoulder of (P2)), the bracketed term is at most $\alpha + B'(\theta_B^*)$, so the uniform-form dominance condition of Proposition 2(iii), $L_r\, r'/u' \geq \alpha + B'(\theta_B^*)$, is sufficient; under a uniform prior it is necessary and sufficient.

(iii) The atomistic device is conservative. Fix π and let θ_{id}^* denote the identity-calibration boundary of Proposition 1 with gap $g_{id}^* = u - \theta_{id}^*$, both boundary pairs interior. Then $\theta_B^* > \theta_{id}^*$ and $g_B^* > g_{id}^*$, so (L2) gives $L_r(g_B^*, r, \sigma) \geq L_r(g_{id}^*, r, \sigma)$; whenever the Bayesian pool lies where f is nonincreasing—in particular under a uniform prior—if the uniform-form dominance condition of Proposition 2(iii) holds at the identity calibration, then (3) holds at the Bayesian equilibrium.

Proof. Online Appendix, §S9.6. The uniform-prior parametric example there, together with a nonuniform (triangular-prior) instance, verifies each claim symbolically over an entire policy corridor. □

Bayesian consistency moves the pooled message, and the movement favors the reversal. An approvable pool must straddle the bar: the below-bar types drag the posterior at the pooled message down, so the message must rise above u until the high types swept into the pool pull the posterior back up to the bar. The equilibrium gap is therefore larger than under the identity calibration—at any given boundary type, Bayesian consistency doubles the lie a uniform-prior pool requires—and by (L2) a larger gap makes the enforcement cost more sensitive to the ratification threshold, which is the lever the coupled regime pulls. The comparative statics keep the structure of Proposition 2, with the bar–boundary race now run between the pool's two endpoints: the top admits mass at a rate governed by $f(\hat{m})$, the bottom sheds it at a rate governed by $f(\theta_B^*)$, and the ratification term contracts the pool exactly as before. When the pool sits on the declining shoulder of the prior—the configuration at the stringent bars that drive the paper's applications—the density race reinforces the enforcement channel, and the operative test remains the familiar

$L_r\, r'/u' \geq \alpha + B'(\theta^*)$. The identity calibration of the core model is thus a tractability device that is conservative for the central result: it understates the gap, and with it the enforcement sensitivity, at which the reversal is evaluated.

Corollary 2 (Sharp ratification-cost dominance condition)**.** *Maintain the hypotheses of Proposition 2 and adopt the ratification-vote microfoundation of Lemma 1 with general resistance distribution G and recovery $R(g,x) = \ell(g)\cdot x$ with extraction x^* held fixed in r (so $R = R(g)$), with $p(g,r)$ differentiable in r, on the feasible region $r(\pi) \leq 1-\sigma$ and the active branch $r > r_0(R(g^*),\sigma)$ (one-sided at the kink, where the inequality holds weakly). Under the carry-forward baseline $r(\pi) = \pi$ (so $r'(\pi) = 1$), uniform prior, and identity beliefs, the ratification-cost dominance condition (2) becomes:*

$$\frac{p_r(g^*,r)\,C(R^*,r,\sigma) \;+\; p(g^*,r)\,\dfrac{R^* + v^*(r,\sigma)}{1-\sigma}}{u'(\pi)} \;\geq\; \alpha + B'(\theta^*), \tag{4}$$

where $g^* = m^{\mathrm{req}}(u) - \theta^*$, $R^* := R(g^*,x^*)$, $v^*(r,\sigma) := G^{-1}\big(r/(1-\sigma)\big)$ is the settlement-support cutoff, and C is the ratification cost from Lemma 1. If $r > 1-\sigma$, cleansing is infeasible in this microfoundation and the effective enforcement burden is unbounded. The displayed expression is evaluated on the active branch of Lemma 1, where the cleansing cost is strictly positive ($C > 0$); below that branch $C = C_r = 0$ and the ratification channel is inactive, so the second numerator term is taken as its positive part.

Under uniform resistance $G(v) = (1+v)/2$ on $[-1,1]$ (Example 2), $v^* = 2r/(1-\sigma) - 1$ and $R^* + v^* = \big(2r - (1-\sigma)(1-R^*)\big)/(1-\sigma)$, so the second term in the numerator reduces to $p(g^*,r)\,\big(2r - (1-\sigma)(1-R^*)\big)_+/(1-\sigma)^2$.

Proof. Proof in Appendix B. □

The left side of (4) decomposes the enforcement tightening into two channels. The first, $p_r\, C$, is the *extensive margin*: a tougher threshold raises the probability of suit. The second, $p\, C_r = p\,(R^* + v^*)/(1-\sigma)$, is the *intensive margin*: conditional on suit, assembling the supermajority costs more because the pivotal voter's reservation price is higher—a distribution-free term from Lemma 1. The right side is the dampening force $\alpha + B'(\theta^*)$. Reversal requires both channels jointly to overcome this dampening, and larger r and σ strengthen both. Because the reversal magnitude scales with the boundary gap g^* while a more enforcement-sensitive cost surface pushes θ^* upward and contracts g^*, the dominance condition does not grow without bound in enforcement sensitivity: it is met most readily when the prize slope $\alpha + B'(\theta^*)$ is small—a controller with a low cash-flow stake and low marginal diversion—and the pool is correspondingly wide, with a high court-credit discount σ or cost convexity κ otherwise required to keep g^* and the sensitivity term large. The reversal is therefore a genuine but parametrically demanding phenomenon rather than an automatic consequence of any positive enforcement sensitivity.

Remark (A primitive-level sufficient condition). The dominance condition (4) can be turned into a sufficient condition stated on primitives. On a stringent corridor with $r(\pi) = \pi \leq 1-\sigma$, $r > r_0 := (1-\sigma)\,G(-\underline{R})$ (the general active-branch threshold of Lemma 1 at which the ratification

channel switches on; under the uniform resistance of Example 2 it equals $(1-\sigma)(1-\underline{R})/2$, ensuring $Z := 2r - (1-\sigma)(1-\underline{R}) > 0$), $u'(\pi) \le \overline{u}'$, and $\alpha + B'(\theta^*) \le \overline{A}$, suppose the suit technology and recovery are bounded below along the corridor: $p(g^*, r) \ge \underline{p}$, $p_r(g^*, r) \ge \underline{p}_r$, $R(g^*) \ge \underline{R}$, and the pivotal reservation price $v^*(r, \sigma) \ge \underline{v}$. Then the reversal condition (2) holds whenever

$$\underline{p}_r \, C_{\min} \;+\; \underline{p}\,\frac{\underline{R} + \underline{v}}{1-\sigma} \;\ge\; \overline{u}'\,\overline{A},$$

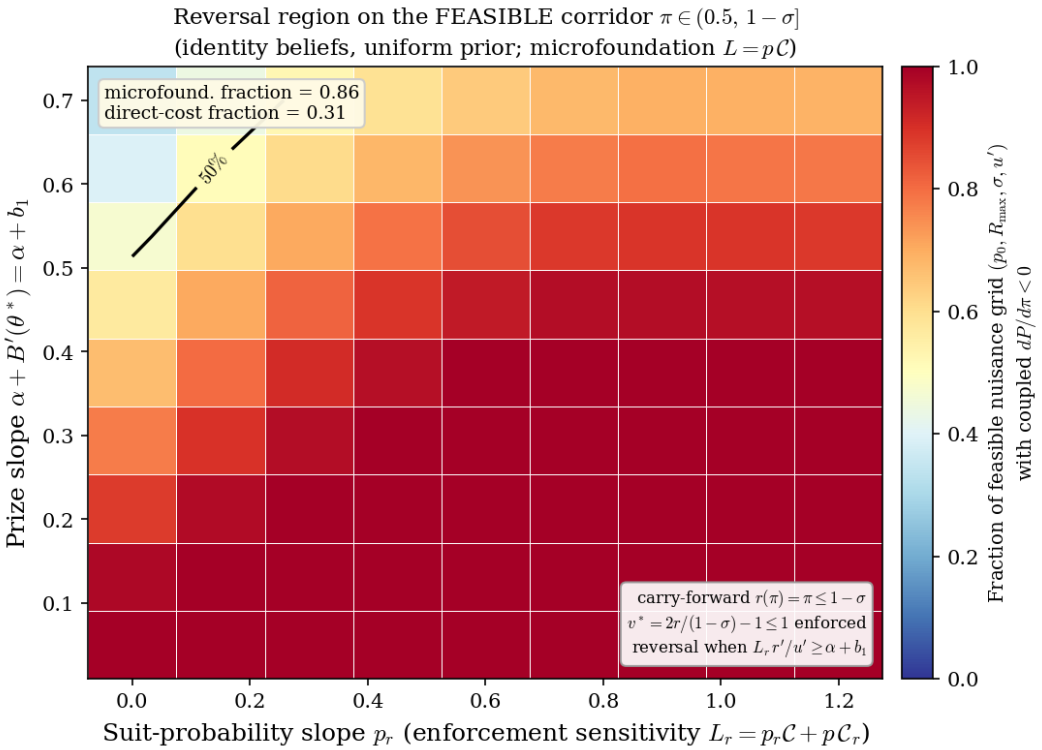
where $C_{\min}$ is the corresponding lower bound on the cleansing cost. Under uniform resistance (Example 2), writing $d = 1 - \sigma$ and $Z = 2r - d(1 - \underline{R})$, the active-branch choice $r > r_0$ makes $Z > 0$, and the condition reduces to the explicit primitive inequality

$$\underline{p}_r \,\frac{Z^2}{4d^2} \;+\; \underline{p}\,\frac{Z}{d^2} \;\ge\; \overline{u}'\,(\alpha + \overline{B'}), \qquad Z > 0.$$

Below the active branch ($Z \le 0$) the ratification channel is inactive, so the left side is read as its positive part and the inequality cannot be met vacuously through the squared term. On any open set of primitives satisfying it, the coupled stricter rule *does* reduce pooling, turning the existence statement of Proposition 2 into a comparative-static prediction on a characterized region.

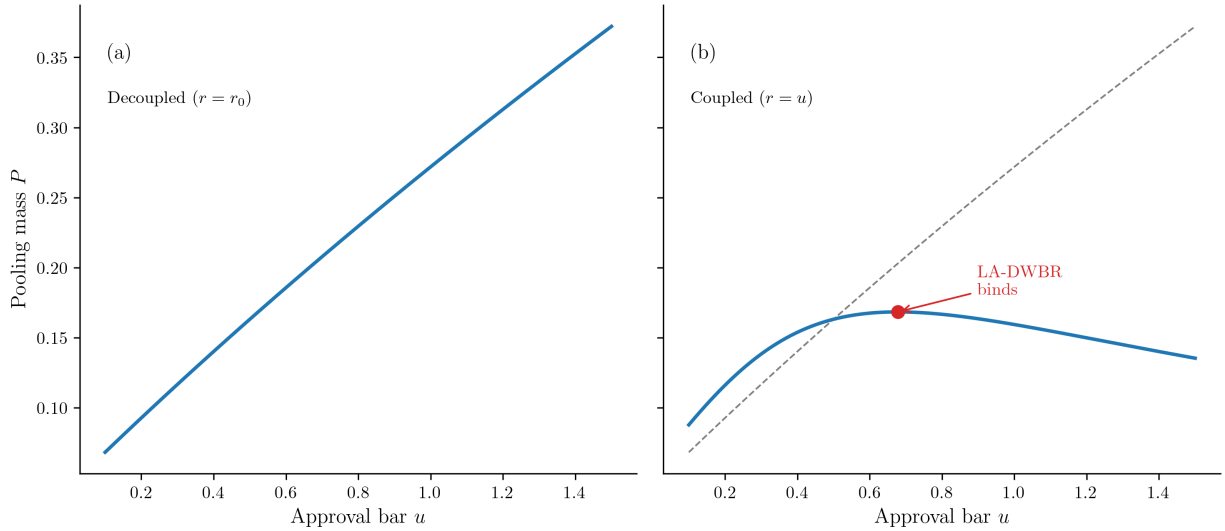
Figure 2 measures the region of Remark 5.4 numerically on the feasible cleansing corridor $\pi \in (0.5, 1-\sigma]$—the stringent range on which $r(\pi) = \pi \le 1 - \sigma$ keeps the settlement-support cutoff $v^* = 2r/(1-\sigma) - 1$ within the resistance support (the corridor is empty for $\sigma \ge \frac{1}{2}$)—under the maintained identity-belief, uniform-prior calibration. The calibration deliberately shuts down the one-stage density channel: identity beliefs and a uniform prior give $f(u) = f(\theta^*)$, so the bar-only channel alone cannot reverse the benchmark and any reversal must come from the Stage-2 enforcement channel. (This is the hardest case for the density channel on the calibration, not globally: under a general monotone belief map the uniform-prior condition reads $L_r\, r'/u' \ge (\mu^{-1})'(u)\,[\alpha + B'(\theta^*)]$, so belief sluggishness $(\mu^{-1})'(u) > 1$ raises the reversal threshold.) Each cell of the figure reports the fraction of a nuisance grid—suit-probability level $p_0 \in [0.1, 0.9]$, recovery scale $R_{\max} \in [0.1, 0.5]$, court-credit discount $\sigma \in \{0.1, 0.2, 0.3, 0.4\}$, and belief-cutoff slope $u'(\pi) \in [0.5, 3]$—on which the coupled pooling derivative is negative somewhere on the corridor. The reversal is non-negligible yet parametrically demanding: it holds on roughly 0.31 of the grid under the gap-self-limiting direct cost $L = \eta(r)|g| + \frac{\kappa(r)}{2} g^2$, whose sensitivity $L_r = \eta_r g^* + \frac{\kappa_r}{2} g^{*2}$ vanishes as the boundary gap collapses, and on about 0.86 under the ratification microfoundation $L = p\,C$ with the concave, self-limiting recovery $R(g) = R_{\max}(1 - e^{-\chi g})$, whose sensitivity instead carries the pivotal-voter price $C_r = (R + v^*)/(1-\sigma)$ (there L need not be globally convex, and the reversal turns on L_r rather than L_{gg}). Under both closures the region recedes as the prize slope $\alpha + B'(\theta^*)$ rises and as litigation becomes weakly threshold-sensitive ($p_r \to 0$), so the reversal is a substantial but conditioned region of primitives rather than either a knife-edge or a generic outcome. The numerical $dP/d\pi$ test and the closed-form condition of Proposition 2(iii) (equivalently the uniform-prior, identity form of (4)) agree in 97% of cells under the microfoundation and 100% under the direct cost—the residual 3% is finite-difference noise in the numerical test near corridor endpoints, where the closed-form condition is exact—and no infeasible point ($v^* > 1$) enters the measure.

Figure 2: Measure of the reversal region on the feasible corridor



Each cell reports the fraction of the nuisance grid on which the coupled pooling derivative is negative somewhere on the feasible corridor $\pi \in (0.5,\ 1-\sigma]$ ($\min_\pi dP/d\pi < 0$), as a function of the suit-probability slope p_r (horizontal; enforcement sensitivity $L_r = p_r C + p\, C_r$) and the prize slope $\alpha + B'(\theta^*) = \alpha + b_1$ (vertical), under the identity-belief, uniform-prior calibration. Under the ratification microfoundation $L = p\, C$ the reversal holds on 0.86 of the feasible grid; under the gap-self-limiting direct cost, on 0.31. The region recedes as the prize slope rises and as $p_r \to 0$; grid and feasibility details in the text.

Figure 3: Pooling mass under decoupled and coupled regimes



Panel (a): under the decoupled regime ($r = r_0 = 0.5$ fixed), pooling mass $P(u)$ is monotonically increasing in u—the benchmark result of Proposition 8. Panel (b): under the coupled regime ($r = u$), $P(u)$ initially rises but then falls once the ratification-cost dominance condition (Proposition 2) binds (red dot). The dashed grey line reproduces the decoupled benchmark for comparison. The reversal occurs because the enforcement channel—harder ratification at higher u—deters enough marginal exaggerators to reduce total pooling despite the higher bar. Uniform prior on $[-1, 2]$, identity beliefs, quadratic enforcement costs.

6 Welfare, institutional design, and comparative implications

The preceding analysis shows that coupling the ratification threshold to the approval threshold can reduce the mass of exaggerating controllers. Whether that reduction improves welfare is a separate question, and its answer is conditional. Tighter thresholds deter some value-destroying deals (false positives) while also deterring value-creating deals that now fall below the higher bar (false negatives), and coupling raises enforcement costs for the surviving poolers (the intensity effect). Whether the mass reduction dominates these countervailing forces depends on four institutional parameters that the welfare analysis identifies.

We organize the analysis around four institutionally interpretable parameters that jointly determine optimal governance design: the deadweight share of enforcement costs (ω), the convexity of enforcement costs in the misrepresentation gap (κ), the sensitivity of enforcement costs to the ratification threshold (L_r), and the quality of the marginal deterred deal (θ^*). The first three govern the enforcement margin, namely how coupling affects the total cost of policing exaggeration; the fourth governs the selection margin, namely whether deterring marginal controllers improves or harms firm value. We begin by deriving a welfare decomposition that isolates the roles of these four parameters in the coupling question—alongside the equilibrium densities and policy slopes that also enter the formal condition—and then use them to organize the institutional features of

a set of selected jurisdictions.

This section develops the welfare implications of the equilibrium objects characterized in Sections 3–5. The core game is unchanged. For each policy parameter π, take as given the induced continuation objects:

$$\big(u(\pi),\ r(\pi),\ \theta^*(u(\pi), r(\pi)),\ m^{\text{req}}(u(\pi))\big).$$

6.1 Welfare criterion

Assumption 4 (W1: Type distribution and integrability)**.** *θ has density f on $[\underline{\theta}_0, \overline{\theta}_0]$ with $\underline{\theta}_0 < 0 < \overline{\theta}_0$, f continuous and $f(\theta) \geq 0$, $\int_{\underline{\theta}_0}^{\overline{\theta}_0} f(\theta)\, d\theta = 1$, and $\int_{\underline{\theta}_0}^{\overline{\theta}_0} |\theta| f(\theta)\, d\theta < \infty$.*

Assumption 5 (W2: Enforcement regularity and feasible policy domain)**.** *$L(g, r, \sigma)$ is continuous on $\{g \geq 0\} \times \mathcal{R} \times \{\sigma\}$ and continuously differentiable in (g, r) on the interior, with L_g and L_r locally bounded along the corridor. Welfare analysis is conducted on a policy region Π^{int} such that $r(\pi) \in \mathcal{R}$ for every $\pi \in \Pi^{\text{int}}$. Under the ratification-vote microfoundation, $\mathcal{R} = (0, 1 - \sigma]$.*

Assumption 6 (W3: Regularity of equilibrium mappings on Π^{int})**.** *On Π^{int}, the policy maps $\pi \mapsto u(\pi)$ and $\pi \mapsto r(\pi)$ are C^1, $\pi \mapsto m^{\text{req}}(u(\pi))$ is C^1, and $\pi \mapsto \theta^*(\pi)$ with*

$$\theta^*(\pi) := \theta^*(u(\pi), r(\pi))$$

is C^1 and satisfies interior partial pooling $\underline{\theta}_0 < \theta^*(\pi) < u(\pi) < \overline{\theta}_0$ (at interior roots, Lemma 3(iii) delivers the C^1 dependence of θ^* on (u, r) and the partials used in Proposition 5).

The welfare analysis is conducted under the identity calibration maintained in the core model, $m^{\text{req}}(u) = u$, so the pool top coincides with the bar (under a general monotone belief map the pool top is $\theta^\top(u) = \min\{m^{\text{req}}(u), \overline{\theta}_0\}$ and the corresponding endpoints replace $u(\pi)$ below; §S4). Define the approval and pooling sets

$$\mathcal{A}(\pi) := [\theta^*(\pi), \overline{\theta}_0], \qquad \mathcal{P}(\pi) := [\theta^*(\pi), u(\pi)).$$

Define the equilibrium misrepresentation gap

$$g(\theta; \pi) := \big(m^{\text{req}}(u(\pi)) - \theta\big)\mathbf{1}\{\theta \in \mathcal{P}(\pi)\},$$

so $g(\theta; \pi) \geq 0$ on $\mathcal{P}(\pi)$ and $g(\theta; \pi) = 0$ outside $\mathcal{P}(\pi)$.

Define expected enforcement burden:

$$\text{EC}(\pi) := \int_{\theta^*(\pi)}^{u(\pi)} L(m^{\text{req}}(u(\pi)) - \theta,\ r(\pi),\ \sigma)\, f(\theta)\, d\theta.$$

The two welfare criteria are:

$$W^{\text{firm}}(\pi) := \mathrm{E}[\mathbf{1}\{\theta \in \mathcal{A}(\pi)\}\theta] - \omega\,\mathrm{EC}(\pi),$$

$$W^{\text{tot}}(\pi) := \mathrm{E}\big[\mathbf{1}\{\theta \in \mathcal{A}(\pi)\}\big(\theta + b_0\big)\big] - \omega\,\mathrm{EC}(\pi).$$

Here $\omega \in [0,1]$ is the deadweight share of enforcement costs, a reduced-form stand-in rather than a microfounded object. The Stage-2 ratification expenditure splits into a vote-buying transfer to minority voters—which nets out in a social aggregate that includes them—and a residual litigation and court loss that is genuine deadweight; because the microfoundation does not by itself pin down the aggregate share ω, the cross-regime welfare ranking is conditional on it, and we treat ω as a parameter of the welfare criterion. Formally, write the enforcement cost as $L = T + K$, where $T(g, r, \sigma)$ is the settlement and vote-buying *transfer* to minority voters and $K(g, r, \sigma)$ is the deadweight process loss (legal fees, court costs, bargaining breakdown); the implied deadweight share is $\omega = K/(T+K)$. Since the microfounded buying-peace expenditure is transfer-dominated (T large relative to K), the institutional reading "coupling improves welfare when enforcement waste is large" applies to jurisdictions where deadweight process costs, rather than transfers, dominate.

Lemma 2 (Exact relation between welfare criteria)**.** *Under Assumptions 4-6,*

$$W^{\text{tot}}(\pi) - W^{\text{firm}}(\pi) = b_0\ \Pr\big(\theta \geq \theta^*(\pi)\big).$$

Proof. Subtract the two definitions. The enforcement terms cancel exactly, leaving

$$W^{\text{tot}}(\pi) - W^{\text{firm}}(\pi) = \mathrm{E}[\mathbf{1}\{\theta \geq \theta^*(\pi)\}b_0] = b_0\ \Pr(\theta \geq \theta^*(\pi)).$$

□

Remark. Lemma 2 implies the two criteria are not generically identical in comparative statics: they coincide only when $b_0 = 0$ or when comparing policies with the same approval probability.

6.2 Welfare decomposition: selection errors vs. enforcement frictions

Take the firm-value benchmark first-best cutoff

$$\theta_{\text{FB}} := 0, \qquad W^{\text{FB}} := \int_0^{\bar{\theta}_0} \theta f(\theta)\, d\theta.$$

Define selection-loss terms

$$\mathrm{FP}(\pi) := \mathbf{1}\{\theta^*(\pi) < 0\} \int_{\theta^*(\pi)}^{0} (-\theta) f(\theta)\, d\theta,$$

$$\mathrm{FN}(\pi) := \mathbf{1}\{\theta^*(\pi) > 0\} \int_0^{\theta^*(\pi)} \theta f(\theta)\, d\theta.$$

These selection-loss terms are defined on firm value θ, and the first-best cutoff $\theta_{\text{FB}} = 0$ approves exactly the value-creating deals. The diversion component $D(\theta)$ of the controller's private benefit, a transfer from minorities to the controller, nets to zero in the utilitarian aggregate and therefore enters neither W^{firm} nor W^{tot}; the cross-regime welfare ranking is driven by the selection of θ and by the deadweight enforcement term $\omega\,\text{EC}$, while diversion remains the economic motive for the conflict that the design must screen.

Proposition 4 (Welfare depends on selection errors and enforcement waste)**.** *Under the welfare regularity assumptions* (W1)–(W3)*, firm-value welfare under rule π equals first-best value less false positives, false negatives, and the deadweight share of enforcement costs:*

$$W^{\text{firm}}(\pi) = W^{\text{FB}} - \text{FP}(\pi) - \text{FN}(\pi) - \omega\,\text{EC}(\pi).$$

Proof. Proof in Appendix B, §B.1. □

The decomposition says that any governance regime's welfare shortfall relative to first-best comes from exactly three sources, and these three sources exhaust the gap. Foremost, false positives are the direct value destruction from approving negative-NPV deals—transactions that a fully informed, conflict-free planner would reject. Next, false negatives are the forgone value from rejecting positive-NPV deals—transactions the planner would approve but that fail to clear the bar because the controller's exaggeration is too costly to sustain. Finally, enforcement costs are the deadweight component of the litigation and ratification process—the legal fees, court costs, and bargaining losses that are burned rather than spent on productive activity; the vote-buying component is instead a transfer that nets out in the social aggregate and enters welfare only through the deadweight share ω (as the treatment of ω above makes explicit). In the first-best world of perfect information and no conflicts, all three terms are zero, and the design question is how to minimize their sum. The division among the three terms is governed by the location of θ^* relative to zero, which is itself determined by the primitives of the enforcement technology: the cost parameters $\eta(r,\sigma)$ and $\kappa(r,\sigma)$ that shape how expensive it is to sustain a given misrepresentation gap, and the ratification threshold r that governs how hard it is to buy peace. When coupling raises r alongside π, it pushes θ^* upward through the enforcement channel of Proposition 2: fewer types find it profitable to lie, so the welfare identity shifts weight from the FP and EC terms toward the FN term.

6.3 Why linking r to π changes welfare predictions

Define

$$\phi(\theta,\pi) := L(m^{\text{req}}(u(\pi)) - \theta,\ r(\pi),\ \sigma)\, f(\theta), \qquad \theta^* := \theta^*(\pi),\ u := u(\pi).$$

Proposition 5 (Stricter coupled rules trade off deterrence and enforcement costs)**.** *Under the welfare regularity assumptions* (W1)–(W3)*, for $\pi \in \Pi^{\text{int}}$ firm-value welfare moves with the rule according to*

$$\frac{d}{d\pi} W^{\text{firm}}(\pi) = -\theta^*(\pi) f(\theta^*(\pi))\, \theta^{*\prime}(\pi) - \omega \frac{d}{d\pi}\text{EC}(\pi), \qquad \frac{d}{d\pi}\text{EC}(\pi) = u'(\pi)\, \mathcal{D}_u(\pi) + r'(\pi)\, \mathcal{D}_r(\pi),$$

where the boundary, interior-gap, and coupling terms comprising the coefficients $\mathcal{D}_u(\pi)$ and $\mathcal{D}_r(\pi)$ are given in Appendix B, §B.2. A stricter coupled rule therefore raises firm-value welfare if and only if the selection effect of raising the boundary outweighs the weighted change in enforcement cost, $-\theta^*(\pi) f(\theta^*(\pi))\, \theta^{*\prime}(\pi) > \omega \frac{d}{d\pi}\mathrm{EC}(\pi)$.

Proof. Full derivation, with the four-term Leibniz expansion of $d\mathrm{EC}/d\pi$ and the explicit $\mathcal{D}_u, \mathcal{D}_r$, in Appendix B, §B.2. □

The four terms in $d\mathrm{EC}/d\pi$ have distinct economic content. The *top-boundary term* $\phi(u, \pi)\, u'(\pi)$ captures the enforcement cost brought in by the new type that enters the pooling set at the top when the bar rises. Since this type sits at the pool top with a vanishing gap ($m^{\mathrm{req}}(u) - u = 0$ under the maintained identity calibration), $L(0, r, \sigma) = 0$ makes its enforcement cost negligible and the top-boundary contribution washes out. The same logic extends to a general monotone belief map on the uncapped branch: the pool top is then $\theta^{\top}(u) = m^{\mathrm{req}}(u)$, the entering type's gap $m^{\mathrm{req}}(u) - \theta^{\top}(u)$ again vanishes, and only the interior-gap term below picks up the extra factor $(m^{\mathrm{req}})'(u)$ (on the capped branch $\theta^{\top}(u) = \overline{\theta}_0$ is fixed and no type enters at the top). The *bottom-boundary term* $-\phi(\theta^*, \pi)\, \theta^{*\prime}(\pi)$ captures the enforcement cost *removed* when the marginal liar at θ^* is deterred. This type has the *largest* gap in the pooling set—$g^* = m^{\mathrm{req}}(u) - \theta^*$—and therefore, by convexity of L in g, bears the highest per-capita enforcement cost of any pooler. Whenever coupling raises θ^* (i.e. $\theta^{*\prime} > 0$), it is precisely this most expensive type that drops out. The *interior-gap term* $\int L_g \cdot (m^{\mathrm{req}})' u'\, f\, d\theta$ reflects the fact that a higher bar forces all surviving poolers to send a larger message, widening each type's gap and raising per-type enforcement costs. This force is present even without coupling and is always welfare-harmful. The *coupling term* $\int L_r \cdot r'\, f\, d\theta$ is the pure effect of tightening the ratification threshold: for every surviving pooler, each unit of gap is now more costly to sustain because the enforcement technology itself has stiffened. This term is identically zero when $r'(\pi) = 0$ and is always welfare-harmful when $r' > 0$ and $L_r > 0$.

The welfare question on the enforcement margin therefore reduces to whether the bottom-boundary saving—removing the most expensive types—dominates the two cost-escalation forces on survivors. When that saving dominates, $d\mathrm{EC}/d\pi$ is itself negative, so a stricter coupled rule lowers enforcement waste; and where the marginal deterred deal destroys value ($\theta^* \leq 0$, with $\theta^{*\prime} > 0$), the selection term moves the same way, so both terms of Proposition 5 then point toward tightening. The bottom-boundary and interior forces map onto recognizable transactions. The bottom of the pooling set—large-gap, high-cost types—corresponds to deals like the *Dell* freeze-out (2013), where the offered price was roughly twenty percent below judicially determined fair value; exaggeration on that scale generates years of appraisal litigation and expert costs, placing these types on the steep, convex region of L. The top of the pool, by contrast—a routine related-party lease priced marginally below market—involves a tiny gap, negligible litigation risk, and near-zero enforcement cost on the flat part of the cost curve.

Corollary 3 (Coupling channel versus decoupling channel)**.** *Under the assumptions of Proposition 5, if $r'(\pi) = 0$ (decoupled regime) then*

$$\frac{d}{d\pi}\mathrm{EC}(\pi) = u'(\pi)\, \mathcal{D}_u(\pi).$$

If $r'(\pi) \neq 0$ (coupled regime), the additional term $r'(\pi)\mathcal{D}_r(\pi)$ enters exactly.

Proof. Immediate from Proposition 5. □

When the ratification threshold is decoupled from the approval threshold ($r' = 0$), raising the approval standard affects enforcement costs only through the bar channel: a higher required message widens every pooler's gap. When r is coupled to π ($r' > 0$), an additional coupling channel enters through $r'(\pi)\,\mathcal{D}_r(\pi)$. This channel operates through two opposing sub-forces. Foremost, a *mass effect*: because a higher ratification threshold deters lying—the boundary type θ^* rises—fewer types pool, directly reducing the enforcement-cost integral. Next, an *intensity effect*: for the types that still pool, each faces a steeper enforcement cost surface because r is now tougher.

Convexity of L in the gap g is what makes the mass effect powerful. As the examples above illustrate, enforcement waste is concentrated at the bottom of the pool, and convexity amplifies the skew: bottom types are disproportionately expensive, not merely proportionally so. Coupling removes precisely these types first, while routine near-market transactions at the top of the pool, sitting on the flat part of the cost curve, are barely affected. The mass effect therefore operates as a targeted filter, removing the types whose enforcement costs are disproportionately large relative to their share of the pool; when ratification-cost dominance holds, this targeted removal from the expensive end dominates the per-unit escalation on survivors.

Combining this enforcement-margin analysis with the selection margin yields the full welfare picture. On the *selection margin*, coupling pushes θ^* upward—welfare-improving when $\theta^* < 0$ (removing negative-NPV deals) and welfare-harmful when $\theta^* > 0$ (over-deterring positive-NPV deals). The two margins align when the pool bottom is populated by negative-NPV types with large gaps: coupling simultaneously reduces false positives and enforcement waste, at the cost of only a modest false-negative increase. Restrepo's (2021) finding that MoM adoption surged after *MFW* without measurable effects on deal premiums or completion rates is consistent with this alignment, since the deterred deals were marginal, near-zero-NPV transactions that were cheap to lose on the selection margin and expensive to sustain on the enforcement margin. The tension arises only when θ^* has crossed zero and coupling is over-deterring genuinely valuable deals while also escalating costs for the remaining poolers.

The deadweight parameter ω governs how much enforcement waste matters for welfare. In the pre-*Trulia* merger-litigation environment, where over ninety percent of deals were challenged and the dominant resolution was a disclosure-only settlement that changed nothing about deal terms, enforcement expenditures were largely waste rather than productive monitoring—a high-ω world in which the enforcement-cost savings from coupling carry substantial welfare weight.

Proposition 2 is a sufficient condition under which the enforcement-feedback component in Proposition 5 dominates the bar-only benchmark component along the relevant policy corridor.

6.4 Example: closed-form welfare under uniform priors and identity beliefs

The uniform-prior, identity-beliefs specialization is more than a tractability device; it isolates the mechanism in its starkest form. Under these assumptions, the density ratio $f(u)/f(\theta^*)$ equals one, so the standard DWBR condition always fails: the one-stage model unambiguously predicts that pooling increases with the threshold. Any reversal must therefore come entirely from the Stage 2 enforcement channel, making this environment the hardest case *for the one-stage density channel* (under a general monotone belief map the condition carries the extra factor $(\mu^{-1})'(u)$, so sluggish beliefs make reversal harder still). The closed-form expressions below make the key welfare tradeoffs algebraically transparent. The pooling width $w = u - \theta^*$ is determined by a quadratic equation whose coefficients are exactly the primitives of the enforcement technology: $\eta(r,\sigma)$ captures the fixed marginal cost of any non-zero misrepresentation, $\kappa(r,\sigma)$ captures the convexity that makes large lies disproportionately expensive, and $\alpha + b_1$ is the prize slope that dampens boundary movement. The enforcement cost integral EC decomposes into a linear term (proportional to w^2, driven by η) and a convex term (proportional to w^3, driven by κ), confirming that the cubic term—which grows fastest as the pool widens—is the dominant welfare-relevant force for wide pools. The closed-form welfare expression $W^{\text{firm}} = (\overline{\theta}_0^2 - \theta^{*2})/(2\Delta_\theta) - \omega\,\text{EC}$ shows that coupling improves welfare precisely when the boundary rise improves selection—the selection term gains when $\theta^* < 0$, since raising a negative boundary shrinks θ^{*2} by deterring value-destroying deals, and loses once θ^* crosses zero—and the decrease in w (which shrinks enforcement costs) outweighs any residual false-negative loss: the same four-parameter condition (ω, κ, L_r, θ^*) now visible in explicit algebra.

Example 1 (Closed-form welfare under uniform prior and identity beliefs)**.** Maintain Proposition 8's environment with

$$\theta \sim U[\underline{\theta}_0, \overline{\theta}_0], \qquad \mu(m) = m, \qquad B(\theta) = b_0 + b_1\theta, \;\; b_1 \in (0, 1-\alpha),$$

and

$$L(g, r, \sigma) = \eta(r,\sigma)\, g + \frac{\kappa(r,\sigma)}{2} g^2, \qquad g \geq 0.$$

Let u be any interior bar at which pooling is interior and partial: $(\alpha + b_1)u + b_0 > 0$ and $w(u,r) < u - \underline{\theta}_0$ for the root defined below. Define $w := u - \theta^*(u,r)$. Then:

$$\frac{\kappa(r,\sigma)}{2} w^2 + \big(\alpha + b_1 + \eta(r,\sigma)\big) w = (\alpha + b_1)u + b_0.$$

If $\kappa(r,\sigma) > 0$:

$$w(u,r) = \frac{-\big(\alpha + b_1 + \eta(r,\sigma)\big) + \sqrt{\big(\alpha + b_1 + \eta(r,\sigma)\big)^2 + 2\kappa(r,\sigma)\big((\alpha + b_1)u + b_0\big)}}{\kappa(r,\sigma)},$$

and $\theta^*(u,r) = u - w(u,r)$. If $\kappa(r,\sigma) = 0$:

$$w(u,r) = \frac{(\alpha + b_1)u + b_0}{\alpha + b_1 + \eta(r,\sigma)}.$$

Let $\Delta_\theta := \overline{\theta}_0 - \underline{\theta}_0$. Then

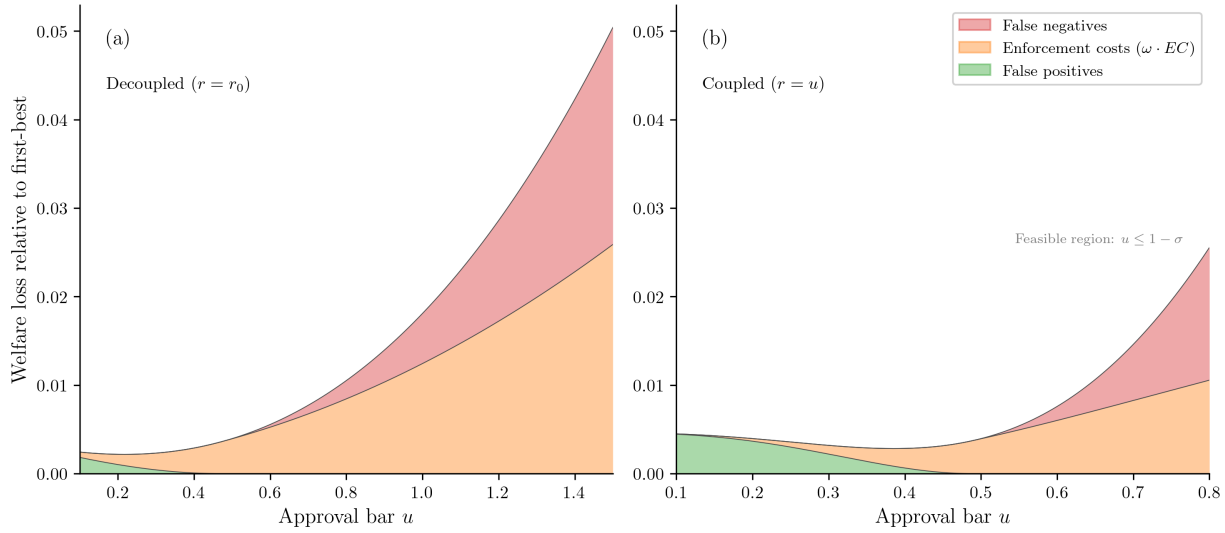
$$\mathrm{EC}(u,r) = \eta(r,\sigma)\frac{w(u,r)^2}{2\Delta_\theta} + \kappa(r,\sigma)\frac{w(u,r)^3}{6\Delta_\theta},$$

and

$$W^{\text{firm}}(u,r) = \frac{\overline{\theta}_0^{\,2} - \theta^*(u,r)^2}{2\Delta_\theta} - \omega\,\mathrm{EC}(u,r).$$

The derivation of these objects is in Appendix B, §B.3.

Figure 4: Welfare loss decomposition under decoupled and coupled regimes



Panel (a): decoupled regime ($r = r_0 = 0.5$). As the bar rises, enforcement costs (orange) grow monotonically because more types pool with larger gaps. Panel (b): coupled regime ($r = u$), plotted over the feasible region $u \leq 1 - \sigma = 0.8$. Enforcement costs are smaller at comparable bar levels—coupling deters the high-gap types that drive enforcement waste (mass effect). False negatives (red) grow more steeply under coupling because the higher enforcement costs deter some value-creating deals (over-deterrence). False positives (green) appear only at low bars where $\theta^* < 0$. The stacked areas sum to the total welfare loss relative to first-best. Shared y-axis scales permit direct cross-panel comparison.

6.5 Comparative institutional implications

The comparative exercise that follows is diagnostic rather than estimative. The model identifies the parameters that matter for the welfare of coupling—the deadweight share of enforcement costs ω, the convexity of those costs in the disclosure gap κ, the sensitivity of enforcement to

the ratification threshold L_r, and the quality of the marginal deterred deal θ^*—but for most jurisdictions these parameters are not directly observed. We therefore use legal architecture, stated policy rationales, and the existing empirical literature to place each jurisdiction in a plausible parameter region; in particular, the sign of θ^* is read off each regime's own stated policy rationale rather than estimated. The resulting classifications are hypotheses generated by the model, each carrying only the claim strength indicated below, and validating any of them would require separate empirical work.

The four welfare parameters $(\omega, \kappa, L_r, \theta^*)$ are institutionally interpretable features of legal systems (cf. Djankov et al., 2008; Dyck and Zingales, 2004; Ouyang and Zhu, 2016; Shleifer and Vishny, 1997; Atanasov et al., 2011), and different jurisdictions occupy different points in this space. Jurisdictions that have tightened controller-transaction requirements (India, Hong Kong) are consistent with a policy belief that marginal deterred deals destroy value (a negative-θ^* diagnosis); those that have loosened requirements (Japan, the UK after 2024) are consistent with the belief that marginal deterred deals create value (a positive-θ^* diagnosis).

In jurisdictions with concentrated ownership and documented tunneling, the selection and enforcement margins point in the same direction. In India, SEBI mandates majority-of-the-minority approval for material related-party transactions (Li, 2021; Fried et al., 2020; Dai et al., 2019), subject to audit-committee review, with promoter groups barred from voting on their own deals, so the procedural hurdle becomes a genuine substantive barrier (L_r high). The case for coupling rests on the selection margin: the marginal deterred deals are value-destroying tunneling transactions ($\theta^* < 0$). Hong Kong presents a similar picture (Kim, 2019; Lim, 2019): Chapter 14A of the HKEX Listing Rules requires an independent financial adviser and a disinterested shareholder vote for connected transactions, and empirical studies document target-firm trading discounts of ten to twenty percent attributable to expropriation risk. In both jurisdictions enforcement costs are low to moderate (ω low–moderate, κ concave), so false-positive reduction rather than enforcement savings drives the welfare case.[13]

In jurisdictions that have diagnosed over-deterrence, the selection margin reverses. Japan's METI has actively encouraged parent-subsidiary consolidation (Kim, 2019), issuing Fair M&A Guidelines (2019) and Guidelines for Corporate Takeovers (2023) that establish procedural safe harbors. The procedural sensitivity L_r is high, and the stated policy aim is consistent with a positive-θ^* diagnosis: if the marginal deterred deal creates value, coupling would over-deter exactly the transactions the state is trying to promote. The United Kingdom moved in a parallel direction (Davies, 2019, 2020; Becht et al., 2016): in July 2024 the FCA removed the mandatory independent shareholder vote for related-party transactions for listed commercial companies—a change to the FCA listing-rule related-party-transaction vote that leaves in force the s. 283 statutory special-resolution ratification discussed in the Introduction—motivated by the assessment that the prior regime was deterring efficient transactions (a positive-θ^* diagnosis).

[13] China is a further negative-θ^* case whose distinctive feature is the enforcement-cost channel. Because its administrative regime attaches multiplier-based monetary sanctions that scale with the illicit gain once the CSRC opens a formal penalty, the enforcement-cost schedule is kinked and convex in the disclosure gap, so that coupling can reduce enforcement waste directly by deterring the high-gap poolers who attract the steepest sanctions. This reading is weaker than the others advanced here, and would require tightening against the enforcement record.

The sharpest comparative test is the India–Japan contrast. Both employ majority-of-the-minority voting with special committee oversight and high L_r, but the plausible θ^* diagnosis differs in sign: negative in India (promoter tunneling), positive in Japan (efficiency-enhancing consolidation) (cf. Goshen and Hamdani, 2016). Under those diagnoses the model implies that the identical procedural tool improves selection in India and worsens it in Japan, with the net welfare verdict then turning on the enforcement margin of Proposition 5—a hypothesis about marginal-transaction returns and deal volume.

Delaware is a useful diagnostic case because several observable features are consistent with the coupling-favorable region. Foremost, enforcement costs have a high deadweight share (ω high): disclosure-only settlements dominated merger litigation until *In re Trulia* (2016). Next, the cost structure is step-function with convexity above the entire-fairness trigger (κ step/convex). Finally, the sensitivity of enforcement to the cleansing threshold was very high (L_r very high) under the *MFW* framework.[14]

S.B. 21 (Senate Substitute 1 for Senate Bill 21, signed March 25, 2025) relaxed the ex ante cleansing requirement for non-going-private controller transactions, which under revised §144(b) may now reach the safe harbor through *either* a disinterested-committee path *or* a disinterested-stockholder vote; going-private transactions under §144(c) retain a both-protections structure. The safe harbor operates on the fiduciary-cleansing margin and the standard of review, and it preserves the separate requirement that the transaction be authorized in compliance with the statute, the certificate of incorporation, and the bylaws, so the coupled authorization baseline of Section 3 is untouched by the reform.[15] The statute is operative law, and the Delaware Supreme Court upheld it against constitutional challenge in *Rutledge v. Clearway Energy Group LLC* (Del. Feb. 27, 2026), so the live question is the design quality of the non-going-private either/or safe harbor rather than its validity. The model asks a bounded question: when, within the relevant parameter region, treating committee approval and minority approval as substitutes is welfare-inferior. A welfare justification for the either/or structure would rest on a positive-θ^* diagnosis, which the available evidence does not clearly establish: Restrepo (2021) finds that MoM adoption surged after *MFW* without measurable effects on deal premiums or completion rates, a pattern consistent with $\theta^* \approx 0$ (see also Gözlügöl, 2022; Boone et al., 2018). Under the model's coupling-favorable parameter region—where $\theta^* \leq 0$ and (ω, κ, L_r) favor coupling—the non-going-private either/or structure forfeits the fixed-allocation procedural complementarity of Corollary 4. That inference is a fixed-allocation reading, subject to the endogenous-boundary qualification of Remark 6.6.2, which binds in this region (with $\theta^* \leq 0$ the boundary term offsets the fixed-allocation complementarity). That pattern is consistent with the governance-externality channel formal-

[14]The conflicted-controller cleansing authorities govern the standard of review rather than corporate authorization, and are analytically separate from the coupled authorization baseline of Section 3. From an ex ante perspective, the *MFW* dual-protection framework requires both an independent special committee and an uncoerced majority-of-the-minority vote (*Kahn v. M&F Worldwide Corp.*, 88 A.3d 635 (Del. 2014)), a framework extended to controller transactions conferring a non-ratable benefit in *In re Match Group, Inc. Derivative Litigation*, 315 A.3d 446 (Del. 2024). Ex post, a later disinterested vote does not by itself cleanse the breach, and at most shifts the burden of proving entire fairness (*Tornetta v. Musk*, 310 A.3d 430 (Del. Ch. 2024); 326 A.3d 1203 (Del. Ch. 2024)), with the rescission remedy later reversed on other grounds in *In re Tesla, Inc. Derivative Litigation* (Del. Dec. 19, 2025).

[15]Since the design question concerns the procedural prongs, the substantive fairness path that revised §144 also retains—a showing that the transaction is fair to the corporation and its stockholders—is set aside here.

ized in the private-ordering extension (§S10.1.1), although the model does not by itself establish legislative motive (cf. Khoo and Tallarita, 2025).

France operates a third regime outside the model's current scope. Under the *conventions réglementées* framework (see Conac et al., 2007; Bianchi et al., 2019; Enriques and Volpin, 2007) (Code de commerce, Art. L225-38 to L225-42), the controller may execute the transaction regardless of minority approval but bears strict liability for prejudicial consequences—an "execute and compensate" architecture that shifts the policy instrument from deterrence to ex post redistribution. Extending the model to accommodate such liability-shifting regimes is a natural direction for future work.[16]

6.6 Extensions: committee screening and private ordering

This section keeps Stages 0-2 unchanged and adds a Stage 0.5 committee filter and a Stage -1 charter choice, mapping committee design and private ordering into the same equilibrium objects used in the core model. We examine the committee filter first, then turn to the private-ordering charter choice.

6.6.1 Committee as a second procedural lever

Committees act as a pre-vote filter: a more independent committee shifts the mix of deals reaching the vote toward higher quality, reducing both the mass of exaggerators and expected enforcement costs. The committee is procedurally complementary to vote and ratification design, addressing a different margin (pre-vote composition) while reinforcing the same disciplinary channel (Gilson and Gordon, 2003; Bebchuk and Hamdani, 2017). This extension adds a committee lever to the Stage 0–2 backbone.

Add Stage 0.5 before Stage 1: the committee either allows the proposal to proceed to the shareholder vote or stops it. Let $\iota \in [0,1]$ denote committee independence and let $Q : [\underline{\theta}_0, \overline{\theta}_0] \to [0,1]$ satisfy $Q'(\theta) > 0$. Define the pass probability

$$q_\iota(\theta) := (1-\iota) + \iota Q(\theta) \in (0,1],$$

and the induced type density among deals reaching the vote

$$f_\iota(\theta) := \frac{q_\iota(\theta) f(\theta)}{Z_\iota}, \qquad Z_\iota := \int_{\underline{\theta}_0}^{\overline{\theta}_0} q_\iota(t) f(t)\, dt.$$

[16] Germany is a further out-of-model, liability-shifting regime of the same type. Under §311 AktG *Konzernrecht*, a disadvantageous transaction is permitted so long as the controller compensates the subsidiary by year-end; failing that, the minority may pursue an ex post *Spruchverfahren* valuation claim. Because the design keeps procedural resistance deliberately low (L_r low) and caps the marginal sanction (a concave or capped κ), discipline here runs through ex post redistribution rather than through approval-coupling.

The Online Appendix (§S10) confirms that higher committee independence shifts the conditional type distribution in the MLR sense, reduces low types reaching the vote, and lowers enforcement costs at any fixed bar. The signal channel is similarly well-behaved. Independence helps; the economically interesting question is whether its benefits interact with enforcement toughness, and at what cost.

The committee observes $s = \theta + \varepsilon$ with $\mathrm{E}[\varepsilon] = 0$ and ε independent of θ. With probability ι, it recommends $y = \mathbf{1}\{s \geq s_0\}$; with probability $1-\iota$, it is captured and sets $y = 1$. The pass probability is $p_\iota(\theta) := 1 - \iota G_\varepsilon(s_0 - \theta)$, and the effective posterior given message m and recommendation y is $\mu_\iota(m, y) = \tau(m)\,\bar{\mu}_\iota(m, y) + (1 - \tau(m))\mu_0$, where $\bar{\mu}_\iota(m, y) := \mathrm{E}[\theta \mid m, y]$ and attention $\tau(m)$ may vary with the message but not with the recommendation (the recommendation-independence is what the signal-channel monotonicity results use). The monotonicity properties of the signal channel—favorable recommendations lower the required persuasive message and weakly reduce enforcement exposure—are established in the Online Appendix (§S10).

6.6.2 Cost of committee independence and procedural complementarity

The preceding results establish that committee independence weakly reduces false positives and enforcement costs at fixed voting and enforcement parameters. A more independent committee, however, also blocks deals that a captured committee would have let through, including deals with positive NPV. This subsection formalizes that cost, then shows that, under the stated conditions, the benefit of committee independence is amplified by tougher enforcement while the cost is invariant, generating a conditional welfare complementarity between committee independence and ratification toughness rather than a claim that joint requirements are always optimal.

Let

$$\mathrm{FN}^{\mathrm{gate}}(\iota) := \int_0^{\bar{\theta}_0} \big(1 - q_\iota(\theta)\big)\,\theta\, f(\theta)\, d\theta$$

denote the expected value of these blocked positive-NPV deals. This gate false-negative cost is strictly increasing and linear in ι, with $\mathrm{FN}^{\mathrm{gate}}(0) = 0$, and—because the gate operates before enforcement—is independent of (u, r, σ) (Online Appendix, Lemma 9). The benefit side, by contrast, depends on (r, σ) through L; this asymmetry is the source of the complementarity result below.

Proposition 6 (Committees and shareholder votes are enforcement complements at a fixed allocation)**.** *Fix $(u, \theta^*, m^{\mathrm{req}}, \sigma)$ with $\theta^* < u \leq m^{\mathrm{req}}$ (so every gap in the integrand is nonnegative) and let $\mathrm{EC}(\iota, r) := \int_{\theta^*}^{u} q_\iota(\theta)\, L\big(m^{\mathrm{req}} - \theta,\ r,\ \sigma\big)\, f(\theta)\, d\theta$ be expected enforcement exposure. If $L_r(g, r, \sigma) \geq 0$ for all $g \geq 0$, then enforcement exposure is submodular in committee independence ι and ratification toughness r,*

$$\frac{\partial^2 \mathrm{EC}}{\partial \iota\, \partial r} \leq 0,$$

strictly so when $L_r(m^{\mathrm{req}} - \theta, r, \sigma) > 0$ and $Q(\theta) < 1$ on a positive-f-measure subset of $[\theta^*, u)$. Equivalently, a more independent committee lowers enforcement exposure by more when ratification is tougher.

Proof. Differentiating $\mathrm{EC}(\iota, r)$ under the integral gives $\partial^2\mathrm{EC}/\partial\iota\partial r = \int_{\theta^*}^{u}(Q(\theta)-1)L_r f\, d\theta$, whose integrand is weakly negative pointwise ($Q \le 1$, $L_r \ge 0$); it is strictly negative under the stated non-degeneracy. See the Online Appendix, §S10.4. □

The sign of $\partial^2\mathrm{EC}/\partial\iota\,\partial r \le 0$ says that the marginal enforcement-cost reduction from a more independent committee is *larger* when the ratification threshold is tougher. Intuitively, when r is high, each remaining pooler bears steep enforcement costs, so removing a type from the pool via the gate saves more per type removed. A tougher enforcement environment therefore amplifies the value of pre-vote screening.

Corollary 4 (Fixed-allocation welfare supermodularity in committee independence and ratification toughness)**.** *Under the conditions of Lemma 9 and Proposition 6, define the committee-augmented welfare at fixed* $(u, \theta^*, m^{\mathrm{req}}, \sigma)$*:*

$$\mathcal{W}(\iota, r) := \underbrace{\int_{\theta^*}^{\overline{\theta}_0} q_\iota(\theta)\,\theta\, f(\theta)\, d\theta}_{\text{expected approved value (net of gate blocking)}} - \omega\,\mathrm{EC}(\iota, r).$$

Then

$$\frac{\partial^2 \mathcal{W}}{\partial\iota\,\partial r} = -\omega\,\frac{\partial^2 \mathrm{EC}}{\partial\iota\,\partial r} \ge 0,$$

since $\partial^2/\partial\iota\,\partial r$ *of the approved-value integral vanishes at fixed* θ^**. Therefore* $\mathcal{W}$ *is supermodular in* (ι, r) at this fixed allocation*: committee independence and ratification toughness are welfare complements when the marginal pooler* θ^* *is held fixed. Remark 6.6.2 treats the endogenous-boundary case.*

Proof. Since the approved-value integral at fixed θ^* is independent of r, the only surviving cross-partial is $-\omega\,\partial^2\mathrm{EC}/\partial\iota\partial r \ge 0$, so $\mathcal{W}$ is supermodular in (ι, r). See the Online Appendix, §S10.4. □

Remark (Conservativeness). The supermodularity is established at a *fixed* allocation $(u, \theta^*, m^{\mathrm{req}})$. When the boundary is endogenized, $\theta^* = \theta^*(r)$ with $\partial\theta^*/\partial r > 0$ (Proposition 2), the total cross-partial acquires a boundary term $-(Q(\theta^*)-1)\,f(\theta^*)\,\theta^*_r\,[\theta^* - \omega L(g^*, r, \sigma)]$, whose sign tracks $\theta^* - \omega L(g^*, r, \sigma)$ (the prefactor $-(Q(\theta^*)-1)f(\theta^*)\theta^*_r \ge 0$). Economically, the committee gate and the coupled vote act on an overlapping margin, since both discipline the same low-quality boundary pooler—the gate by stopping it before the vote, coupling by deterring it through enforcement. When that pooler destroys value ($\theta^* < 0$) and is expensive to enforce against ($\omega L > \theta^*$), the boundary term is negative and the two instruments behave as partial substitutes on this marginal type, offsetting part of the fixed-allocation complementarity once the boundary is allowed to move. The fixed-allocation complementarity is therefore conservative only when this term does not offset it—for instance when $\theta^* \ge \omega L(g^*, r, \sigma)$, or when the fixed-allocation enforcement cross-partial dominates the boundary term. Absent such a condition the full-equilibrium cross-partial sign is not established, so we read Corollary 4 as an enforcement complementarity *at a fixed allocation*—suggestive for, rather than dispositive of, the full-equilibrium comparison of joint requirements against corner solutions.

The economic logic of Corollary 4 is an asymmetry: the *cost* of committee independence (blocking good deals) is invariant to enforcement parameters, while the *benefit* (reducing enforcement waste) scales with r through L_r. Since tougher enforcement raises the benefit without raising the cost, the marginal return to committee quality is increasing in enforcement toughness.

This is a conditional supermodularity result. Under the conditions studied here—a gate cost invariant to enforcement toughness and an enforcement-cost saving that rises with the ratification threshold—committees and minority votes are complements. It is a local property of the welfare surface and does not imply that joint procedural requirements are optimal across all parameter regions.

Under *MFW*, Delaware required both an independent committee and MoM approval, raising ι and r jointly, consistent with the conditional complementarity established above. S.B. 21's *either/or* safe harbor for non-going-private transactions permits corner solutions, resting on the premise that the two instruments are substitutes. They address *different* margins, however—composition (committee) versus discipline (vote/ratification)—and under the stated conditions these margins are complements. Within the coupling-favorable parameter region, the conditional complementarity therefore favors the joint requirement over either corner at a fixed allocation, and—when welfare is in addition (weakly) increasing in each instrument over the relevant range—makes the corner solutions welfare-inferior to it, subject to the endogenous-boundary qualification of Remark 6.6.2, which binds in this region.

Where a negative-θ^* diagnosis is plausible (India, Hong Kong, China), the gate's false-negative cost is small and both forces favor high (ι, r). Where a positive-θ^* diagnosis is plausible (Japan, the UK), the optimal *level* of both instruments is lower, but the complementarity still holds locally: it governs the *shape* of the welfare surface, while the sign of θ^* governs the *location* of the optimum.

Figure 7 in the Online Appendix illustrates the fixed-allocation welfare surface: the interior optimum, the *MFW* point near it, and the two S.B. 21 corners welfare-inferior to it.

6.6.3 Private ordering: the Stage -1 charter choice

At a Stage -1 charter choice the controller selects a lax/decoupled regime L or a protective/coupled regime H,[17] each inducing continuation primitives $\Gamma(G) = (\pi_G, r_G, \sigma_G)$ and the associated continuation-equilibrium objects $(u_G, \theta^*_G, m^{\text{req}}_G)$, pooling mass $P(G)$, and expected enforcement burden $EC(G)$. Competitive, risk-neutral pricing capitalizes ex ante minority value $V_M(G)$ into the price $p(G) = V_M(G)$, and the controller's Stage -1 value is $V_C(G; \alpha, \nu) = \alpha\, p(G) + \Pi(G; \nu)$, where $\Pi(G; \nu)$ collects private-benefit rents net of the controller-borne enforcement share. The full charter-choice construction ($\Gamma(G)$, the message policy m_G, and the V_M, Π, V_C objects) is given in the Online

[17] As used here, the labels *coupled* and *decoupled* index the strength of the approval-to-enforcement linkage that each charter induces—the continuation bundle (π_G, r_G, σ_G), including whether the ratification standard tracks the approval threshold—rather than the doctrinal authorization rule of Section 3. On this reading, representing a cleansing-margin reform such as revised §144 as a move toward the lax regime captures the weakening of the effective ex post discipline a controller faces; it does not imply that the reform alters the authorization coupling $r = \pi$, which it leaves intact.

Appendix, §S10.1.

Assumption 7 (Two-regime ordering for Stage -1)**.** *There exist constants* $\Delta_M > 0$ *and a function* $\Delta_\Pi(v)$ *such that*

$$\Delta_M := V_M(H) - V_M(L) > 0, \qquad \Delta_\Pi(v) := \Pi(H; v) - \Pi(L; v) < 0$$

for relevant v*, and* $\Delta'_\Pi(v) \leq 0$*.*

The conditions in Assumption 7 correspond to the $(\omega, \kappa, L_r, \theta^*)$ region where coupling is welfare-improving: $\Delta_M > 0$ requires that false-positive reduction and enforcement savings outweigh false-negative costs, while $\Delta_\Pi < 0$ holds when coupling constrains extraction. Outside this region (typically $\theta^* > 0$), the welfare ranking can reverse and the governance externality need not arise, since controller and planner may agree on the lax regime.

The controller trades off a higher share price (capturing $\alpha \cdot \Delta_M$ via competitive pricing) against lost extraction rents ($|\Delta_\Pi|$), adopting protection only when $\alpha \geq \alpha^*(v) := -\Delta_\Pi(v)/\Delta_M$. The wedge that drives the externality is one of incidence: competitive pricing returns to the controller only his cash-flow share α of the minority-value gain Δ_M that coupling creates, whereas he bears the entire loss of rents $|\Delta_\Pi|$, so his private marginal benefit of protection falls short of the social marginal benefit by the fraction $1 - \alpha$ of Δ_M that accrues to the minority. The governance externality is therefore *most severe where the welfare case for mandatory coupling is strongest*: high private-benefit intensity raises $\alpha^*(v)$ (Proposition 7) and, where it also widens the welfare gap Δ_W, the two move together. Markets with concentrated ownership and large extraction rents—India, Hong Kong, China—face the widest gap between socially optimal governance and what private ordering delivers.

Assumption 8 (Planner ranking)**.**

$$\Delta_W := W(H) - W(L) > 0.$$

Proposition 7 (Private ordering may underselect protection)**.** *Maintain Assumption 7, and let* $\alpha^*(v) := -\Delta_\Pi(v)/\Delta_M > 0$ *denote the cash-flow-alignment threshold at which the controller is indifferent between the two regimes. The controller adopts the protective regime* H *if and only if* $\alpha \geq \alpha^*(v)$*, and this threshold is weakly increasing in private-benefit intensity* v*. Consequently, whenever welfare ranks* H *above* L *(Assumption 8) but cash-flow alignment is insufficient,* $\alpha < \alpha^*(v)$*, private ordering selects the lax regime even though firm-value welfare is maximized by the protective one:*

$$\arg\max_{G \in \{L,H\}} V_C(G; \alpha, v) \neq \arg\max_{G \in \{L,H\}} W(G).$$

The region of inefficient lax choice expands weakly with v *and shrinks with* α*.*

Proof. The regime-choice threshold and its monotonicity in v are established as Propositions 10 and 11 in the Online Appendix (§S10); combined with Assumption 8, they yield the externality, as shown in §S10.2.3 there. □

Higher ν weakly raises $\alpha^*(\nu)$, so more controllers choose the lax regime; and where it also raises Δ_W (coupling is more socially valuable), the result is a vicious circle. The controllers who would benefit society most from adopting protection are precisely those who will not do it voluntarily, which formalizes the case for mandatory procedural standards.

The controller-borne share of enforcement costs ξ mediates the externality's severity. Under the English Rule (UK, Australia, Singapore), the losing party bears both sides' costs (ξ high), partially correcting the externality. Under the American Rule (ξ lower, with D&O insurance absorbing exposure), the externality is exacerbated. This connects the private ordering model to cross-jurisdictional variation in cost-shifting rules.

Figure 6 in the Online Appendix illustrates the externality zone in (ν, α) space, with approximate jurisdiction placements (Japan above the adoption boundary, Delaware near it, India deep in the zone).

SB 21 amended DGCL Section 144 to permit business judgment protection for non-going-private controller transactions (revised §144(b)) with *either* disinterested-stockholder approval *or* independent-committee approval; going-private transactions under §144(c) continue to require both.[18] The either/or structure interacts with Corollary 4: under the stated conditions, and where welfare is also increasing in each instrument over the relevant range, the conditional complementarity makes the non-going-private safe harbor's corner solutions welfare-inferior to the joint requirement that *MFW* imposed, within the coupling-favorable parameter region. When $\alpha < \alpha^*(\nu)$, controllers select the procedural prong that constrains extraction least (Proposition 7). In that parameter region the non-going-private either/or structure is therefore costly on two margins: it forfeits procedural complementarity, and private ordering lets controllers select the less protective corner. Khoo and Tallarita (2025) reports evidence consistent with the governance-externality channel in the reform's legislative history, though the model does not by itself establish legislative motive.

7 Conclusion

This paper began from a puzzle: if tougher shareholder votes are meant to discipline controllers, why does a standard signaling model predict that they increase misrepresentation? The paper has shown that the answer lies in what the one-stage model leaves out, since that model treats approval as the end of the story. Once the difficulty of buying post-vote peace is tied to the approval threshold, a tougher vote raises both the price of persuasion and the price of peace. Because the second channel can dominate, stricter voting contracts the pooling region rather than expanding it.

Whether that reversal is desirable is a conditional question. The welfare effect of coupling turns

[18] Delaware General Assembly, SB21: https://legis.delaware.gov/BillDetail/141930; implementation context: https://www.reuters.com/legal/legalindustry/delaware-law-changes-parameters-transactions-involving-interested-directors-2025-05-08/.

on four institutionally interpretable parameters—the deadweight share of enforcement costs ω, the convexity of those costs in the disclosure gap κ, the sensitivity of enforcement to the ratification threshold L_r, and the quality of the marginal deterred deal θ^* (together with the equilibrium densities and policy slopes that also enter the formal condition)—and different jurisdictions plausibly occupy different regions of this space. Coupling is welfare-improving where the marginal deterred deal destroys value and welfare-reducing where the deal creates value, so the comparative discussion treats each jurisdiction's placement as a model-generated diagnosis rather than as an estimate.

Two institutional implications follow, each conditional on the same parameters. The first concerns the relationship between screening instruments. Since the cost of an independent committee is largely independent of the enforcement environment while its benefit is increasing in ratification toughness, committee independence and ratification toughness reinforce one another under the conditions studied here, which gives a qualified reason to treat them as complements rather than as interchangeable substitutes. The second concerns private ordering. Because a controller who selects its own governance arrangements internalizes the resulting gains to minority value only in proportion to its stake, private ordering can underprovide protection precisely in the concentrated-ownership, high-tunneling settings where coupling is most valuable, and the allocation of litigation costs between the parties mediates how severe that shortfall becomes.

The analysis does not imply that tougher votes are always preferable, or that joint procedural requirements should be universal. Its claim is narrower. A voting rule cannot be evaluated solely by how it changes the approval margin, because when the same rule also changes the cost of post-vote peace, the enforcement margin can dominate. In short, that interaction is the reason to be cautious about treating independent committees and minority votes as simple substitutes, as the either/or safe harbor for non-going-private transactions under revised Section 144 now permits. Extending the framework to the liability-shifting architectures of French and German law, where compensation rather than approval does the disciplining work, is a natural next step for future research.

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Appendix

Roadmap. The print appendix has two sections. Appendix A proves the fixed-bar equilibrium characterization, records the consolidated boundary-machinery lemma used throughout, and proves the litigation-adjusted comparative static. Appendix B contains the welfare derivations, the primitive-beliefs remark, and the sharp LA-DWBR proof. Proofs for the committee and private-ordering extensions, together with the ratification-cost microfoundation, are in the Online Appendix, which also develops the one-stage benchmark against which the litigation-adjusted result is compared and reports the robustness checks (endogenous court credit, continuous extraction, settlement bargaining, bounded-gap message spaces) that are not needed for the baseline equilibrium or welfare results.

A Core equilibrium and comparative-static proofs

A.1 Deferred proof of Proposition 1

Proof. Throughout, we impose the maintained identity calibration $\mu(m) = m$ of Proposition 1, so that $m_{\text{req}}(u) = u$ and every equality $\mu(\theta) = \theta$ below holds by the calibration rather than by Bayes' rule; the general monotone-belief case is treated in the Online Appendix. Preliminaries (voting and shape): By (M1), μ is continuous, strictly increasing, and invertible, so $m_{\text{req}}(u)$ exists

and is unique and satisfies $\mu(m_{\text{req}}(u)) = u$. Define $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(m_{\text{req}}(u), \theta)$. The wedge properties (S3) and Assumption (P3) imply:

(a) $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing because

$$\Lambda_\theta(\theta; u) = \alpha + B'(\theta) - C_\theta\big(m_{\text{req}}(u), \theta\big) \;\geq\; \alpha + B'(\theta) \;>\; 0;$$

(b) by Lemma 3(ii), $\Lambda_u(\theta; u) \;=\; -\,C_m(m_{\text{req}}(u), \theta)\,\mu^{-1\prime}(u) \;\leq\; 0$, with strict inequality whenever $m_{\text{req}}(u) > \theta$.

(i) Existence. We verify sender optimality type-by-type and then receivers' best responses.

Low types $\theta < \theta^*(u)$. By definition of $\theta^*(u)$, $\Lambda(\theta; u) < 0$. If a low type deviates to the least-cost approved message $m_{\text{req}}(u)$, her payoff is $\Lambda(\theta; u) < 0$, strictly below truthful rejection (0 by (S2)). Any other approved message $m > m_{\text{req}}(u)$ is strictly more costly because $C_m(m, \theta) > 0$ and is increasing on the up-wedge (S3), hence strictly worse. Any under-report $m < \theta$ is rejected since μ is increasing and $m < m_{\text{req}}(u)$ implies $\mu(m) < u$; moreover $C(m, \theta) > C(\theta, \theta) = 0$ by (S2)-(S3). Thus truth maximizes payoff for $\theta < \theta^*(u)$.

Persuaders $\theta \in [\theta^*(u), u)$. Truth would be rejected ($\mu(\theta) < u$), delivering 0. Sending $m_{\text{req}}(u)$ guarantees approval with payoff $\Lambda(\theta; u) \geq 0$. Among approved messages, $m_{\text{req}}(u)$ minimizes signaling cost by the same up-wedge argument as above, hence is strictly optimal.

High types $\theta \geq u$. Under the candidate strategy, high types separate truthfully on $[u, \overline{\theta}_0]$. By the identity calibration along this separating segment, $\mu(m = \theta) = \theta \geq u$, so truth is approved at zero cost (S2). Any $m > \theta$ raises cost (no incremental benefit), and any $m < \theta$ either remains approved yet adds strictly positive cost on the down-wedge (S3), or falls below u and is rejected. Hence $m = \theta$ is optimal.

Receiver best responses and beliefs. Given the message strategy $m(\cdot)$ above, the on-path messages are (i) $m = \theta$ for $\theta \geq u$, (ii) $m = m_{\text{req}}(u)$ for $\theta \in [\theta^*(u), u)$, and (iii) $m = \theta$ for $\theta < \theta^*(u)$ (truthful, rejected). On the separating segment, the identity calibration gives $\mu(\theta) = \theta \geq u$, so truth is approved. At the pooling message, $m_{\text{req}}(u)$ meets the message threshold by construction (see Remark B.4), so minorities approve.

For off-path messages we assign GP-credible beliefs (the low separating messages $m = \theta < \theta^*(u)$ are on-path, read off the primitive belief μ, and rejected). For $m' < m_{\text{req}}(u)$, any sender is rejected under the primitive perception, so a deviator earns 0 by (C1); the weak-gainer set consists of the types whose equilibrium payoff is 0—$[\underline{\theta}_0, \theta^*(u)]$, including the indifferent boundary type—and the prior restricted to that set has mean below $\theta^*(u) < u$, so rejection is sustained by GP-consistent beliefs. Approved off-path messages arise only above the support ($m' > \overline{\theta}_0$, where the primitive perception still clears the bar); such a message carries a strictly larger gap than $m_{\text{req}}(u)$ for every type, so no type profits from it. No type benefits from any deviation, so the stated profile is an AVE.

(ii) Allocation uniqueness for fixed u. Let $(\hat{m}, \widehat{\text{beliefs}})$ be any AVE with the same cut-off u. We show the implementation set is $\{\theta \geq \theta^*(u)\}$.

Claim A. No $\theta < \theta^*(u)$ can be implemented. If $\hat{m}(\theta)$ induces approval, then among approved messages the least costly is $m_{\text{req}}(u)$ (by S3), giving payoff $\Lambda(\theta; u) < 0$, strictly below truthful rejection 0.

Claim B. Every $\theta > \theta^*(u)$ must be implemented. We distinguish two sub-cases.

Sub-case B1: $\theta \in (\theta^*(u), u)$. If $\hat{m}(\theta)$ is rejected, deviating to $m_{\text{req}}(u)$ yields payoff $\Lambda(\theta; u) > 0$ (by single-crossing, Lemma 3(i), and $\theta > \theta^*(u)$), a strict improvement. Among approved messages, $m_{\text{req}}(u)$ minimizes cost on the up-wedge (S3), so $m_{\text{req}}(u)$ is strictly optimal. Hence θ must be implemented.

Sub-case B2: $\theta \geq u$. Truthful reporting $m = \theta$ is approved under GP-credible beliefs: on the separating segment $[u, \overline{\theta}_0]$, the identity calibration gives $\mu(\theta) = \theta \geq u$. The payoff from truth is $\alpha\theta + B(\theta) - C(\theta, \theta) = \alpha\theta + B(\theta) \geq \alpha u + B(u) \geq 0$, where the first inequality uses $\alpha + B' > 0$ (P3) and the second is (CPC). If instead $\hat{m}(\theta)$ is rejected, payoff is 0, strictly worse. If $\hat{m}(\theta)$ induces approval via some $m \neq \theta$, then $C(m, \theta) > 0 = C(\theta, \theta)$ by (S2), so truth at zero cost strictly dominates.

Thus all $\theta > \theta^*(u)$ are implemented.

At $\theta = \theta^*(u)$, $\Lambda(\theta^*(u); u) = 0$; the type is indifferent between truthful rejection and sending $m_{\text{req}}(u)$, and either tie-break leaves all other allocations unchanged. (In the boundary cases $\theta^*(u) = u$ or $\theta^*(u) = \underline{\theta}_0$, indifference is vacuous: the equilibrium reduces to full separation or full pooling below u, as in part (iii).) Combining the claims gives the stated allocation uniqueness.

(iii) Pooling vs. separation. Part (iii) is the endpoint trichotomy of Lemma 3(iv), applied at the identity calibration ($m_{\text{req}}(u) = u$, so $\theta^\top(u) = u$ and $\Psi_h(u) = \alpha u + B(u)$). □

A.2 General lemmas

The following lemma is the proof scaffolding for the fixed-bar equilibrium characterization of Proposition 1 and for the litigation-adjusted comparative static: it records the single-crossing property of the lying-profit function, its sensitivity to the voting bar and the ratification standard, the induced boundary slopes, and the endpoint pooling test. It is used only within the appendix proofs and is not an independent result.

Lemma 3 (Boundary machinery for the lying-profit function)**.** *Maintain* (P1)–(P3), (S1)–(S2), *and* (M1). *Fix an interior belief cut-off* $u \in (\underline{\theta}_0, \overline{\theta}_0)$ *and write*

$$m^{\text{req}}(u) := \mu^{-1}(u), \qquad \theta^\top(u) := \min\{m^{\text{req}}(u), \overline{\theta}_0\}, \qquad \Lambda(\theta; u, r) := \alpha\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big),$$

where $C(m, \theta) \equiv L(m - \theta, r, \sigma)$ *and* $\Delta := m^{\text{req}}(u) - \theta \geq 0$ *on the up-wedge, so that* $C_m = \eta + \kappa\Delta$ *and* $C_\theta = -C_m$ *there. Then:*

(i) (Single crossing.) *On* $[\underline{\theta}_0, \theta^\top(u)]$, $\Lambda_\theta = \alpha + B'(\theta) + \eta + \kappa\Delta > 0$, *so* $\theta \mapsto \Lambda(\theta; u, r)$ *is continuous*

and strictly increasing and $\Lambda = 0$ has at most one solution there; if $\Lambda(\theta_1; u, r) < 0 < \Lambda(\theta_2; u, r)$, a unique root $\theta^*(u, r) \in (\theta_1, \theta_2)$ exists.

(ii) (Bar and ratification sensitivity.) *Off the kink $\Delta = 0$ (one-sided there), $\Lambda_u = -C_m\,(\mu^{-1})'(u)$, so* $\operatorname{sign}(\Lambda_u) = \operatorname{sign}\big(\theta - m^{\mathrm{req}}(u)\big)$*; on the up-wedge $\Lambda_u = -(\eta + \kappa\Delta)\,(\mu^{-1})'(u) < 0$. Holding (θ, u) fixed, $\Lambda_r = -L_r \le 0$ by (L2).*

(iii) (Boundary slopes.) *Where it is interior, the root $\theta^*(u, r)$ is C^1 with*

$$\frac{\partial\theta^*}{\partial u} = \frac{(\eta + \kappa\Delta)\,(\mu^{-1})'(u)}{\alpha + B'(\theta^*) + \eta + \kappa\Delta} \in \big(0, (\mu^{-1})'(u)\big), \qquad \frac{\partial\theta^*}{\partial r} = \frac{L_r}{\alpha + B'(\theta^*) + \eta + \kappa\Delta} \;\ge\; 0.$$

(iv) (Endpoint trichotomy.) *Define $\Psi_\ell(u) := \Lambda(\underline{\theta}_0; u, r)$ and $\Psi_h(u) := \Lambda\big(\theta^\top(u); u, r\big)$; under the maintained lower-endpoint bound $\alpha\underline{\theta}_0 + B(\underline{\theta}_0) < 0$, $\Psi_\ell(u) < 0$ for every interior u. Exactly one of the following obtains:* partial pooling *($\Psi_\ell < 0 < \Psi_h$), with a unique $\theta^*(u, r) \in (\underline{\theta}_0, \theta^\top(u))$ and strictly positive pooling mass $F(\theta^\top(u)) - F(\theta^*)$;* full separation *($\Psi_h \le 0$), with $\theta^* = \theta^\top(u)$ and an empty pooling set; or* full pooling below the top *($\Psi_\ell \ge 0$), with $\theta^* = \underline{\theta}_0$. On the uncapped branch $m^{\mathrm{req}}(u) \le \overline{\theta}_0$, $\Psi_h(u) = \alpha m^{\mathrm{req}}(u) + B(m^{\mathrm{req}}(u))$; on the capped branch, $\Psi_h(u) = \alpha\overline{\theta}_0 + B(\overline{\theta}_0) - C\big(m^{\mathrm{req}}(u), \overline{\theta}_0\big)$.*

Proof. By (S1), C is C^1 off the diagonal $\{\Delta = 0\}$, with one-sided derivatives (limits $\pm\eta$) at the diagonal; on the up-wedge, (S2)–(S3) give $C_m\big(m^{\mathrm{req}}(u), \theta\big) = \eta + \kappa\Delta > 0$ and $C_\theta = -C_m < 0$; on the down-wedge, $C_m = -\eta + \kappa(m - \theta) < 0$. By (M1), $m^{\mathrm{req}} = \mu^{-1}$ is strictly increasing with $(\mu^{-1})'(u) > 0$. These computations are made once here and used throughout.

(i) On $[\underline{\theta}_0, \theta^\top(u)]$ every type lies weakly below the required message, so $\Lambda_\theta(\theta; u, r) = \alpha + B'(\theta) - C_\theta = \alpha + B'(\theta) + \eta + \kappa\Delta \ge \alpha + B'(\theta) + \eta > 0$ by (P3). Strict monotonicity and continuity give at most one zero; with a sign change, the intermediate value theorem gives existence and uniqueness.

(ii) The chain rule applied to $u \mapsto C\big(m^{\mathrm{req}}(u), \theta\big)$ gives $\Lambda_u = -C_m\,(\mu^{-1})'(u)$, and the sign claims follow from the wedge signs of C_m above (at $\Delta = 0$ the one-sided derivatives carry the corresponding signs). Holding (θ, u) fixed, $\Lambda_r = -\partial L/\partial r = -L_r \le 0$ by (L2).

(iii) At an interior root the gap is strictly positive, so Λ is C^1 in a neighborhood with $\Lambda_\theta > 0$ by (i); the Implicit Function Theorem gives C^1 dependence and

$$\frac{\partial\theta^*}{\partial u} = -\frac{\Lambda_u}{\Lambda_\theta} = \frac{(\eta + \kappa\Delta)\,(\mu^{-1})'(u)}{\alpha + B'(\theta^*) + \eta + \kappa\Delta}, \qquad \frac{\partial\theta^*}{\partial r} = -\frac{\Lambda_r}{\Lambda_\theta} = \frac{L_r}{\alpha + B'(\theta^*) + \eta + \kappa\Delta},$$

the first lying strictly below $(\mu^{-1})'(u)$ because the denominator exceeds the numerator's factor $\eta + \kappa\Delta$ by $\alpha + B' > 0$ (P3), and the second nonnegative by (L2).

(iv) $\Psi_\ell(u) = \alpha\underline{\theta}_0 + B(\underline{\theta}_0) - C\big(m^{\mathrm{req}}(u), \underline{\theta}_0\big) \le \alpha\underline{\theta}_0 + B(\underline{\theta}_0) < 0$, using $C \ge 0$ and the maintained bound. Given the strict monotonicity in (i): if $\Psi_\ell < 0 < \Psi_h$, the intermediate value theorem delivers a unique interior root, and the pooling mass is strictly positive because $f > 0$ on the support (P2); if $\Psi_h \le 0$, then $\Lambda \le 0$ on the whole interval, no type finds shading to $m^{\mathrm{req}}(u)$

profitable, and the pooling set is empty; if $\Psi_\ell \geq 0$, then $\Lambda \geq 0$ throughout and all types below $\theta^\top(u)$ weakly prefer shading to the approving message. The endpoint expressions follow from $C\big(m^{\text{req}}(u), m^{\text{req}}(u)\big) = 0$ on the uncapped branch. □

Degenerate all-Yes/all-No cut-offs and the no-single-pool uniqueness lemma are in the Online Appendix.

A.3 Proof of Proposition 2

Proof. We prove each part in turn.

Part (i): decomposition of $dP/d\pi$. The boundary $\theta^*(u, r)$ solves $\Lambda(\theta^*; u, r) = 0$ and, by Lemma 3(iii), is C^1 with $\partial\theta^*/\partial u \in \big(0, (\mu^{-1})'(u)\big)$ and $\partial\theta^*/\partial r = L_r(g^*, r, \sigma)/\big(\alpha + B'(\theta^*) + C_m\big) \geq 0$, strict whenever $L_r(g^*, r, \sigma) > 0$—as on the active branch of the ratification microfoundation (Lemma 6)—whereas (L2) alone gives only the weak inequality. Economically, harder ratification raises the marginal cost of lying at every gap and pushes the boundary θ^* upward, so more types separate. Under the maintained identity calibration $(\mu^{-1})'(u) = 1$, so $\partial\theta^*/\partial u = C_m/(\alpha + B' + C_m) \in (0, 1)$; the general monotone-belief case is recorded in Corollary 1.

Differentiation of $P(\pi)$. The pooling mass is

$$P(\pi) \;=\; \int_{\theta^*(u(\pi),\, r(\pi))}^{u(\pi)} f(\theta)\, d\theta.$$

By Leibniz's rule and the chain rule, with $\theta^* = \theta^*(u(\pi), r(\pi))$:

$$\begin{aligned}
\frac{dP}{d\pi} &= \frac{\partial}{\partial u}\left[\int_{\theta^*}^{u} f(\theta)\, d\theta\right] u'(\pi) \;+\; \frac{\partial}{\partial r}\left[\int_{\theta^*}^{u} f(\theta)\, d\theta\right] r'(\pi) \\
&= \left[f(u) - f(\theta^*)\,\frac{\partial\theta^*}{\partial u}\right] u'(\pi) \;-\; f(\theta^*)\,\frac{\partial\theta^*}{\partial r}\, r'(\pi) \\
&= u'(\pi)\left[f(u) - f(\theta^*)\,\frac{\partial\theta^*}{\partial u}\right] \;-\; f(\theta^*)\,\frac{\partial\theta^*}{\partial r}\, r'(\pi),
\end{aligned}$$

where the first term is the standard bar–boundary effect and the second term is the ratification-channel effect. This establishes part (i).

Part (ii): LA-DWBR condition. Suppose $dP/d\pi \leq 0$. Since $f(\theta^*) > 0$ (interior support) and $u'(\pi) > 0$ (the affine baseline gives $u'_{\text{SM}}(\pi) = (2/\lambda)/(1 - \alpha - b_1) > 0$ on the whole pooling corridor, sharpening the stringent-branch statement of Lemma 5(i)), we may divide both sides of $dP/d\pi \leq 0$ by $f(\theta^*)\, u'(\pi) > 0$ without changing the inequality direction:

$$\frac{f(u)}{f(\theta^*)} \;-\; \frac{\partial\theta^*}{\partial u} \;-\; \frac{\partial\theta^*}{\partial r}\,\frac{r'(\pi)}{u'(\pi)} \;\leq\; 0.$$

Rearranging:

$$\frac{\partial\theta^*}{\partial u} + \frac{\partial\theta^*}{\partial r}\frac{r'(\pi)}{u'(\pi)} \geq \frac{f(u)}{f(\theta^*)},$$

which is exactly (2). Conversely, if (2) holds, the same algebra in reverse gives $dP/d\pi \leq 0$.

Part (iii): uniform prior and identity beliefs. Assume f is uniform on $[a, b]$ (so $f(\theta) = 1/(b-a)$ for all $\theta \in [a, b]$) and $\mu(m) = m$ (identity beliefs, so $m^{\text{req}}(u) = u$ and $(\mu^{-1})'(u) = 1$). Then:

(a) $f(u)/f(\theta^*) = 1$;
(b) From Lemma 3(iii), $\partial\theta^*/\partial u = C_m/(\alpha + B' + C_m) \in (0, 1)$, since the denominator exceeds the numerator by $\alpha + B' > 0$.

The standard (no-litigation) DWBR condition requires $\partial\theta^*/\partial u \geq f(u)/f(\theta^*) = 1$, which fails because $\partial\theta^*/\partial u < 1$. However, the LA-DWBR condition (2) requires only

$$\frac{C_m}{\alpha + B' + C_m} + \frac{L_r(g^*, r, \sigma)}{\alpha + B' + C_m} \cdot \frac{r'(\pi)}{u'(\pi)} \geq 1,$$

where we used the boundary slopes of Lemma 3(iii) with $(\mu^{-1})'(u) = 1$. Subtracting the first term from both sides:

$$\frac{L_r(g^*, r, \sigma)\, r'(\pi)}{(\alpha + B' + C_m)\, u'(\pi)} \geq 1 - \frac{C_m}{\alpha + B' + C_m} = \frac{\alpha + B'}{\alpha + B' + C_m}.$$

Multiplying both sides by $(\alpha + B' + C_m) > 0$:

$$\frac{L_r(g^*, r, \sigma)\, r'(\pi)}{u'(\pi)} \geq \alpha + B'(\theta^*).$$

This is the stated condition: the marginal tightening of Stage-2 enforcement from ratification must dominate the marginal prize slope. □

B Welfare derivations and deferred proofs

B.1 Proof of Proposition 4

Proof. By definition,

$$W^{\text{FB}} - W^{\text{firm}}(\pi) = \int_0^{\overline{\theta}_0} \theta f(\theta)\, d\theta - \int_{\theta^*(\pi)}^{\overline{\theta}_0} \theta f(\theta)\, d\theta + \omega\, \text{EC}(\pi) = \int_{\theta^*(\pi)}^{0} (-\theta)\, f(\theta)\, d\theta + \omega\, \text{EC}(\pi),$$

where the signed integral is read as $\int_0^{\theta^*} \theta f\, d\theta$ when $\theta^*(\pi) > 0$. That term equals $\text{FP}(\pi)$ when $\theta^*(\pi) < 0$ (with $\text{FN} = 0$), equals $\text{FN}(\pi)$ when $\theta^*(\pi) > 0$ (with $\text{FP} = 0$), and vanishes when $\theta^*(\pi) = 0$: the three cases are the sign cases of θ^* and routine interval algebra. Rearranging gives

$W^{\text{firm}}(\pi) = W^{\text{FB}} - \text{FP}(\pi) - \text{FN}(\pi) - \omega\,\text{EC}(\pi).$ □

B.2 Proof of Proposition 5

Proof. Under Assumptions 4-6,

$$\text{EC}(\pi) = \int_{\theta^*(\pi)}^{u(\pi)} \phi(\theta, \pi)\, d\theta, \quad \phi(\theta, \pi) = L(m^{\text{req}}(u(\pi)) - \theta, r(\pi), \sigma)\, f(\theta),$$

and Leibniz' rule, with the chain rule $\frac{d}{d\pi} m^{\text{req}}(u(\pi)) = (m^{\text{req}})'(u)u'(\pi)$ applied inside the integrand, gives the first derivative formula in the proposition:

$$\frac{d}{d\pi}\text{EC}(\pi) = \phi(u, \pi)u'(\pi) - \phi(\theta^*, \pi)\theta^{*\prime}(\pi) + \int_{\theta^*}^{u} \Big[L_g\,(m^{\text{req}})'(u)\,u'(\pi) + L_r\,r'(\pi)\Big] f(\theta)\, d\theta.$$

For welfare, $W^{\text{firm}}(\pi) = \int_{\theta^*(\pi)}^{\overline{\theta}_0} \theta f(\theta)\, d\theta - \omega\,\text{EC}(\pi)$, so again by Leibniz

$$\frac{d}{d\pi} W^{\text{firm}}(\pi) = -\theta^*(\pi) f(\theta^*(\pi))\theta^{*\prime}(\pi) - \omega\,\frac{d}{d\pi}\text{EC}(\pi).$$

Since $\theta^*(u, r)$ is C^1 with the slopes of Lemma 3(iii), $\theta^{*\prime}(\pi) = \theta^*_u\,u'(\pi) + \theta^*_r\,r'(\pi)$; substituting into $\text{EC}'(\pi)$ and collecting coefficients of $u'(\pi)$ and $r'(\pi)$ yields $\frac{d}{d\pi}\text{EC}(\pi) = u'(\pi)\mathcal{D}_u(\pi) + r'(\pi)\mathcal{D}_r(\pi)$ with

$$\mathcal{D}_u(\pi) := \phi(u, \pi) - \phi(\theta^*, \pi)\theta^*_u + \int_{\theta^*}^{u} L_g(m^{\text{req}}(u) - \theta, r(\pi), \sigma)\,(m^{\text{req}})'(u) f(\theta)\, d\theta,$$

$$\mathcal{D}_r(\pi) := -\phi(\theta^*, \pi)\theta^*_r + \int_{\theta^*}^{u} L_r(m^{\text{req}}(u) - \theta, r(\pi), \sigma)\, f(\theta)\, d\theta.$$

□

B.3 Derivation for Example 1

Proof. Under identity beliefs $m^{\text{req}}(u) = u$; write $w := u - \theta^*$. Boundary indifference $\Lambda(\theta^*; u) = 0$ with $B(\theta) = b_0 + b_1\theta$ rearranges to

$$\frac{\kappa(r, \sigma)}{2} w^2 + \big(\alpha + b_1 + \eta(r, \sigma)\big) w = (\alpha + b_1)u + b_0,$$

whose positive root is

$$w(u, r) = \frac{-\big(\alpha + b_1 + \eta(r, \sigma)\big) + \sqrt{\big(\alpha + b_1 + \eta(r, \sigma)\big)^2 + 2\kappa(r, \sigma)\big((\alpha + b_1)u + b_0\big)}}{\kappa(r, \sigma)}$$

for $\kappa(r,\sigma) > 0$, and $w(u,r) = \big((\alpha+b_1)u + b_0\big)/\big(\alpha + b_1 + \eta(r,\sigma)\big)$ in the linear case $\kappa(r,\sigma) = 0$; hence $\theta^*(u,r) = u - w(u,r)$. With uniform density $1/\Delta_\theta$ on $[\underline{\theta}_0, \overline{\theta}_0]$ and the substitution $t = u - \theta$,

$$\mathrm{EC}(u,r) = \frac{1}{\Delta_\theta}\int_0^w \left[\eta(r,\sigma)\,t + \frac{\kappa(r,\sigma)}{2}t^2\right] dt = \eta(r,\sigma)\frac{w^2}{2\Delta_\theta} + \kappa(r,\sigma)\frac{w^3}{6\Delta_\theta},$$

and

$$W^{\mathrm{firm}}(u,r) = \frac{1}{\Delta_\theta}\int_{\theta^*(u,r)}^{\overline{\theta}_0} \theta\, d\theta - \omega\,\mathrm{EC}(u,r) = \frac{\overline{\theta}_0^{\,2} - \theta^*(u,r)^2}{2\Delta_\theta} - \omega\,\mathrm{EC}(u,r).$$

□

B.4 Deferred derivations from the main text

This subsection collects material moved from the main text to streamline the exposition. Cross-references in Sections 4–5 point here.

The buying-peace game underlying Lemma 1, the proof that it implies (L1)–(L3), and the uniform-resistance closed form are in the Online Appendix.

Remark (Primitive beliefs at the pooling message). The voting outcome in this model is governed by the *primitive* belief function μ satisfying (M1), not by the equilibrium Bayesian posterior that a fully strategic receiver would compute: atomistic minority shareholders observe the disclosed message m, form the belief $\mu(m)$, and vote according to the aggregation rule (V2), without reverse-engineering the controller's equilibrium strategy (Levit et al., 2024). Partial attention ($\tau < 1$ in the notation of Section 5.1.3) pulls the effective belief toward the prior mean, which can sustain approval of the pooling message under Bayesian updating only where $\mathrm{E}[\theta] > u$ and attention is low; at the stringent bars $u \geq \mathrm{E}[\theta]$ that drive the reversal, approval rests on the atomistic-voter assumption itself; the reversal itself survives without the device (Proposition 3). The supporting threshold algebra is in the Online Appendix (§S9.5).

B.4.1 Proof of Corollary 2 (sharp LA-DWBR under ratification microfoundation)

Proof. Under the microfoundation, the enforcement cost is $L(g,r,\sigma) = p(g,r)\cdot C\big(R(g,x^*),r,\sigma\big)$. Differentiating with respect to r:

$$L_r \;=\; p_r \cdot C \;+\; p \cdot C_r.$$

By Lemma 1(c), on the active branch $r > r_0(R,\sigma)$:

$$C_r(R,r,\sigma) \;=\; \frac{R + v^*(r,\sigma)}{1-\sigma} \;>\; 0,$$

where $v^*(r,\sigma) = G^{-1}\big(r/(1-\sigma)\big)$. Substituting into Proposition 2(iii), with $r'(\pi) = 1$, yields (4). At the kink $r = r_0$ we have $R + v^* = 0$, so $C_r = 0$ and the inequality holds weakly. □

Online Appendix

S1 Credible-beliefs refinement (Grossman–Perry 1986)

S1.1 Definition

Consider any signalling game with Perfect Bayesian Equilibria (PBE). A PBE satisfies the credible-beliefs refinement if, for every message m_{off} sent with zero probability in the candidate equilibrium, the receiver's belief $\mu(\cdot \mid m_{\text{off}})$ assigns probability 1 to the Grossman–Perry credible-beliefs set $T^{\text{best}}(m_{\text{off}}) := \{\theta \in \Theta : \; U_S(m_{\text{off}}, \theta) \geq U_S^*(\theta)\}$ of sender types that would (weakly) gain by deviating relative to their equilibrium payoff $U_S^*(\theta)$; if T^{best} is empty, beliefs are unrestricted. When a specific supporting belief is exhibited, we use the canonical conditional-prior selection: the belief at m_{off} is the prior restricted to $T^{\text{best}}(m_{\text{off}})$. The receiver thus deems the unexpected signal most likely to come from the types with the strongest incentive to deviate—usually the *worst* types from the receiver's standpoint—and this pessimism discourages profitable deviations and helps select a unique equilibrium.

S1.2 Illustration in our partial-pooling equilibrium

Equilibrium recap. For a fixed belief cut-off u, the equilibrium of Proposition 1 partitions types into three regions. Types $\theta < \theta^*(u)$ report truthfully and are rejected; types $\theta \in [\theta^*(u), u)$ send the cheapest persuasive message $m_{\text{req}}(u)$ and are approved (the pooling band); and types $\theta \geq u$ report truthfully and are approved. Minorities approve a message m exactly when $\mu(m) \geq u$, equivalently when $(1-\alpha)\mu(m) - \beta^{\text{Div}}(m) \geq \underline{\kappa}^{(\lambda)}$.

Off-path deviations. Two classes of deviation must be deterred. Consider first an *upward* deviation to an off-path message $m_H > m_{\text{req}}(u)$. Pooling types in $[\theta^*(u), u)$ are already approved at $m_{\text{req}}(u)$, and separating types $\theta \geq u$ are already approved at zero signalling cost; for both, raising the message only adds cost, so neither gains. Consider next a *downward* deviation to an off-path message $m_L < m_{\text{req}}(u)$. Such a message yields voting statistic $\mu(m_L) < u$, so minorities reject; a pooling type that deviates down is therefore rejected and earns at most 0, weakly below its equilibrium payoff $\Lambda(\theta; u) \geq 0$. Since $\Lambda(\theta; u) < 0$ for types $\theta < \theta^*(u)$, those types strictly prefer truthful rejection to pooling and so do not mimic the band.

The Grossman–Perry credible-beliefs construction makes these conclusions belief-proof. As shown in Lemma 4, an off-path message is attributed to the set of types that would gain most from sending it, and under the resulting posterior no type's deviation is profitable. Thus the partial-pooling *allocation* is the unique atomistic-voter equilibrium outcome for the given voting rule (up to the boundary tie-break), without recourse to arbitrary off-path beliefs.

S2 Auxiliary uniqueness results

Corollary 5 (Degenerate voting rules)**.** *Let $u = \underline{\theta}$ be the rule-specific belief cut-off.*

(i) Trivially easy rule $\left[u < \underline{\theta}_0\right]$. Every primitive belief satisfies $\mu(m) \geq u$, hence minorities always vote "Yes". For the controller every message is accepted; because $C(m, \theta)$ is uniquely minimised at $m = \theta$ (the gap cost $\eta|m - \theta| + \frac{\kappa}{2}(m - \theta)^2$ is strictly positive for $m \neq \theta$), truthful reporting is the unique best reply. The unique AVE is therefore fully separating with universal approval, $m(\theta) = \theta$.

(ii) Trivially tough rule $\left[u > \overline{\theta}_0\right]$. No primitive belief can reach the hurdle, so minorities always vote "No". By (C1) every message then pays 0, so universal rejection is the unique AVE allocation*, and every message profile is a best reply; we select truthful reporting $m(\theta) = \theta$ as the cost-free tie-break.*

Lemma 4 (No single-message pooling)**.** *Fix an interior belief cut-off $u \in (\underline{\theta}_0, \overline{\theta}_0)$, and maintain Assumptions* P1-P3, S1-S2, M1*; assume in addition that the minimal persuasive message satisfies $m^{\text{req}} := \mu^{-1}(u) \in (\underline{\theta}_0, \overline{\theta}_0]$ and $\alpha\, m^{\text{req}} + B(m^{\text{req}}) > 0$ (under the identity calibration, (CPC) holding with slack at u). Off the equilibrium path, beliefs satisfy the Grossman-Perry credible-beliefs refinement. Then there is* no *atomistic-voter equilibrium (AVE) in which every $\theta \in (\underline{\theta}_0, \overline{\theta}_0)$ sends the same message $m^\flat$, regardless of whether minorities would approve or reject $m^\flat$.*

Proof. Let $m^{\text{req}} := \mu^{-1}(u)$ denote the minimal persuasive message (V3). Two cases:

(a) Common message is rejected: $\mu(m^\flat) < u$. Then minorities always vote "No" and, by (C1), every type's equilibrium payoff is $U^{\text{eq}}(\theta) = 0$. The message m^{req} is approved under the primitive perception ($\mu(m^{\text{req}}) = u$), so the type $\bar{\theta} := m^{\text{req}}$—in the support by hypothesis—deviating to m^{req} obtains

$$U^{\text{dev}} = \alpha\bar{\theta} + B(\bar{\theta}) - C\big(m^{\text{req}}, \bar{\theta}\big) = \alpha\, m^{\text{req}} + B(m^{\text{req}}) \;>\; 0 \;=\; U^{\text{eq}}(\bar{\theta}),$$

using $C(m^{\text{req}}, m^{\text{req}}) = 0$ (S1) and the maintained hypothesis $\alpha\, m^{\text{req}} + B(m^{\text{req}}) > 0$, contradicting sequential rationality.

(b) Common message is approved: $\mu(m^\flat) \geq u$. We split by the location of $m^\flat$.

(b1) $m^\flat < \overline{\theta}_0$ (some types under-report). Choose any type $\theta^+ \in (m^\flat, \overline{\theta}_0]$ and fix a small $\varepsilon > 0$; consider the deviation $m^+ := m^\flat + \varepsilon$. Because the deviator is under-reporting at $m^\flat$, (S3) implies $C_m(m, \theta^+) = -\eta + \kappa\,(m - \theta^+) \leq -\eta < 0$ for all $m < \theta^+$, hence

$$C(m^+, \theta^+) < C(m^\flat, \theta^+).$$

Under the Grossman-Perry refinement, upon observing the *higher* off-path message m^+, minorities assign probability 1 to the set of types who benefit most from this deviation. Only *high* types with $\theta \geq m^\flat$ strictly benefit from moving up, so the posterior at m^+ is supported on $\{\theta \geq m^\flat\}$. Since the minority's approval statistic

$$\psi(\theta) := (1 - \alpha)\theta - D(\theta)$$

is strictly increasing in θ (P3), the posterior at m^+ first-order stochastically *dominates* the pooling posterior at $m^\flat$, and thus $\mathbb{E}[\psi(\theta)\mid m^+] \geq \mathbb{E}[\psi(\theta)\mid m^\flat]$. Because the project is approved at $m^\flat$, it remains approved at m^+ as well. The deviator's benefit is unchanged (approval), while her cost is strictly lower, so $U^{\text{dev}}(\theta^+) > U^{\text{eq}}(\theta^+)$–a contradiction.

(b2) $m^\flat \geq \overline{\theta}_0$ (all types over-report). Evaluate the pooling payoff at the bottom type. Under the maintained lower-endpoint bound $\alpha\underline{\theta}_0 + B(\underline{\theta}_0) < 0$ and since $m^\flat > \underline{\theta}_0$, (S2)-(S3) give $C(m^\flat, \underline{\theta}_0) > 0$. Because $\underline{\theta}_0 < 0$,

$$U^{\text{eq}}(\underline{\theta}_0) = \alpha\,\underline{\theta}_0 + B(\underline{\theta}_0) - C(m^\flat, \underline{\theta}_0) \;<\; 0.$$

This type strictly prefers truthful rejection (payoff 0), so a single-message approved pooling profile cannot be an AVE.

Combining (a), (b1), and (b2), no AVE can feature a single message used by all types when the cut-off u is interior. □

S3 Benchmark one-stage model and bar-boundary results

This subsection reports the one-stage fixed-enforcement benchmark and is included only to anchor the reversal result. The paper's central mechanism is the additional enforcement-feedback term under coupled rules $r(\pi)$. At a fixed belief cut-off, the payoff-side comparative statics are routine: an upward quality shift or a higher private-benefit level raises the approval prize while the cost of the same minimal lie is unchanged, so the pooling band widens, whereas a larger cash-flow stake α makes the controller internalize more of a negative θ and shrinks it. Detailed bar-boundary algebra, thin-tail conditions, general monotone-belief bar–boundary variants, and supplementary figures are collected in §S4.

S3.1 Majority-of-Minority versus Supermajority Rules

The following results summarize the fixed-enforcement benchmark comparison between MoM and supermajority rules. Partial pooling under MoM and lenient rules follows directly from the endpoint test; the stringent case $\pi > 0.5$, where sufficiently high quotas can induce full separation, is characterized in Lemma 5.

S3.1.1 Partial Pooling and Separation under MoM and Supermajority Rules

Fix any interior belief cut–off u (either u_{MoM} or $u_{\text{SM}}(\pi)$ with $\pi < \frac{1}{2}$). If a local strengthened participation condition holds at the bar, $\alpha u + B(u) > C\big(\mu^{-1}(u), u\big)$, the fixed-$u$ equilibrium exhibits *partial pooling*: the lower endpoint of the lying profit is negative by the maintained bound $\alpha\underline{\theta}_0 + B(\underline{\theta}_0) < 0$; on the branch $\theta^\top(u) := \min\{m^{\text{req}}(u), \overline{\theta}_0\} \geq u$ (automatic under the identity calibration and under sluggish beliefs at stringent bars), the upper endpoint test $\Psi_h(u) = \Lambda(\theta^\top(u); u)$ is positive because Λ is increasing and the displayed condition is $\Lambda(u; u) > 0$; the endpoint trichotomy

(Lemma 3(iv)) then delivers a unique boundary $\theta^*(u) \in (\underline{\theta}_0, \theta^\top(u))$—types in $[\theta^*(u), \theta^\top(u))$ shade up to the minimal persuasive message and pass, while types above $\theta^\top(u)$ separate truthfully (allocation uniqueness for a given u follows from Proposition 1(ii); under the identity calibration $\theta^\top(u) = u$ and the pool is $[\theta^*(u), u)$). If the inequality fails—for instance, under a lenient SM that drives u sufficiently left so the approval gain shrinks while $C\big(\mu^{-1}(u), u\big) > 0$ (since $\mu^{-1}(u) \neq u$ in general)—the entire pooling region collapses and the fixed-u equilibrium is *separating* at that bar.

Lemma 5 (Stringent SM: partial pooling below a critical quota, full separation above)**.** *Maintain* P1-P3, S1-S2, M1, *and the tail-growth property of the cost specification. Fix a* stringent *quota* $\pi \in (\frac{1}{2}, 1]$ *and let* $u_{\mathrm{SM}}(\pi)$ *denote the induced (rule-specific) belief cut-off defined by* (V2)*; assume it is interior whenever invoked. Write* $m^{\mathrm{req}}(u) := \mu^{-1}(u)$,

$$\theta^\top(u) := \min\{m^{\mathrm{req}}(u), \overline{\theta}_0\}, \qquad \Lambda(\theta; u) := \alpha\theta + B(\theta) - C\big(m^{\mathrm{req}}(u), \theta\big),$$

and $\theta^*(u) := \sup\{\theta < \theta^\top(u) : \Lambda(\theta; u) < 0\}$.

(i) Interior mapping. For any stringent π *with interior* $u_{\mathrm{SM}}(\pi)$*, the interior cut-off is unique. Moreover, on its domain, the map* $\pi \mapsto u_{\mathrm{SM}}(\pi)$ *is strictly increasing; if* $\beta \circ \mu^{-1}$ *is* C^1*, then* u_{SM} *is* C^1 *with*

$$\frac{du_{\mathrm{SM}}}{d\pi} = \frac{2/\lambda}{(1-\alpha) - \beta_u\big(u_{\mathrm{SM}}(\pi)\big)} > 0.$$

(ii) Dichotomy for stringent SM under the corrected endpoint test. Define

$$\Psi_h(u) := \Lambda\big(\theta^\top(u); u\big) = \alpha\theta^\top(u) + B\big(\theta^\top(u)\big) - C\big(m^{\mathrm{req}}(u), \theta^\top(u)\big).$$

Provided Ψ_h *is single-crossing on the stringent corridor, is positive at the lower end of the corridor, and becomes negative on the capped branch* $m^{\mathrm{req}}(u) > \overline{\theta}_0$*, there exists* $\bar{\pi} \in (\frac{1}{2}, 1]$ *such that:*

$$\begin{cases} \text{If } \frac{1}{2} < \pi < \bar{\pi}, \implies \underline{\theta}_0 < \theta^*\big(u_{\mathrm{SM}}(\pi)\big) < \theta^\top\big(u_{\mathrm{SM}}(\pi)\big) \text{ (partial pooling);} \\ \text{If } \pi \geq \bar{\pi}, \implies \theta^*\big(u_{\mathrm{SM}}(\pi)\big) = \theta^\top\big(u_{\mathrm{SM}}(\pi)\big) \text{ and the pooling set is empty (full separation).} \end{cases}$$

If, in addition, $w(u) := \theta^\top(u) - \theta^*(u)$ *is strictly decreasing on the stringent corridor, then* $\bar{\pi}$ *is unique.*

Proof. (i) By (V2), an interior cut-off for a given π solves an equation in u whose left side has slope $(1-\alpha) - \beta_u > 0$ under the aggregator condition maintained at (V3) (under the affine baseline with attentive share τ, $\beta_u = b_1/\tau$, so the condition is $(1-\alpha)\tau > b_1$, which reduces to (P3) at full attention); this pins down $u_{\mathrm{SM}}(\pi)$ uniquely, and differentiating the (V2) identity yields the displayed derivative when $\beta \circ \mu^{-1}$ is C^1, with positive denominator by the same condition.

(ii) Fix a stringent π with interior $u := u_{\mathrm{SM}}(\pi)$. Since $m^{\mathrm{req}}(u) > \underline{\theta}_0$, $C(m^{\mathrm{req}}(u), \underline{\theta}_0) > 0$, so the maintained lower-endpoint bound gives $\Psi_\ell(u) := \Lambda(\underline{\theta}_0; u) < 0$; by the endpoint trichotomy (Lemma 3(iv)), partial pooling obtains exactly when $\Psi_h(u) > 0$. On the uncapped branch $m^{\mathrm{req}}(u) \leq$

$\overline{\theta}_0$ we have $\Psi_h(u) = \alpha m^{\text{req}}(u) + B(m^{\text{req}}(u)) > 0$ (as $C(m^{\text{req}}(u), m^{\text{req}}(u)) = 0$), so full separation cannot occur there. On the capped branch, $m^{\text{req}}(u) \uparrow +\infty$ as $u \uparrow \overline{\theta}_0$ while $\theta^\top(u) = \overline{\theta}_0$, so tail growth forces $\Psi_h(u) = \alpha\overline{\theta}_0 + B(\overline{\theta}_0) - C\big(m^{\text{req}}(u), \overline{\theta}_0\big) \to -\infty$. By continuity and the assumed single crossing there is a unique $\bar{u}$ with $\Psi_h(\bar{u}) = 0$, necessarily on the capped branch; let $\bar{\pi}$ solve (V2) at $\bar{u}$. Since $\pi \mapsto u_{\text{SM}}(\pi)$ is strictly increasing by (i), quotas $\frac{1}{2} < \pi < \bar{\pi}$ give $\Psi_h > 0$ and partial pooling, while $\pi \geq \bar{\pi}$ gives $\Psi_h \leq 0$, $\theta^* = \theta^\top$, and an empty pooling set. If $w(u) := \theta^\top(u) - \theta^*(u)$ is strictly decreasing on the corridor, the threshold quota is unique. □

The intuition behind Lemma 5 is as follows. Tightening the super-majority quota raises the belief cut-off u required for approval. To clear a higher bar, the controller must send a larger persuasive message $m_{\text{req}}(u) = \mu^{-1}(u)$, which increases the lie $m_{\text{req}}(u) - \theta$ for types just below u. Once the required message exceeds the type support, $m_{\text{req}}(u) = \mu^{-1}(u) > \overline{\theta}_0$, even the highest type $\overline{\theta}_0$ must send the large lie $m_{\text{req}}(u) - \overline{\theta}_0$; with tail-growth in signaling costs its expense eventually overwhelms the bounded prize $\alpha\overline{\theta}_0 + B(\overline{\theta}_0)$. At that point—corresponding to a critical quota $\bar{\pi}$—even the top type will not misreport, so the equilibrium becomes fully separating. For less strict quotas, the local strengthened participation condition ensures that the prize at the bar still exceeds the minimal signaling cost, so a nonempty set of near-threshold types shade up and partial pooling obtains.

S3.1.2 Benchmark: Pooling under Uniform Prior and Identity Beliefs

As a benchmark, we establish the pooling ranking under the simplest information environment: uniform f and identity beliefs $\mu(m) = m$.

Proposition 8 (Uniform prior & identity beliefs: pooling rises with the bar; hence $P(u_{\text{SM}}^{\text{len}}) < P(u_{\text{MoM}}) < P(u_{\text{SM}}^{\text{str}})$). *Maintain* (P1)-(P3), (S1)-(S2), *and* (M1) *in the payoff-dominant (up-talk) regime* $0 < B'(\theta) < 1 - \alpha$. *Assume the prior is uniform on* $[a, b]$ *with density* $f(\theta) = \frac{1}{b-a}$ *and identity beliefs* $\mu(m) = m$, *so the least-cost approved message at bar* u *is* $m^{\text{req}}(u) = u$. *For each interior bar* $u \in (a, b)$ *define*

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C(u, \theta), \qquad \theta^*(u) \;:=\; \sup\{\theta < u \,:\, \Lambda(\theta; u) < 0\},$$

and the pooling mass

$$P(u) \;:=\; F(u) - F\big(\theta^*(u)\big) \;=\; \int_{\theta^*(u)}^{u} f(\theta)\, d\theta \;=\; \frac{u - \theta^*(u)}{b - a}.$$

Suppose the rule-induced cut-offs are interior and ordered as $u_{\text{SM}}^{\text{len}} < u_{\text{MoM}} < u_{\text{SM}}^{\text{str}}$, *and that partial pooling persists for every* u *on the two corridors* $[u_{\text{SM}}^{\text{len}}, u_{\text{MoM}}]$ *and* $[u_{\text{MoM}}, u_{\text{SM}}^{\text{str}}]$ *(so* $\theta^*(u) \in (a, u)$ *there). Then* $P(u)$ *is strictly increasing in* u *on each corridor. Consequently,*

$$P\big(u_{\text{SM}}^{\text{len}}\big) \;<\; P\big(u_{\text{MoM}}\big) \;<\; P\big(u_{\text{SM}}^{\text{str}}\big).$$

Proof. By Lemma 3(iii), $0 < \theta^{*\prime}(u) < 1$ on any corridor with partial pooling (the denominator exceeds the numerator because $\alpha + B' > 0$). The key accounting identity is the variable-limits

derivative

$$P'(u) \;=\; f(u) \;-\; f\big(\theta^*(u)\big)\,\theta^{*\prime}(u). \tag{5}$$

Under the uniform prior $f(u) = f(\theta^*(u)) = \frac{1}{b-a}$, so $P'(u) = \frac{1-\theta^{*\prime}(u)}{b-a} > 0$. Since $u_{\mathrm{SM}}^{\mathrm{len}} < u_{\mathrm{MoM}} < u_{\mathrm{SM}}^{\mathrm{str}}$ and P is strictly increasing along both corridors, the ranking follows. □

S3.1.3 Economic intuition: threshold-bunching in shareholder approval votes

A shareholder vote creates a cliff: private benefits arrive only if the deal passes, so when disclosure and deal marketing can shift perceived quality, a wide range of borderline transactions converge on the *minimum* persuasive story that just clears the pivotal shareholders' bar. In real-world proxy practice, that minimum persuasive story maps to a fairly standard approval package—a familiar bundle of banker analyses, management projections, comparables, and deal rationales calibrated to look *fair enough* to the voters who matter (Bhattarai et al., 2022). High-quality deals do not need to lean as hard on that playbook, and very low-quality deals cannot profitably (or credibly) spin enough to get over the cliff; the strategic action happens in the middle, where many different underlying qualities rationally choose the same "just-passing" disclosure package, so the observed messages look more alike even though fundamentals differ.

This is the same economics as *threshold-bunching* in corporate reporting and regulation: when rewards jump at a benchmark, agents manage the observable metric to finish just above it, producing a pile-up right at the cutoff and "missing mass" just below (Burgstahler and Dichev, 1997; Degeorge et al., 1999).[19] A tougher voting rule moves the benchmark: transactions that would have passed on straightforward disclosure become borderline and acquire the strongest incentive to adopt the standardized just-passing package, so in a one-stage benchmark with fixed enforcement costs, tightening the vote can *increase* pooling. All of the above holds enforcement-cost parameters fixed ($r'(\pi) = 0$); when the approval rule also governs settlement and ratification, the cost of just-passing persuasion rises with the threshold, which is the enforcement channel the paper develops.

S4 Bar–boundary tradeoff analysis

This appendix collects the detailed bar–boundary comparison formulae, thin-tail sufficient conditions, general monotone-belief bar–boundary microfoundations, and illustrative figures that complement the benchmark result (Proposition 8) and the litigation-adjusted analysis (Proposition 2) in the main text.

[19] The general bunching methodology is developed in Saez (2010) and surveyed by Kleven (2016). We draw on the analogy only at the level of economic mechanism—a cliff incentive generates bunching—and do not import the estimation toolkit.

S4.1 Bar-Boundary Comparison Formula

Because the sign of $P'(u)$ depends crucially on the "bar-boundary" comparison outlined in Proposition 8 (Milgrom and Shannon (1994)), it is instructive to consider its extension to a general density, retaining the identity calibration $m^{\text{req}}(u) = u$ so that the pool's upper endpoint is u; the general monotone-belief version, with the support-truncated endpoint $\min\{m^{\text{req}}(u), \overline{\theta}_0\}$ and the normalized responsiveness, is developed in the bar–boundary microfoundations below:

$$\theta^{*\prime}(u) \gtrless \frac{f(u)}{f(\theta^*(u))}$$

where, from Lemma 3(iii) and $C_\theta = -C_m$ under gap costs,

$$\theta^{*\prime}(u) = -\frac{\Lambda_u(\theta^*(u); u)}{\Lambda_\theta(\theta^*(u); u)} = \frac{C_m\big(\mu^{-1}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)}{\alpha + B'\big(\theta^*(u)\big) + C_m\big(\mu^{-1}(u), \theta^*(u)\big)}$$

is the behavioral responsiveness of the marginal type to a small change in the bar, and

$$\chi(u) := \frac{f(u)}{f(\theta^*(u))}$$

is the density ratio at the two endpoints of the pooling band.

Both terms are economically meaningful. The numerator of $\theta^{*\prime}(u)$, $C_m\,(\mu^{-1})'(u)$, is the marginal cost pressure of a bar move du: a higher bar enlarges the required lie, so shading becomes more expensive per unit tightening. The denominator, $\alpha + B' + C_m$, is the total marginal benefit of the restoring boundary move $d\theta^*$: the truthful payoff rises at slope $\alpha + B'(\theta^*)$, and, because $C_\theta = -C_m$, a higher boundary shrinks the lie and relieves signaling cost at rate C_m. The ratio $\chi(u)$ is the purely mechanical side of the comparison, benchmarking the mass the bar picks up per unit shift, $f(u)$, against the mass the boundary would release per unit shift, $f(\theta^*(u))$, holding behavior fixed. Under a log–concave (unimodal) density with monotone shoulders, $\chi(u) \geq 1$ on the left (increasing) shoulder and $\chi(u) \leq 1$ on the right (decreasing) shoulder, where the bar sits in a thinner tail; this ratio is exactly the benchmark in the comparison $P'(u) = f(u) - f(\theta^*(u))\,\theta^{*\prime}(u)$.

S4.2 Intuition behind Proposition 8

In the special case of Proposition 8, $\mu(m) = m$ and $(\mu^{-1})'(u) = 1$, so $\theta^{*\prime}(u) = \frac{C_m(u,\theta^*)}{\alpha + B'(\theta^*) + C_m(u,\theta^*)} \in (0, 1)$: the behavioral response moves the boundary by less than one-for-one when the bar shifts. Under a uniform prior the mechanical side is neutral, $\chi(u) = 1$, so $\theta^{*\prime}(u) < \chi(u)$ and the pooling interval $[\theta^*(u), u]$ widens as the bar rises. The same logic applies on any corridor where partial pooling persists, so the pooling mass $P(u)$ grows monotonically from a lenient SM through MoM and toward stricter supermajorities.

S4.3 Centered Log-Concave (Truncated-Normal) Densities and Thin Tails

We now turn to the comparative statics of the pooling region under single–peaked log–concave priors (for instance the truncated normal). With identity beliefs $\mu(m) = m$ and a log–concave density f whose mass is centered at its mode (which in our normalization coincides with the MoM bar, $u_{\text{MoM}} = \theta_{\text{mode}} = 0$), the idea is that most deals are middling while truly bad and truly great deals are rare (thin tails). A simple sufficient condition that formalizes "thin right tails"—and thus guarantees that the *boundary outruns the bar* once we tighten beyond MoM—is:

$$\sup_{u\in[0,\,u^{\text{str}}_{\text{SM}}]} \frac{f(u)}{f(\theta^*(u))} \;\le\; \underline{\gamma} \quad \text{where} \quad \underline{\gamma} := \frac{\eta}{\alpha + \overline{B'} + \eta} \in (0,1), \;\; \overline{B'} := \sup_{\theta} B'(\theta).$$

Intuitively, the thin–tail bound makes the density at the bar uniformly small relative to the density at the moving boundary on the right shoulder, while the responsiveness bound $\theta^{*\prime}(u) \ge \underline{\gamma}$ (from gap costs and $0 < B'(\theta) < 1-\alpha$) keeps the boundary moving, so the density–weighted comparison $P'(u) = f(u) - f(\theta^*(u))\,\theta^{*\prime}(u)$ turns negative to the right of MoM; on the left corridor the density is weakly increasing and $\theta^{*\prime}(u) < 1$, so $P'(u) > 0$. Because the bound necessarily fails at the mode ($\chi(0) \ge 1$), pooling keeps rising just *past* MoM before the thin right tail turns P' negative, so the pooling mass peaks at a bar strictly above MoM. The lying band of a near-central rule sits in the thick part of the quality distribution while the lie needed to reach the bar is still affordable—ironically, a near-central rule can be the noisiest in terms of pooling.

Proposition 9 (Up–talk, centered case: pooling peaks at an interior bar above MoM under thin right tails)**.** *Maintain* (P1)–(P3), (S1)–(S2)*, and* (M1) *in the payoff–dominant (up–talk) regime* $0 < B'(\theta) < 1 - \alpha$*. Assume identity beliefs* $\mu(m) = m$ *and affine private benefits* $B(\theta) = b_0 + b_1\theta$ *with* $b_1 \in (0, 1-\alpha)$*. Work under the standing conventions of the paper that the MoM bar and the mode of* f *coincide at* $u_{\text{MoM}} = \theta_{\text{mode}} = 0$*, so* f *is (weakly) increasing on* $(-\infty, 0]$ *and (weakly) decreasing on* $[0, \infty)$*. Let* $u^{\text{len}}_{\text{SM}} < 0 < u^{\text{str}}_{\text{SM}}$ *be interior bars and suppose partial pooling persists on both corridors* $[u^{\text{len}}_{\text{SM}}, 0]$ *and* $[0, u^{\text{str}}_{\text{SM}}]$*.*

Define the lower–side pooling mass by $P(u) := \int_{\theta^*(u)}^{u} f(\theta)\,d\theta$*, where* $\theta^*(u) \in (\underline{\theta}_0, u)$ *is the unique boundary that solves* $\Lambda(\theta^*(u); u) = 0$*. Then:*

(a) Left corridor ($u < 0$*). P is strictly increasing as* $u \uparrow 0$*, i.e.*

$$P'(u) \;=\; f(u) \;-\; f(\theta^*(u))\,\theta^{*\prime}(u) \;>\; 0.$$

(b) Right corridor ($u > 0$*). If the right shoulder is* sufficiently thin away from the mode*—there is* $u_0 \in (0, u^{\text{str}}_{\text{SM}}]$ *such that*

$$\sup_{u\in[u_0,u^{\text{str}}_{\text{SM}}]} \frac{f(u)}{f(\theta^*(u))} \;\le\; \underline{\gamma} \quad \textit{with} \quad \underline{\gamma} := \frac{\eta}{\alpha + \overline{B'} + \eta} \in (0,1), \;\; \overline{B'} := \sup_{\theta} B'(\theta),$$

then $P'(u) \le 0$ *for* $u \ge u_0$*, while* $P'(0) = f(0) - f(\theta^*(0))\,\theta^{*\prime}(0) > 0$ *(the mode bound fails at*

$u = 0$*); i.e.*

$$P'(u) \;=\; f(u) \;-\; f(\theta^*(u))\,\theta^{*\prime}(u) \;\le\; 0.$$

Consequently P is strictly increasing through the MoM bar ($P'(0) > 0$) and weakly decreasing on $[u_0, u_{\mathrm{SM}}^{\mathrm{str}}]$, so it attains a maximum at some bar strictly above MoM (interior to the corridor when $u_0 < u_{\mathrm{SM}}^{\mathrm{str}}$):

$$u^* \in (0, u_0] \;\; \textit{with} \;\; P'(u^*) = 0, \;\; \textit{i.e.} \;\; f(u^*) = f(\theta^*(u^*))\,\theta^{*\prime}(u^*).$$

The thin–tail bound is imposed only on $[u_0, u_{\mathrm{SM}}^{\mathrm{str}}]$, so we do not claim a unique *maximizer on $(0, u_0]$; the content is that the pooling–maximizing bar is interior and strictly stricter than MoM. Pooling is therefore maximized by a rule slightly stricter than MoM, lying close to—but not at—the MoM bar when the right tail is thin.*

Proof. By the variable–limits rule,

$$P'(u) \;=\; f(u) \;-\; f(\theta^*(u))\,\theta^{*\prime}(u).$$

Under identity beliefs ($m^{\mathrm{req}}(u) = u$) and gap costs (S3), Lemma 3(iii) gives

$$\theta^{*\prime}(u) \;=\; \frac{C_m(u, \theta^*(u))}{\alpha + B'(\theta^*(u)) + C_m(u, \theta^*(u))} \;\in\; (0, 1).$$

Since on the up–wedge $C_m(u, \theta^*) = \eta + \kappa\Delta(u) \ge \eta$ with $\Delta(u) := u - \theta^*(u) > 0$ and $B'(\theta^*) \le \overline{B'}$, we obtain the uniform lower bound

$$\theta^{*\prime}(u) \;\ge\; \underline{\gamma} \;:=\; \frac{\eta}{\alpha + \overline{B'} + \eta} \in (0, 1).$$

(a) Left corridor. For $u < 0$, the band $[\theta^*(u), u]$ sits on the increasing shoulder, so $f(u) \ge f(\theta^*(u))$. With $0 < \theta^{*\prime}(u) < 1$,

$$P'(u) \;=\; f(u) - f(\theta^*(u))\,\theta^{*\prime}(u) \;\ge\; f(\theta^*(u))\big(1 - \theta^{*\prime}(u)\big) \;>\; 0.$$

(b) Right corridor. For $u > 0$, write $\chi(u) := f(u)/f(\theta^*(u)) \in (0, 1]$. Then

$$P'(u) \;=\; f(\theta^*(u))\big[\chi(u) - \theta^{*\prime}(u)\big] \;\le\; f(\theta^*(u))\big[\underline{\gamma} - \theta^{*\prime}(u)\big] \;\le\; 0,$$

whenever $\chi(u) \le \underline{\gamma}$. Thus P (weakly) decreases on $[u_0, u_{\mathrm{SM}}^{\mathrm{str}}]$, while $P'(0) > 0$ because $\chi(0) \ge 1 > \underline{\gamma}$.

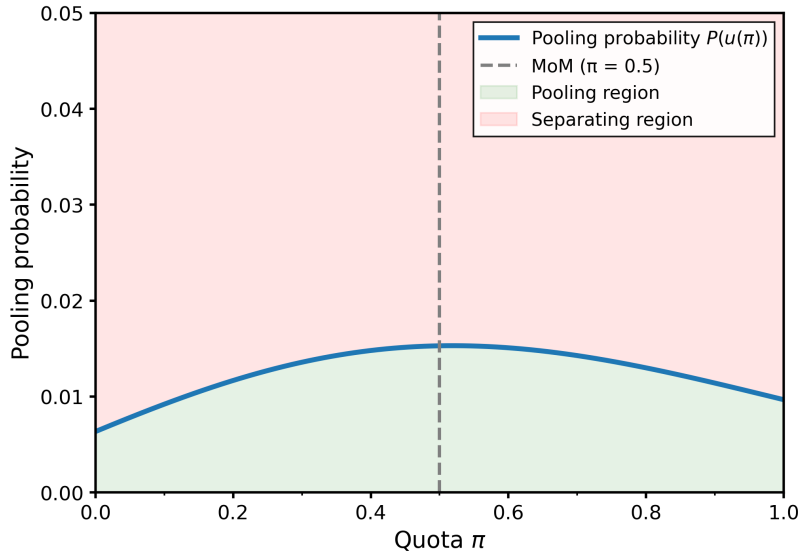
Combining (a) and (b), P rises through $u = 0$ and falls on $[u_0, u_{\mathrm{SM}}^{\mathrm{str}}]$, so by continuity it attains an interior maximum at some $u^* \in (0, u_0]$ with $f(u^*) = f(\theta^*(u^*))\,\theta^{*\prime}(u^*)$, strictly above the MoM bar. □

As shown in Figure 5, once the bar moves well into the right corridor the boundary advances

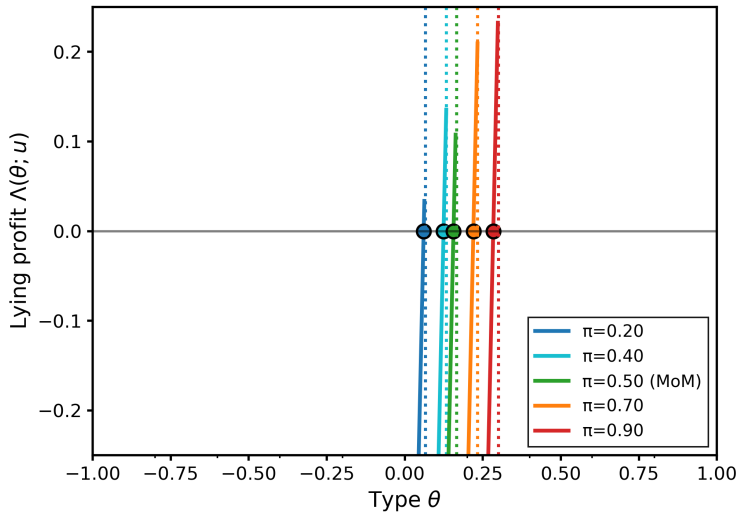
faster than the bar, so pooling declines as the bar moves from a near-central rule toward the strict SM regime.

Figure 5: Pooling Probabilities and Lying-Profit Curves (Logconcave, Identity Beliefs)

(a) Panel A: Pooling Probabilities



(b) Panel B: Lying-Profit Curves



Panel A plots the simulated pooling probability $P(u(\pi))$ (solid blue) where the controller's cost is $C(m, \theta) = \eta|m - \theta| + \frac{\kappa}{2}(m-\theta)^2$; parameter values are $(\alpha, b_0, b_1, \lambda, K, \eta, \kappa) = (0.50, 0.12, 0.20, 0.20, 0.60, 20, 0.10)$. The density function is $f(\theta) = \dfrac{1}{0.20\sqrt{2\pi}\,[\Phi(1/0.20) - \Phi(-1/0.20)]} \exp\!\left(-\theta^2/(2 \cdot 0.20^2)\right) \mathbf{1}_{[-1,1]}(\theta)$. The grey dashed line marks the majority-of-the-minority threshold $\pi = 0.5$ and the green (pink) shading indicates the pooling (separating) region. Panel B shows the lying-profit function on the up-wedge, $\Lambda(\theta, u) = \alpha\theta + B(\theta) - C(u)$, for $\pi \in \{0.20, 0.40, 0.50, 0.70, 0.90\}$; solid curves correspond to quotas that still permit pooling, dashed/dotted curves to quotas with complete separation, vertical dotted lines mark the induced belief cut-off $u = u_{\text{rule}}(\pi)$, and open markers indicate the type cutoff θ^* where $\Lambda(\theta, u) = 0$.

S4.4 Sluggish beliefs reproduce the interior peak

Remark (Belief sluggishness is a special case of Proposition 9). Replacing identity beliefs with a sluggish belief map $\mu(m) = Km$, $K \in (0, 1]$, so that minorities underreact to disclosure, does not change the qualitative comparative static. Under the turnout microfoundation of §5.1.3, $K = \tau$ is the attentive-participation share (Blume et al., 2007; Sims, 2003; Matějka and Tabellini, 2021; Palfrey and Rosenthal, 1983; Osborne et al., 2000; Levine and Palfrey, 2007). Two objects change relative to Proposition 9: the truthful-approval boundary and the upper limit of the pool become $m^{\text{req}}(u) = \mu^{-1}(u) = u/K$, and the boundary slope picks up the factor $(\mu^{-1})'(u) = 1/K$,

$$\theta^{*\prime}(u) = \frac{\eta + \kappa\,\Delta(u)}{\alpha + b_1 + \eta + \kappa\,\Delta(u)} \cdot \frac{1}{K}, \qquad \Delta(u) = m^{\text{req}}(u) - \theta^*(u).$$

The pooling mass is $P(u) = \int_{\theta^*(u)}^{\min\{m^{\text{req}}(u),\,\overline{\theta}_0\}} f$, with $P'(u) = f(m^{\text{req}}(u))\,(\mu^{-1})'(u) - f(\theta^*)\,\theta^{*\prime}(u)$ on the interior region $m^{\text{req}}(u) < \overline{\theta}_0$. Because $(\mu^{-1})'(u)$ multiplies the movement of *both* the bar's image and the boundary, it cancels from the comparison: the sign of $P'(u)$ is governed by the *normalized* responsiveness $\rho(u) := \theta^{*\prime}(u)/(\mu^{-1})'(u) = C_m/(\alpha + B' + C_m) \in (0, 1)$ against the density ratio $f(m^{\text{req}}(u))/f(\theta^*(u))$—exactly the test of Proposition 9. Sluggishness therefore does *not* make pooling monotone in the bar, nor does it supply a separate reversal channel; the pattern remains an interior peak, with its location *shifted*. A smaller K pushes the required message $m^{\text{req}}(u) = u/K$ into the thin right tail—and against the support top $\overline{\theta}_0$—at a more lenient rule, so the peak moves toward more lenient quotas as beliefs become more sluggish. Under a uniform prior pooling *rises* on the interior region (Corollary 1), and a thin right tail is required for the stringent-side decline, just as under identity beliefs. The genuinely distinct driver of the stringent-side reversal is the litigation channel of Proposition 2, which operates through $r'(\pi) > 0$ rather than through the belief map. Whereas the density mechanism of Proposition 9, and with it the sluggish-belief special case above, delivers the stringent-side decline only when the prior tail is thin enough that the density ratio falls below the normalized responsiveness, $f(m^{\text{req}}(u))/f(\theta^*) \le \rho(u)$, the litigation channel carries no such distributional restriction: with $r'(\pi) > 0$ and the ratification-cost dominance condition (2) in force, the pool contracts on the stringent side even under a uniform prior—the very configuration in which both the belief and density forces above leave pooling monotone.

The remaining subsections collect robustness extensions referenced in the main text. Each formalizes an alternative specification or generalization; none is required for the baseline results.

S5 Endogenous cleansing scrutiny: $\sigma = \sigma(g)$

The baseline analysis treats court-credit risk σ as exogenous. As a tractable strengthening extension, suppose cleansing scrutiny depends on the disclosure gap. On the up-talk wedge, where $g \ge 0$, define

$$\sigma : [0, \infty) \to [0, 1), \qquad \sigma'(g) \ge 0, \qquad \sigma(0) = \sigma_0,$$

and in a two-sided version replace g by $|g|$. The legal interpretation is direct: as the disclosure gap grows, courts are more likely to find that the vote was not fully informed or was effectively coerced, so they are less likely to credit ratification as cleansing. With a slight abuse of notation, σ denotes both the scalar court-credit parameter and, in this subsection, the schedule $\sigma(\cdot)$; the partials L_σ below are taken with respect to the third argument of L and then evaluated at $\sigma(g)$.

Define the induced reduced-form enforcement cost

$$\widetilde{L}(g, r) \;:=\; L\big(g, r, \sigma(g)\big).$$

By the chain rule,

$$\frac{d}{dg}\widetilde{L}(g, r) \;=\; L_g\big(g, r, \sigma(g)\big) \;+\; L_\sigma\big(g, r, \sigma(g)\big)\, \sigma'(g).$$

Under Assumption 2, $L_\sigma \geq 0$ and $L_{g\sigma} \geq 0$, so any $\sigma'(g) \geq 0$ adds the nonnegative term $L_\sigma\, \sigma'(g)$ to the marginal enforcement cost of the gap. Economically, deception now raises expected enforcement through two channels: direct exposure and a lower probability that courts credit vote-based cleansing. This extension therefore amplifies the litigation channel and makes the ratification-cost-dominance reversal easier to obtain. Relative to a one-stage threshold-bunching benchmark, pooling is less likely to survive once cleansing becomes endogenously harder for larger misrepresentations.

S6 Continuous-extraction formulation

When extraction is endogenous, the controller chooses $x \in [0, B(\theta)]$ and faces an expected enforcement cost $K(g, x, r, \sigma) = p(g, r) \cdot x + \frac{\varrho}{2} x^2$ with $\varrho > 0$, where $p(g, r)$ is the probability of suit. The controller's extraction margin nets the benefit $(1 - \alpha)x$ against K, so the optimal extraction is

$$x^*(g, r, \sigma) \;=\; \max\Big\{0,\; \min\Big\{B(\theta),\; \tfrac{(1-\alpha)-p(g,r)}{\varrho}\Big\}\Big\},$$

which is continuous in (g, r, σ). Define the reduced-form net cost as $L(g, r, \sigma) := K\big(g, x^*(g, r, \sigma), r, \sigma\big) - (1-\alpha)\, x^*(g, r, \sigma)$, so that on approval the controller's Stage-1 payoff is $\alpha\theta - L(g, r, \sigma)$—the extraction benefit $(1 - \alpha)x^*$ netted against the enforcement cost K—which maps to the fixed-extraction formulation (C1) with L as the cost term. Fixing $x^* = B(\theta)$ recovers the lump-sum specification.

S7 Alternative settlement-bargaining microfoundation

This subsection provides a reduced-form settlement foundation for $L(g, r, \sigma)$ in the spirit of Bebchuk (1984). Let $p(g)$ be the probability that minority plaintiffs file suit, with $p(0) = 0$ and $p'(|g|) \geq 0$ (truthful disclosure attracts no suit, giving $L(0, r, \sigma) = 0$ as (L1) requires). Conditional on filing, the controller makes a take-it-or-leave-it settlement offer s (or equivalently chooses settlement intensity), and if settlement is rejected or if ratification is not credited, the case proceeds

with expected trial loss $D(g)$ plus litigation cost k, where $D'(|g|) \geq 0$ and $D''(|g|) \geq 0$.

Let $\phi(r, \sigma) \in [0, 1]$ denote the probability that a settlement/ratification package is court-creditable and effectively extinguishes claims, with $\phi_r \leq 0$ and $\phi_\sigma \leq 0$ because stricter vote thresholds and stricter cleansing scrutiny make release harder. A compact expected-cost representation is

$$L(g, r, \sigma) \;=\; p(g)\Big[s(g, r, \sigma) \;+\; \big(1 - \phi(r, \sigma)\big)D(g) \;+\; k\Big].$$

Under standard monotonicity conditions ($p'(|g|) \geq 0$, $s, D, k \geq 0$ with $s_g, s_r, s_\sigma \geq 0$, $s_{gr}, s_{g\sigma} \geq 0$, $D'(|g|) \geq 0$, and $\phi_r, \phi_\sigma \leq 0$), L is increasing in $|g|$, increasing in r, and increasing in σ. If in addition p, D, and s are convex in $|g|$ (settlement terms worsen convexly with case strength), then L is a product of nonnegative, increasing, convex factors and is therefore convex in $|g|$. Supermodularity ($L_{gr} \geq 0, L_{g\sigma} \geq 0$) follows because larger gaps both raise trial stakes and reduce the practical ability to buy a release through ratification.

S8 Tail-growth: bounded-gap caveat

The signaling cost depends on the gap $\Delta := m - \theta$ via a convex, wedge-symmetric map $C(\Delta)$ with $C(0) = 0$ and C strictly increasing in $|\Delta|$.

(a) *Unbounded-gap case.* If for every θ the feasible message set allows a sequence $\{m_n\}$ with $|m_n - \theta| \to \infty$, then

$$C(\Delta) \xrightarrow[|\Delta| \to \infty]{} \infty,$$

so sufficiently large lies are eventually prohibitive.

(b) *Bounded-gap case (when tail-growth may not bite).* If the feasible set and posteriors imply a *gap envelope*

$$\bar{\Delta}(u) \;=\; m^{\text{req}}(u) - u \quad \text{is bounded on } u \in (\underline{\theta}_0, \overline{\theta}_0),$$

then $|\Delta|$ cannot grow arbitrarily and the "tail-growth" force in (a) need not trigger separation. In this case, separation requires the capped-branch endpoint test of Lemma 3(iv) to fail at the support top,

$$\Psi_h(u) \;=\; \alpha\,\overline{\theta}_0 + B(\overline{\theta}_0) - C\big(m^{\text{req}}(u) - \overline{\theta}_0\big) \;\leq\; 0 \quad \text{for some interior } u \text{ on the capped branch } m^{\text{req}}(u) > \overline{\theta}_0,$$

rather than appeal to $|\Delta| \to \infty$; with a bounded envelope the capped gap $m^{\text{req}}(u) - \overline{\theta}_0 \leq \bar{\Delta}(u)$ stays bounded as well, so the test can fail nowhere and pooling can persist at every bar.

S9 Ratification microfoundation: buying-peace derivations

S9.1 Proof of Lemma 1 (ratification cost)

We first restate the explicit buying-peace game that underlies the cost $C(R, r, \sigma)$, then compute the expenditure and its derivatives.

We now derive the ratification cost from an explicit buying-peace game. Suppose each minority shareholder i has privately known ratification-resistance type $v_i \overset{\text{iid}}{\sim} G$ on $[\underline{v}, \overline{v}]$, where G admits a continuous density $g_v > 0$ on $(\underline{v}, \overline{v})$ and satisfies $G(\underline{v}) = 0$, $G(\overline{v}) = 1$. Let $R \in [0, 1]$ denote the per-share recovery at stake if litigation continues. If shareholder i votes to ratify, she forgoes continuation value and bears resistance cost; with transfer $t_i \geq 0$, utility from ratifying is

$$u_i^Y = t_i - (R + v_i),$$

while utility from rejecting is normalized to $u_i^N = 0$. Hence i votes to ratify iff

$$t_i \geq R + v_i.$$

The reservation price $R+v_i$ has two economically distinct components. The common component R compensates for the litigation value that every shareholder forgoes by ratifying—the expected per-share recovery if the lawsuit continued to judgment, which depends on the strength of the legal case and hence on the misrepresentation gap g. The idiosyncratic component v_i compensates for shareholder i's private resistance to settlement. The additive structure means that an increase in R shifts every shareholder's reservation price upward by the same amount—a parallel shift of the supply curve—while the heterogeneity in v_i generates the curve's upward slope.

The controller therefore buys support from the lowest-reservation-price voters first (lowest v_i), with minimum transfer schedule

$$t(v) = \max\{0,\ R + v\}.$$

The resistance type v_i captures idiosyncratic reluctance to ratify: passive index funds have low v_i; activist hedge funds have high v_i. The distribution G reflects shareholder-base composition (cf. Hamdani and Yafeh, 2013; Hu and Black, 2006). Crucially, the qualitative results below do not depend on the shape of G—the key derivative signs are determined at the pivotal voter, with the single kink exception noted in Remark 4.

For tractability, we adopt a deterministic court-credit discount: only a fraction $(1-\sigma)$ of purchased votes is credited as cleansing, with $\sigma \in [0, 1)$. If the controller buys fraction q of minority votes, the credited fraction is $(1 - \sigma)q$. To satisfy ratification threshold r, he must buy at least

$$q \ \geq \ \frac{r}{1 - \sigma}.$$

Purchasing fraction q means buying all types $v \le v^*$ where $G(v^*) = q$, i.e. $v^* = G^{-1}(q)$. Thus ratification is feasible only on

$$0 < r \le 1 - \sigma.$$

When $r > 1-\sigma$, no amount of settlement-support purchasing can produce enough credited votes.

Proof. Step 1: vote cutoff and feasibility. If the controller buys all voters with $v \le v^*$, the purchased fraction is $G(v^*)$. The credited fraction is $(1-\sigma)G(v^*)$. Meeting threshold r requires

$$(1-\sigma)\,G(v^*) \;=\; r \qquad\Longrightarrow\qquad v^* \;=\; G^{-1}\!\left(\frac{r}{1-\sigma}\right).$$

Since $G(\overline{v}) = 1$, feasibility requires $r/(1-\sigma) \le 1$, i.e. $r \le 1-\sigma$. If $r > 1-\sigma$, then $v^* > \overline{v}$ and ratification is infeasible, so $C = \infty$. Below we work on $r \le 1-\sigma$.

Step 2: total expenditure. The minimum transfer for voter type v is $t(v) = (R+v)_+$. The total expenditure is therefore

$$C \;=\; \int_{\underline{v}}^{v^*} (R+v)_+\, g_v(v)\, dv \;=\; \int_{\max\{\underline{v},\,-R\}}^{v^*} (R+v)\, g_v(v)\, dv,$$

where the second equality uses $(R+v)_+ = 0$ for $v < -R$. If $v^* \le -R$ (equivalently, $G(v^*) \le G(-R)$, i.e. $r \le (1-\sigma)G(-R) = r_0$), the integration interval is empty and $C = 0$. If $v^* > -R$ (i.e. $r > r_0$), we are on the active branch and $C > 0$.

Step 3: partial derivatives. Throughout, write $d := 1-\sigma \in (0,1]$ and work on the active interior $r_0 < r < d$ where $v^* > -R$. Note that $v^* = G^{-1}(r/d)$, so

$$\frac{\partial v^*}{\partial r} \;=\; \frac{1}{d\; g_v(v^*)}, \qquad \frac{\partial v^*}{\partial \sigma} \;=\; \frac{r}{d^2\; g_v(v^*)}. \tag{6}$$

Because $g_v(v^*) > 0$ on the interior, both partials are well defined and strictly positive. Crucially, v^* does not depend on R.

(a) C_R. Write $\ell(R) := \max\{\underline{v}, -R\}$ for the effective lower limit. Differentiate $C = \int_{\ell(R)}^{v^*} (R+v)\, g_v(v)\, dv$ with respect to R. By Leibniz' rule, v^* does not depend on R, so there is no upper-boundary contribution. The integrand derivative is $\partial(R+v)/\partial R = 1$. If $-R > \underline{v}$, the lower limit is $\ell = -R$ and the boundary evaluation is $(R + (-R))\, g_v(-R) \cdot (-1) = 0$. Therefore in all cases:

$$C_R \;=\; \int_{\ell(R)}^{v^*} g_v(v)\, dv \;=\; G(v^*) - G(\ell(R)) \;>\; 0,$$

since $v^* > \ell(R)$ on the active region.

(b) C_{RR}. Differentiate $C_R = G(v^*) - G(\ell(R))$ with respect to R. Since v^* is independent of R, only the lower-limit term contributes. If $-R > \underline{v}$, then $\ell(R) = -R$ and $\partial \ell/\partial R = -1$, giving

$C_{RR} = g_v(-R) > 0$. If $-R \le \underline{v}$, then $\ell(R) = \underline{v}$ is constant in R, so $C_{RR} = 0$. Hence

$$C_{RR} \;=\; g_v(-R)\,\mathbf{1}\{-R > \underline{v}\} \;\ge\; 0.$$

(c) C_r. By Leibniz' rule at the upper limit (the lower limit does not depend on r):

$$C_r \;=\; (R+v^*)\,g_v(v^*)\,\frac{\partial v^*}{\partial r} \;=\; (R+v^*)\,g_v(v^*)\,\frac{1}{d\,g_v(v^*)} \;=\; \frac{R+v^*}{d} \;>\; 0,$$

where the density $g_v(v^*)$ cancels. The strict inequality uses $R + v^* > 0$ on the active region.

(d) C_σ. Similarly, by Leibniz at the upper limit:

$$C_\sigma \;=\; (R+v^*)\,g_v(v^*)\,\frac{\partial v^*}{\partial \sigma} \;=\; (R+v^*)\,g_v(v^*)\,\frac{r}{d^2\,g_v(v^*)} \;=\; \frac{r\,(R+v^*)}{d^2} \;>\; 0.$$

(e) C_{Rr}. Differentiate $C_r = (R + v^*)/d$ with respect to R. Since v^* and d are independent of R:

$$C_{Rr} \;=\; \frac{1}{d} \;=\; \frac{1}{1-\sigma} \;>\; 0.$$

(f) $C_{R\sigma}$. Differentiate $C_R = G(v^*) - G(\ell(R))$ with respect to σ. Only v^* depends on σ:

$$C_{R\sigma} \;=\; g_v(v^*)\,\frac{\partial v^*}{\partial \sigma} \;=\; g_v(v^*)\,\frac{r}{d^2\,g_v(v^*)} \;=\; \frac{r}{d^2} \;=\; \frac{r}{(1-\sigma)^2} \;>\; 0.$$

Kink. At $r = r_0(R, \sigma)$, we have $v^* = -R$ and $R + v^* = 0$, so $C_r = 0$ and $C_\sigma = 0$. Right-derivatives in r are zero, and all stated inequalities hold weakly. □

S9.2 Ratification microfoundation implies (L1)–(L3)

Lemma 6 (Ratification microfoundation implies (L1)–(L3) on the feasible region)**.** *Fix $\sigma \in [0, 1)$ and define $d := 1 - \sigma$. On the up-wedge $g \ge 0$, consider $L(g, r, \sigma) := p(g, r)\,C\big(R(g), r, \sigma\big)$ for $0 < r \le d$, where $C(R, r, \sigma)$ is the ratification cost from Lemma 1 under a general resistance distribution G with continuous positive density g_v on $(\underline{v}, \overline{v})$. Assume:* (a) *$p \in C^2$, $p(0, r) = 0$, $p_g, p_{gg}, p_r, p_{gr} \ge 0$ on $g \ge 0$, $0 < r \le d$, with $p_g(g, r) > 0$ for $g > 0$;* (b) *$R \in C^2$, $R(0) = 0$, and $R_g, R_{gg} \ge 0$ on $g \ge 0$; and* (c) *either $r > r_0(R(g), \sigma)$ (active branch of C) or one-sided derivatives at the kink $r = r_0$ are used (weak inequalities). Then on $g \ge 0$, $0 < r \le d$, L satisfies* (L1)–(L3)*:*

$$L(0, r, \sigma) = 0, \quad L_g \ge 0, \quad L_{gg} \ge 0, \quad L_r \ge 0, \quad L_{gr} \ge 0, \quad L_\sigma \ge 0, \quad L_{g\sigma} \ge 0.$$

On the active branch $r > r_0(R(g), \sigma)$ one has $C > 0$, so $p_g(g, r) > 0$ yields the strict condition $L_g > 0$ for $g > 0$ required by (L1); further strict inequalities hold there whenever a corresponding primitive

derivative (p_{gg}, p_r, p_{gr}, R_g, or R_{gg}) is strictly positive. At the kink and on the inactive branch ($r \le r_0$, where $C = 0$) the inequalities hold weakly.

Proof. We use the sign conditions from Lemma 1. On the active branch $r > r_0(R, \sigma)$:

$$C_R > 0, \quad C_{RR} \ge 0, \quad C_r > 0, \quad C_\sigma > 0, \quad C_{Rr} > 0, \quad C_{R\sigma} > 0. \tag{7}$$

When $r \le r_0$, $C = 0$ and one-sided derivatives are weakly nonnegative. Recall that $C_{RR} = g_v(-R)\,\mathbf{1}\{-R > \underline{v}\} \ge 0$, where g_v denotes the resistance density; all other signs in (7) are distribution-free (Remark S9.4).

Verification of (L1). At $g = 0$: $L(0, r, \sigma) = p(0, r)\, C(R(0), r, \sigma) = 0$ by assumptions (a)–(b) ($p(0, r) = 0$ and $R(0) = 0$).

For $L_g > 0$ on $g > 0$, differentiate the product $L = p\, C$:

$$L_g \;=\; p_g\, C \;+\; p\, C_R\, R_g \;\ge\; 0,$$

since $p_g \ge 0$ (assumption (a)), $C \ge 0$, $p \ge 0$, $C_R \ge 0$ (Lemma 1(a)), and $R_g \ge 0$ (assumption (b)). On the active branch $C(R(g), r, \sigma) > 0$, so $p_g(g, r) > 0$ for $g > 0$ (assumption (a)) gives $L_g \ge p_g\, C > 0$.

For $L_{gg} \ge 0$, differentiate again:

$$L_{gg} \;=\; p_{gg}\, C \;+\; 2\, p_g\, C_R\, R_g \;+\; p\left(C_{RR}\, R_g^2 \;+\; C_R\, R_{gg}\right) \;\ge\; 0,$$

using $p_{gg} \ge 0$ (assumption (a)), $C_{RR} \ge 0$ (Lemma 1(b)), and $R_{gg} \ge 0$ (assumption (b)). Each of the four terms is individually nonnegative.

Verification of (L2). For $L_r \ge 0$:

$$L_r \;=\; p_r\, C \;+\; p\, C_r \;\ge\; 0,$$

using $p_r \ge 0$ (assumption (a)) and $C_r \ge 0$ (Lemma 1(c)).

For $L_{gr} \ge 0$:

$$L_{gr} \;=\; p_{gr}\, C \;+\; p_g\, C_r \;+\; p_r\, C_R\, R_g \;+\; p\, C_{Rr}\, R_g \;\ge\; 0,$$

using $p_{gr} \ge 0$ (assumption (a)) and $C_{Rr} \ge 0$ (Lemma 1(e)). Each term is individually nonnegative.

Verification of (L3). Since p and R do not depend directly on σ:

$$L_\sigma \;=\; p\, C_\sigma \;\ge\; 0,$$

using $C_\sigma \ge 0$ (Lemma 1(d)). And:

$$L_{g\sigma} \;=\; p_g\, C_\sigma \;+\; p\, C_{R\sigma}\, R_g \;\ge\; 0,$$

using $C_{R\sigma} \ge 0$ (Lemma 1(f)).

Thus (L1)–(L3) hold on the feasible set $0 < r \le 1-\sigma$ under any continuous resistance distribution with positive density. Weak inequalities obtain at the kink $r = r_0$; strict inequalities hold on the active interior whenever at least one corresponding primitive slope is strictly positive. □

S9.3 Closed-form ratification cost under uniform resistance

Example 2 (Closed-form ratification cost under uniform resistance)**.** If $v_i \sim \text{Unif}[-1,1]$, then $G(v) = (1+v)/2$, $g_v(v) = 1/2$, $v^* = 2r/(1-\sigma) - 1$, and the kink threshold is $r_0(R,\sigma) = (1-\sigma)(1-R)/2$. On the feasible region $r \le 1-\sigma$,

$$C(R,r,\sigma) \;=\; \frac{\big(2r-(1-\sigma)(1-R)\big)_+^2}{4(1-\sigma)^2}.$$

On the active interior,

$$C_R = \frac{Z}{2d} > 0, \qquad C_{RR} = \frac{1}{2} > 0,$$
$$C_r = \frac{Z}{d^2} > 0, \qquad C_\sigma = \frac{rZ}{d^3} > 0,$$

and

$$C_{Rr} = \frac{1}{d} > 0, \qquad C_{R\sigma} = \frac{r}{d^2} > 0,$$

where $Z := 2r-(1-\sigma)(1-R) > 0$ and $d := 1-\sigma$.

Derivation. Substitute $G(v) = (1+v)/2$ and $g_v(v) = 1/2$ into Lemma 1. The cutoff is $v^* = 2r/d - 1$ and $R + v^* = R - 1 + 2r/d = Z/d$. On the active branch,

$$C = \frac{1}{2}\int_{-R}^{v^*} (R+v)\,dv = \frac{1}{4}(R+v^*)^2 = \frac{Z^2}{4d^2},$$

and the derivative expressions follow from Lemma 1 with $R + v^* = Z/d$, or directly from $C = Z^2/(4d^2)$.

S9.4 Distribution-free signs of the ratification cost

Remark (Distribution-free signs). The sign conditions $C_r > 0$, $C_\sigma > 0$, $C_{Rr} > 0$, and $C_{R\sigma} > 0$ hold for any continuous resistance distribution G with positive density. The density $g_v(v^*)$ cancels in every Leibniz evaluation at the upper limit $v^*(r,\sigma)$, and this cancellation reflects the economics of procurement along a supply curve.

When the ratification threshold rises by dr, the controller must buy a thin additional slice of shareholders near the current pivotal voter v^*. If the density is high at v^* (many shareholders

clustered near the margin), each one is barely more expensive than the last, but a large number must be purchased to cover the increment. If the density is low (few shareholders near the margin), each one is substantially more expensive, but only a few are needed. In both cases the total incremental cost is the same: $(R+v^*)\,dr/(1-\sigma)$. The number of additional voters, $g_v(v^*)\cdot dv^*$, and the rate at which the cutoff moves, $\partial v^*/\partial r = 1/\big((1-\sigma)\,g_v(v^*)\big)$, always offset. What determines the marginal cost is the *price* at the margin—the pivotal voter's reservation value $R+v^*$—not the *density* at the margin.

The exception is $C_{RR} = g_v(-R)\,\mathbf{1}\{-R > \underline{v}\}$, which asks a different economic question: not "what does the next vote cost at the top of the supply curve?" but "how many shareholders just crossed from the zero-cost region into the active region as R increased?" This is a question about the density at the *kink*, not the density at the *pivotal voter*, so no offsetting inverse-CDF derivative arises. The curvature of the cost in recovery stakes is the one property that depends on the distributional shape of shareholder heterogeneity.

S9.5 Primitive beliefs and Bayesian updating at the pooling message

The voting input in the atomistic-voter framework is the primitive belief $\mu(m)$ of (M1); this subsection records the threshold algebra behind Remark B.4.

This distinction matters for the partial-pooling equilibrium of Proposition 1. Types $[\theta^*(u), u)$ pool at $m_{\text{req}}(u)$, and types $\theta \geq u$ separate truthfully. A receiver who solved the full signalling game and observed $m = m_{\text{req}}(u)$ would compute $\mathrm{E}[\theta \mid \theta \in [\theta^*(u), u)] < u$ and vote to reject. However, in the atomistic-voter framework of Levit et al. (2024), the voting input is $\mu(m)$—the primitive effective posterior from (M1)—and the approval condition reduces to the message threshold $m \geq m_{\text{req}}(u) := \mu^{-1}(u)$. Since the pooling message equals $m_{\text{req}}(u)$ by construction, the threshold is met and the proposal is approved.

When attention is partial ($\tau < 1$ in the notation of Section 5.1.3), the effective posterior $\mu(m) = \tau\,\mathrm{E}[\theta \mid m] + (1-\tau)\mathrm{E}[\theta]$ is pulled toward the prior mean. Because the pooling posterior $\mathrm{E}[\theta \mid \theta \in [\theta^*(u), u)]$ lies below u, this convex combination can exceed u only when the prior mean itself lies above the bar, $\mathrm{E}[\theta] > u$: there a low enough τ sustains approval under Bayesian updating. At the stringent bars $u > \mathrm{E}[\theta]$ that drive the reversal, however, both the pooling posterior and the prior mean lie below u, so no degree of inattention rescues Bayesian approval, and approval rests on the atomistic-voter assumption itself. The benchmark $\tau = 1$ is the limiting case in which that assumption does all the work.

S9.6 Fully Bayesian voters: the reversal without the atomistic device

The atomistic-voter device is not necessary for the reversal of Proposition 2. This subsection proves Proposition 3: the receiver is fully Bayesian—the approval test is applied to the equilibrium posterior rather than to the primitive perception μ—and the coupled comparative static is governed by the dominance condition (3) in the general environment of the main text (log-concave

prior, affine baseline, general enforcement cost). The unraveling logic that defeats an approved pool *below* the bar (§S9.5) is respected throughout, since Bayesian consistency forces the pool to straddle the bar. A parametric example with corridor-wide symbolic verification closes the subsection.

S9.6.1 Environment and equilibrium notion.

The prior F has a continuous, positive, log-concave density f on $[\underline{\theta}_0, \overline{\theta}_0]$ (P1)–(P2); the baseline is affine, $B(\theta) = b_0 + b_1\theta$ with $b_1 \in (0, 1-\alpha)$ (P3), and we write $A := \alpha + b_1$ for the prize slope. Affinity makes the voting reduction of §5.2.3 exact for any posterior: with $D(\theta) = b_1\theta$, the aggregator (V2) approves message m if and only if $(1-\alpha-b_1)\,\mathrm{E}[\theta \mid m] \geq \underline{\kappa}^{(\lambda)}_{\text{rule}}$, i.e. $\mathrm{E}[\theta \mid m] \geq u$, so the induced bar $u(\pi)$ and slope $u'(\pi) > 0$ are those of (V4). The enforcement cost L satisfies (L1)–(L3), is C^1 in (g, r) on the up-wedge $g > 0$, and is nonnegative on the down-wedge $g \leq 0$ (the symmetric specification (S1) is one instance; the down-wedge value serves only as off-path discipline). Along the policy corridor, $u = u(\pi)$ and $r = r(\pi)$ are C^1 with $u' > 0$ and $r' \geq 0$, and σ is fixed.

A *threshold-pooling profile* is parameterized by a boundary pair $(\theta^*, \hat{m})$ with $\theta^* < \hat{m}$: types $\theta < \theta^*$ disclose truthfully and are rejected; types $\theta \in [\theta^*, \hat{m})$ pool at $\hat{m}$; types $\theta \geq \hat{m}$ disclose truthfully and are approved. Beliefs follow Bayes' rule on path (a point mass at m for separating messages; the prior truncated to $[\theta^*, \hat{m}]$ at the pooled message); off-path messages in $[\theta^*, \hat{m})$ carry a point mass on θ^* and are rejected, messages below $\underline{\theta}_0$ carry a point mass on $\underline{\theta}_0$, and messages above $\overline{\theta}_0$ carry a point mass on $\overline{\theta}_0$. Tying the pooled message to the pool's upper endpoint keeps every on-path gap nonnegative (up-talk, as the enforcement microfoundation prices), and the equilibrium selected is sender-preferred in the sense of the inf-selection (V3): the pooled message is the lowest approvable one. Lowering the pool-top message lowers every remaining pooler's gap, so the sender-preferred pool top is the smallest approvable one. For a candidate top b, the boundary indifference $\alpha a + B(a) = L(b-a, r, \sigma)$ determines the pool bottom $a(b)$ (its left side minus its right side is strictly increasing in a, with slope $A + L_g > 0$), and implicit differentiation gives $a'(b) = L_g/(A + L_g) \in (0, 1)$; the pooled posterior $T(a(b), b) = \mathrm{E}[\theta \mid a(b) \leq \theta \leq b]$ is therefore strictly increasing in the candidate top, $dT/db = T_a\, a'(b) + T_b > 0$, with $T_a, T_b > 0$ from Step 1 below. Approval requires $T \geq u$, so at the smallest approvable pool top the constraint binds: the consistency condition holds with equality,

$$\mathrm{E}\big[\theta \,\big|\, \theta^* \leq \theta \leq \hat{m}\big] \;=\; u, \tag{B1}$$

and the boundary type is indifferent between pooling and truthful rejection,

$$\alpha\theta^* + B(\theta^*) \;=\; L\big(\hat{m} - \theta^*,\; r, \sigma\big). \tag{B2}$$

The interiority condition invoked by Proposition 3 is

$$\alpha u + B(u) \;>\; 0 \qquad \text{and} \qquad \Psi_\pi\big(\theta^*_{\min}(u)\big) \;<\; 0, \tag{BI}$$

where $\theta^*_{\min}(u) := \inf\{a : \mathrm{E}[\theta \mid a \le \theta \le \overline{\theta}_0] \ge u\}$ is the lowest pool bottom at which (B1) is feasible within the support, $M(a;u)$ denotes the unique solution $\hat{m}$ of (B1) at $\theta^* = a$ (Step 1 of the proof), and $\Psi_\pi(a) := \alpha a + B(a) - L\big(M(a;u) - a,\ r(\pi), \sigma\big)$. The first inequality keeps the pool nonempty at the top (it is the endpoint value $\Psi_h(u)$ of Proposition 1 and holds under (CPC) with slack); the second rules out the capped branch, on which the pool top would sit at $\overline{\theta}_0$ with a slack posterior—the analogue of the capped branch in Corollary 1, which we do not use.

Proof of Proposition 3. Step 1: the consistency locus. Write $T(a,b) := \mathrm{E}[\theta \mid a \le \theta \le b]$ for $\underline{\theta}_0 \le a < b \le \overline{\theta}_0$. Leibniz differentiation gives

$$T_a = \frac{f(a)\big(T(a,b) - a\big)}{F(b) - F(a)} \ > \ 0, \qquad T_b = \frac{f(b)\big(b - T(a,b)\big)}{F(b) - F(a)} \ > \ 0,$$

both continuous, the signs holding because a doubly truncated mean lies strictly between the truncation points. Fix $a \in [\theta^*_{\min}(u), u)$. Then $T(a,\cdot)$ increases strictly and continuously from a (as $b \downarrow a$) to $T(a,\overline{\theta}_0) \ge u$, so (B1) has a unique solution $\hat{m} = M(a;u)$, and $a < u < M(a;u)$: any pool satisfying (B1) straddles the bar. By the implicit function theorem, M is C^1 with $M'(a) = -T_a/T_b < 0$.

Step 2: existence and uniqueness of the boundary pair. On $[\theta^*_{\min}(u), u)$ the map $\Psi_\pi(a) = \alpha a + B(a) - L(M(a;u) - a, r, \sigma)$ has slope

$$\Psi'_\pi(a) \ = \ A + L_g \cdot \big(1 - M'(a)\big) \ = \ A + L_g\left(1 + \frac{T_a}{T_b}\right) \ > \ A \ > \ 0,$$

so it is strictly increasing. As $a \uparrow u$, monotonicity forces $M(a;u) \downarrow u$ (a limit strictly above u would contradict (B1), since $T(a,\bar{m}) > u$ for a near u and $\bar{m} > u$), the gap vanishes, and $\Psi_\pi \to \alpha u + B(u) > 0$ by (BI); at the lower endpoint, $\Psi_\pi(\theta^*_{\min}(u)) < 0$ by (BI). There is therefore exactly one root $\theta^*_{\mathrm{B}} \in (\theta^*_{\min}(u), u)$, and $\hat{m} = M(\theta^*_{\mathrm{B}}; u) < M(\theta^*_{\min}(u); u) \le \overline{\theta}_0$ is strictly interior.

Step 3: the profile is a perfect Bayesian equilibrium. Beliefs satisfy Bayes' rule on path by construction. Receiver optimality: separating messages reveal the type, and approved separating types satisfy $\theta \ge \hat{m} > u$ while rejected ones satisfy $\theta < \theta^*_{\mathrm{B}} < u$; at the pooled message the posterior mean equals u by (B1), so approval is consistent with the weak inequality in (V2); every off-path belief is a point mass on a type strictly below u except above the support, where approval of $m > \overline{\theta}_0$ attracts no deviation because such a message carries a strictly larger gap than every on-path option. Sender optimality: on the pool, $\Lambda(\theta) := \alpha\theta + B(\theta) - L(\hat{m} - \theta, r, \sigma)$ has slope $A + L_g > 0$ and $\Lambda(\theta^*_{\mathrm{B}}) = 0$ by (B2), so every pooler weakly prefers pooling to any rejected message (which pays 0); a deviation to an approved separating message $m > \hat{m}$ enlarges the gap at no benefit; types $\theta < \theta^*_{\mathrm{B}}$ have $\Lambda(\theta) < 0$, and any approved message pays even less—approval requires $m \ge \hat{m}$, whose gap $m - \theta \ge \hat{m} - \theta$ makes the payoff at most $\Lambda(\theta) < 0$ by (L1)—so truthful rejection (payoff 0) is optimal; and types $\theta \ge \hat{m}$ obtain the approved zero-cost payoff $\alpha\theta + B(\theta) > 0$ (positive by (BI) and $A > 0$), which no deviation improves upon since $L \ge 0$ everywhere. Together with Steps 1–2 this proves part (i).

Step 4: comparative statics. Differentiating the system (B1)–(B2) along the corridor gives

$$T_a\,\dot{\theta}^* + T_b\,\dot{\hat{m}} = u', \qquad (A + L_g)\,\dot{\theta}^* - L_g\,\dot{\hat{m}} = L_r\,r',$$

with Jacobian determinant $-\big(T_a L_g + T_b(A + L_g)\big) \neq 0$, so $\theta^*_{\mathrm{B}}(\pi)$ and $\hat{m}(\pi)$ are C^1 with

$$\dot{\theta}^*_{\mathrm{B}} = \frac{L_g\,u' + T_b\,L_r\,r'}{T_a L_g + T_b(A + L_g)} \;>\; 0, \qquad \dot{\hat{m}} = \frac{(A + L_g)\,u' - T_a\,L_r\,r'}{T_a L_g + T_b(A + L_g)},$$

all cost derivatives evaluated at $(g^*_{\mathrm{B}}, r, \sigma)$ with $g^*_{\mathrm{B}} = \hat{m} - \theta^*_{\mathrm{B}}$.

Step 5: the pooling derivative and the collapse identity. Leibniz differentiation of $P = F(\hat{m}) - F(\theta^*_{\mathrm{B}})$ and substitution of Step 4 yield

$$\frac{dP}{d\pi} = f(\hat{m})\,\dot{\hat{m}} - f(\theta^*_{\mathrm{B}})\,\dot{\theta}^*_{\mathrm{B}} = \frac{u'\big[f(\hat{m})(A + L_g) - f(\theta^*_{\mathrm{B}})L_g\big] - L_r\,r'\big[f(\hat{m})T_a + f(\theta^*_{\mathrm{B}})T_b\big]}{T_a L_g + T_b(A + L_g)}.$$

At the equilibrium, (B1) pins $T = u$, so $T_a = f(\theta^*_{\mathrm{B}})(u - \theta^*_{\mathrm{B}})/P$ and $T_b = f(\hat{m})(\hat{m} - u)/P$, and the ratification weight collapses:

$$f(\hat{m})\,T_a + f(\theta^*_{\mathrm{B}})\,T_b = \frac{f(\theta^*_{\mathrm{B}})\,f(\hat{m})}{P}\big[(u - \theta^*_{\mathrm{B}}) + (\hat{m} - u)\big] = \frac{f(\theta^*_{\mathrm{B}})\,f(\hat{m})\,g^*_{\mathrm{B}}}{P}.$$

Dividing the numerator through by the positive quantity $u'\,f(\theta^*_{\mathrm{B}})f(\hat{m})\,g^*_{\mathrm{B}}/P$ shows that $dP/d\pi \leq 0$ if and only if (3) holds, with $\bar{f} = P/g^*_{\mathrm{B}}$.

Step 6: shoulder bound and uniform tightness. The average density satisfies $\min_{[\theta^*_{\mathrm{B}},\hat{m}]} f \leq \bar{f} \leq \max_{[\theta^*_{\mathrm{B}},\hat{m}]} f$. When f is nonincreasing on the pool, $f(\theta^*_{\mathrm{B}}) \geq \bar{f} \geq f(\hat{m}) > 0$, so

$$\bar{f}\left[\frac{A + L_g}{f(\theta^*_{\mathrm{B}})} - \frac{L_g}{f(\hat{m})}\right] = A\,\frac{\bar{f}}{f(\theta^*_{\mathrm{B}})} + L_g\left(\frac{\bar{f}}{f(\theta^*_{\mathrm{B}})} - \frac{\bar{f}}{f(\hat{m})}\right) \;\leq\; A,$$

because the first ratio is at most one and the parenthetical difference is nonpositive. Under a uniform prior all three densities coincide, the display holds with equality, and (3) reduces to $L_r\,r'/u' \geq A$. This completes part (ii).

Step 7: comparison with the identity calibration. The identity boundary solves $\Psi_{\mathrm{id}}(a) = 0$ with $\Psi_{\mathrm{id}}(a) := \alpha a + B(a) - L(u - a, r, \sigma)$. For every $a \in [\theta^*_{\min}(u), u)$—the domain on which the consistency locus is defined—Step 1 gives $M(a; u) > u$, so $L(M(a; u) - a, r, \sigma) > L(u - a, r, \sigma)$ by (L1), and hence $\Psi_\pi(a) < \Psi_{\mathrm{id}}(a)$ pointwise there. Both maps are strictly increasing, so if $\theta^*_{\mathrm{id}} \geq \theta^*_{\min}(u)$ the roots rank $\theta^*_{\mathrm{B}} > \theta^*_{\mathrm{id}}$; if instead $\theta^*_{\mathrm{id}} < \theta^*_{\min}(u)$, the ranking is immediate from $\theta^*_{\mathrm{B}} > \theta^*_{\min}(u)$. Evaluating the two boundary conditions and using that $\alpha a + B(a)$ increases in a,

$$L\big(g^*_{\mathrm{B}}, r, \sigma\big) = \alpha\theta^*_{\mathrm{B}} + B(\theta^*_{\mathrm{B}}) \;>\; \alpha\theta^*_{\mathrm{id}} + B(\theta^*_{\mathrm{id}}) = L\big(g^*_{\mathrm{id}}, r, \sigma\big),$$

so $g^*_{\mathrm{B}} > g^*_{\mathrm{id}}$ by strict monotonicity of L in the gap, and (L2) ($L_{gr} \geq 0$) delivers $L_r(g^*_{\mathrm{B}}, r, \sigma) \geq L_r(g^*_{\mathrm{id}}, r, \sigma)$. The affine baseline makes $B' \equiv b_1$, so the identity-calibration condition of Proposi-

tion 2(iii) reads $L_r(g^*_{\mathrm{id}}, r, \sigma)\, r'/u' \geq A$. When the Bayesian pool lies where f is nonincreasing, Step 6 bounds the right side of (3) by A, and the chain

$$\frac{L_r(g^*_{\mathrm{B}}, r, \sigma)\, r'}{u'} \;\geq\; \frac{L_r(g^*_{\mathrm{id}}, r, \sigma)\, r'}{u'} \;\geq\; A \;\geq\; \bar{f}\left[\frac{A + L_g}{f(\theta^*_{\mathrm{B}})} - \frac{L_g}{f(\hat{m})}\right]$$

delivers (3); under a uniform prior the Step-6 bound holds with equality, so (3) is exactly the uniform-form test $L_r(g^*_{\mathrm{B}}, r, \sigma)\, r'/u' \geq A$, the identity-form condition evaluated at the Bayesian gap. This proves part (iii). □

Remark (Credible beliefs off path). Under the symmetric specification (S1), the rejection of off-path messages $m \in (\theta^*_{\mathrm{B}}, \hat{m})$ is credible in the sense of §S1. Were such an m approved, the set of types gaining weakly relative to equilibrium would be $[\theta^\circ(m), (m + \hat{m})/2]$, where $\theta^\circ(m)$ is the root of $\alpha\theta + B(\theta) = L(m - \theta, r, \sigma)$, truncated at $\underline{\theta}_0$ if the root falls below the support; $\theta^\circ(m) < \theta^*_{\mathrm{B}}$ because $\alpha\theta + B(\theta)$ is strictly increasing while $L(m - \theta^*_{\mathrm{B}}, r, \sigma) < L(\hat{m} - \theta^*_{\mathrm{B}}, r, \sigma)$ by (L1). Rejected types down to $\theta^\circ(m)$ gain approval, poolers below the midpoint $(m+\hat{m})/2 < \hat{m}$ trade their gap for a smaller one by symmetry of (S1), and poolers above the midpoint and all types above $\hat{m}$ lose (the two indifferent boundary types are included weakly). Because a type's gain depends only on the approval decision, this set is the only candidate for a credible interpretation of the deviation, and it is the equilibrium pool with both endpoints shifted strictly down; by the strict monotonicity of the truncated mean in both endpoints (Step 1), the prior truncated to it has mean strictly below $\mathrm{E}[\theta \mid \theta^*_{\mathrm{B}} \leq \theta \leq \hat{m}] = u$, so the receiver's best response to that interpretation is rejection, under which no type gains: the deviation admits no credible interpretation, and rejection stands.

S9.6.2 A parametric example, verified corridor-wide.

Let $\theta \sim \mathrm{Unif}[-1, 2]$, $B(\theta) = \frac{1}{10} + \frac{1}{5}\theta$, and $\alpha = \frac{2}{5}$, so the prize slope is $A = \frac{3}{5}$; let $L(g, r) = \eta(r)\, g + \frac{\kappa(r)}{2} g^2$ with $\eta(r) = \frac{1}{20} + \eta_1 r$, $\kappa(r) = \frac{1}{20} + \kappa_1 r$, $\eta_1 = 3$, and $\kappa_1 = 1$, which satisfies (L1)–(L3). The rule induces the required posterior u; we normalize $u(\pi) = \pi$ on the corridor $\pi \in [0.55, 0.95]$ and compare the carry-forward coupling $r(\pi) = \pi$ ($r' = 1$) with a decoupled regime ($r' = 0$). Under the uniform prior the consistency condition (B1) reads $(\theta^* + \hat{m})/2 = u$, so that

$$\hat{m} \;=\; 2u - \theta^*, \qquad g^* \;=\; \hat{m} - \theta^* \;=\; 2\,(u - \theta^*),$$

and the boundary indifference (B2) becomes, writing $d := u - \theta^*$,

$$\left(\tfrac{3}{5} + 2\eta(r)\right) d \;+\; 2\kappa(r)\, d^2 \;=\; \tfrac{3}{5}\, u + \tfrac{1}{10}.$$

At any given boundary type, Bayesian consistency thus doubles the lie the pool requires relative to the identity calibration, where $g^* = u - \theta^*$.

The pooling mass is $P = (\hat{m} - \theta^*)/3 = 2d/3$, and the formulas of Steps 4–5 specialize to

$$\left.\frac{dP}{d\pi}\right|_{r'=0} = \frac{2}{3} \cdot \frac{3/5}{3/5 + 2\eta + 4\kappa d} \;>\; 0, \qquad \left.\frac{dP}{d\pi}\right|_{r'=1} = \frac{2}{3} \cdot \frac{3/5 - L_r(g^*, r)}{3/5 + 2\eta + 4\kappa d},$$

with $L_r(g^*, r) = \eta_1 g^* + \frac{\kappa_1}{2} g^{*2} = 2\eta_1 d + 2\kappa_1 d^2$. The coupled derivative is negative exactly when $L_r(g^*, r) > A$: the condition (3) in its uniform form. With the parameters above, solving the boundary equation gives $g^* \approx 0.20$–0.21 and $L_r(g^*, r) \in (0.62, 0.65)$ across the corridor, so the coupled derivative is negative throughout ($dP/d\pi \approx -0.007$ at $\pi = 0.55$, -0.003 at $\pi = 0.95$) while the decoupled derivative stays positive (+0.09 to +0.06). The existence of the boundary root, the interiority of the pool ($\theta^* > -1$, $\hat{m} < 2$), and both sign claims were verified symbolically over the entire corridor rather than pointwise (a `Reduce` quantifier-elimination check returns `True` for each claim).

Because Bayesian consistency enlarges the equilibrium gap (part (iii)), the ratification channel binds *harder* under a fully Bayesian receiver: at the same primitives, the identity-calibration benchmark has $L_r(g) = \eta_1 g + \frac{\kappa_1}{2} g^2 \in (0.56, 0.58) < \frac{3}{5}$ on the corridor and does not reverse.

The same behavior obtains away from the uniform prior. Under the triangular log-concave density $f(\theta) = \frac{2}{9}(2 - \theta)$ on $[-1, 2]$—strictly decreasing, so every pool sits on a declining shoulder—with the same cost primitives, a unique interior boundary pair exists and the coupled derivative is negative on the whole corridor (both statements again verified by quantifier elimination), with $g^*_{\mathrm{B}} \approx 0.20$–$0.21$ and $L_r(g^*_{\mathrm{B}}) \approx 0.63$–$0.65 > \frac{3}{5}$, consistent with the shoulder sufficiency of part (ii). Near the top of that corridor the decoupled derivative also turns slightly negative through the pure density channel (the thin-right-tail effect of Proposition 9); the coupled derivative is roughly an order of magnitude more negative throughout.

The atomistic-voter device remains load-bearing for approval of a pool lying entirely below the bar (§S9.5); for the reversal itself, Proposition 3 shows it is conservative rather than necessary.

S10 Private-ordering and committee proofs

S10.1 Private-ordering construction

The following charter-choice apparatus underlies the private-ordering result (Proposition 7) and its components (Propositions 10 and 11).

S10.1.1 Private ordering: Stage -1 charter choice

At Stage -1, the controller chooses a governance regime

$$G \in \mathcal{G} := \{L, H\}.$$

Regime L is a lax/decoupled package and regime H is a protective/coupled package. Formally, each G induces continuation primitives

$$\Gamma(G) := (\pi_G, r_G, \sigma_G),$$

with $r_L = r_0$, $r_H = \pi_H$, and $\pi_H \geq \pi_L$.

For each G, let $(u_G, \theta^*_G, m^{\text{req}}_G)$ denote the continuation equilibrium objects from Sections 5.2.3 and 5.3.2, where $m^{\text{req}}_G := m^{\text{req}}(u_G)$. Define

$$I_G(\theta) := \mathbf{1}\{\theta \geq \theta^*_G\}, \qquad g_G(\theta) := \big(m_G(\theta) - \theta\big)_+,$$

with equilibrium message policy

$$m_G(\theta) = \begin{cases} \theta, & \theta < \theta^*_G, \\ m^{\text{req}}_G, & \theta \in [\theta^*_G, u_G), \\ \theta, & \theta \geq u_G. \end{cases}$$

Define pooling mass and expected enforcement burden:

$$P(G) := \int_{\theta^*_G}^{u_G} f(\theta)\, d\theta, \qquad EC(G) := \int_{\theta^*_G}^{u_G} L\big(m^{\text{req}}_G - \theta, r_G, \sigma_G\big) f(\theta)\, d\theta.$$

S10.1.2 Private ordering: pricing and controller objective

Let $V_M(G)$ be ex ante minority value under G, normalized per unit minority claim:

$$V_M(G) := \mathrm{E}\big[I_G(\theta)\big(\theta - D(\theta)\big)\big] - \omega\, EC(G),$$

where $\omega \in [0,1]$ is the deadweight share from Section 6.1. Competitive, risk-neutral pricing implies

$$p(G) = V_M(G).$$

Let $\nu \geq 0$ index the level of private-benefit rents and let $\xi \in [0,1]$ be the controller-borne share of enforcement burden. Define

$$\Pi(G; \nu) := \mathrm{E}[I_G(\theta)\, \nu B(\theta)] - \xi\, EC(G),$$

and Stage -1 controller value

$$V_C(G; \alpha, \nu) := \alpha\, p(G) + \Pi(G; \nu).$$

Social value is $W(G)$ from Section 6.

S10.2 Private-ordering proofs

S10.2.1 Cash-flow alignment and the regime-choice threshold

Proposition 10 (Cash-flow alignment and regime choice threshold)**.** *Under Assumption 7, define* $\alpha^*(\nu) := -\Delta_\Pi(\nu)/\Delta_M > 0$*. Then* $V_C(H;\alpha,\nu) - V_C(L;\alpha,\nu) = \alpha\Delta_M + \Delta_\Pi(\nu)$*, so the controller chooses the protective regime* H *if and only if* $\alpha \geq \alpha^*(\nu)$ *(ties at* $\alpha = \alpha^*(\nu)$ *broken toward* H*); the chosen regime is weakly more protective as* α *increases.*

Proof. By definition,

$$V_C(H;\alpha,\nu) - V_C(L;\alpha,\nu) = \alpha\big(V_M(H) - V_M(L)\big) + \big(\Pi(H;\nu) - \Pi(L;\nu)\big) = \alpha\Delta_M + \Delta_\Pi(\nu).$$

Assumption 7 gives $\Delta_M > 0$. Hence $\alpha \mapsto V_C(H;\alpha,\nu) - V_C(L;\alpha,\nu)$ is affine and strictly increasing. The unique zero is

$$\alpha^*(\nu) := -\frac{\Delta_\Pi(\nu)}{\Delta_M} > 0,$$

because $\Delta_\Pi(\nu) < 0$. Therefore

$$V_C(H;\alpha,\nu) \geq V_C(L;\alpha,\nu) \iff \alpha \geq \alpha^*(\nu).$$

So the controller chooses H iff $\alpha \geq \alpha^*(\nu)$, proving monotone comparative statics in α. □

S10.2.2 Private benefits and the regime-choice threshold

Proposition 11 (Private benefits tilt choice toward decoupling)**.** *Under Assumption 7,* $\alpha^*(\nu)$ *is weakly increasing in* ν*; equivalently, for any fixed* α*, an increase in private-benefit intensity weakly expands the parameter region in which the controller chooses the lax regime* L*.*

Proof. From Proposition 10,

$$\alpha^*(\nu) = -\frac{\Delta_\Pi(\nu)}{\Delta_M}, \qquad \Delta_M > 0.$$

Take $\nu_2 > \nu_1$. Assumption 7 implies $\Delta_\Pi(\nu_2) \leq \Delta_\Pi(\nu_1)$. Multiplying by -1 and dividing by positive Δ_M yields

$$\alpha^*(\nu_2) \geq \alpha^*(\nu_1).$$

Hence $\alpha^*(\nu)$ is weakly increasing in ν. Equivalently, fixing α, the set

$$\{\nu \,:\, \alpha < \alpha^*(\nu)\}$$

is increasing in ν, so larger private-benefit intensity weakly expands the region where L is chosen. □

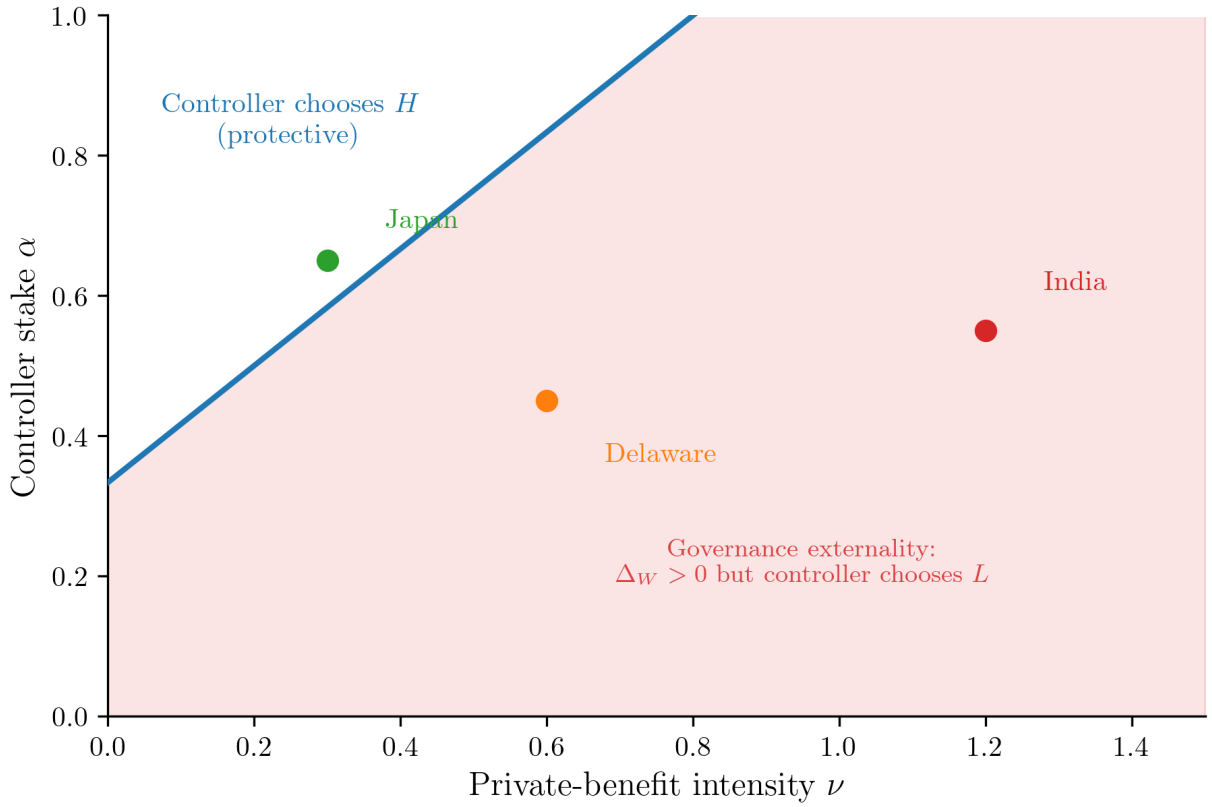
S10.2.3 Governance externality under partial alignment

Proof. Assumption 8 gives $W(H) > W(L)$, so welfare chooses H. If $\alpha < \alpha^*(\nu)$, Proposition 10 implies $V_C(L; \alpha, \nu) > V_C(H; \alpha, \nu)$, so the controller chooses L. Therefore controller choice and welfare choice differ:

$$\arg \max_{G \in \{L,H\}} V_C(G; \alpha, \nu) \neq \arg \max_{G \in \{L,H\}} W(G).$$

□

Figure 6: Governance externality in (ν, α) space



The boundary $\alpha^*(\nu) = -\Delta_\Pi(\nu)/\Delta_M$ (blue line) separates controllers who voluntarily adopt protective governance (above) from those who choose the lax regime (below). The shaded region below the boundary is the governance externality zone: coupling is welfare-improving ($\Delta_W > 0$) but the controller chooses L. The boundary is increasing in ν—higher private-benefit intensity raises the controller's resistance to protection—so the externality is most severe where it matters most (vicious circle). Approximate jurisdiction placements: Japan (green, low ν, high α, above the line—voluntary adoption); Delaware (orange, moderate ν, moderate α, below the line—borderline); India (red, high ν, moderate α, deep in the externality zone). Stylized parameterization for illustration.

S10.3 Committee monotonicity results

The following two lemmas support the committee discussion in Section 6.6.1.

Lemma 7 (Committee screening: gate channel)**.** *For $0 \le \iota_1 < \iota_2 \le 1$:*

(a) the conditional density satisfies the monotone likelihood ratio order, $f_{\iota_2}(\theta)/f_{\iota_1}(\theta)$ increasing in θ, so f_{ι_2} first-order stochastically dominates f_{ι_1};
(b) for every cutoff $c \in (\underline{\theta}_0, \overline{\theta}_0)$ the mass $F_\iota(c) := \Pr_\iota(\theta \le c)$ is weakly decreasing in ι, so at any voting bar u the needs-persuasion mass $\Pr_\iota(\theta < u)$ falls with committee independence; and
(c) if $L_g(g, r, \sigma) \ge 0$ on $g \ge 0$, then at any fixed (u, r, σ) the expected enforcement burden $\mathrm{E}_\iota\big[\mathbf{1}\{\theta < u\}\, L(u - \theta, r, \sigma)\big]$ is weakly decreasing in ι.

Proof. Part (a). For $j \in \{1, 2\}$, write $q_j(\theta) := q_{\iota_j}(\theta) = 1 - \iota_j + \iota_j Q(\theta)$ and $f_j(\theta) := q_j(\theta) f(\theta)/Z_j$ with $Z_j := \int q_j(t) f(t)\, dt > 0$. Then $f_2(\theta)/f_1(\theta) = (Z_1/Z_2)\, q_2(\theta)/q_1(\theta)$, and since Z_1/Z_2 is constant in θ, monotonicity is governed by $h(\theta) := q_2(\theta)/q_1(\theta)$. Differentiating in Q,

$$\frac{d}{dQ}\left(\frac{1 - \iota_2 + \iota_2 Q}{1 - \iota_1 + \iota_1 Q}\right) = \frac{\iota_2 - \iota_1}{(1 - \iota_1 + \iota_1 Q)^2} > 0$$

since $\iota_2 > \iota_1$. Because $Q'(\theta) > 0$, h is increasing in θ, so f_2/f_1 is MLR-increasing, which implies f_{ι_2} FOSD-dominates f_{ι_1}.

Part (b). By part (a) and the definition of FOSD, $F_{\iota_2}(c) = \Pr_{\iota_2}(\theta \le c) \le \Pr_{\iota_1}(\theta \le c) = F_{\iota_1}(c)$ for every c; hence $F_\iota(c)$ is weakly decreasing in ι, and $c = u$ gives the needs-persuasion claim.

Part (c). Set $\Psi_u(\theta) := \mathbf{1}\{\theta < u\}\, L(u - \theta, r, \sigma)$. For $\theta < u$, $\Psi_u'(\theta) = -L_g(u - \theta, r, \sigma) \le 0$ by $L_g \ge 0$, and $\Psi_u(\theta) = 0$ for $\theta \ge u$, so Ψ_u is weakly decreasing. By part (a), larger ι shifts θ up in the FOSD order, and the expectation of a weakly decreasing function is weakly lower; hence $\mathrm{E}_\iota[\Psi_u(\theta)]$ is weakly decreasing in ι. □

Lemma 8 (Committee information and message discipline: signal channel)**.** *For the recommendation experiment with pass probability $p_\iota(\theta) = 1 - \iota G_\varepsilon(s_0 - \theta)$ and posterior means $\mu_\iota(m, y)$ affine in $\bar{\mu}_\iota(m, y) := \mathrm{E}[\theta \mid m, y]$ with a recommendation-independent weight $\tau(m, y) = \tau(m) \in [0, 1]$:*

(a) for fixed $\iota > 0$, p_ι is increasing in θ, the binary recommendation is monotone informative, and $\bar{\mu}_\iota(m, 1) \ge \bar{\mu}_\iota(m, 0)$ for every message m;
(b) for $0 \le \iota_1 < \iota_2 \le 1$, the experiment at ι_2 Blackwell-dominates the experiment at ι_1;
(c) if $m \mapsto \mu_\iota(m, y)$ is strictly increasing for each y, then $m_\iota^{\mathrm{req}}(u, 1) \le m_\iota^{\mathrm{req}}(u, 0)$, where $m_\iota^{\mathrm{req}}(u, y) := \inf\{m : \mu_\iota(m, y) \ge u\}$, so a favorable recommendation weakly lowers the minimum persuasive message; and
(d) at any fixed (m, r, σ), the expected enforcement exposure $\mathcal{E}(\iota; m) := \int_{\underline{\theta}_0}^{\overline{\theta}_0} L\big((m - \theta)_+, r, \sigma\big)\, p_\iota(\theta) f(\theta)\, d\theta$ is weakly decreasing in ι.

Proof. Part (a). Write $p(\theta) := p_\iota(\theta) = 1 - \iota G_\varepsilon(s_0 - \theta)$, so $p'(\theta) = \iota g_\varepsilon(s_0 - \theta) \ge 0$, strict where $g_\varepsilon > 0$. For a binary signal MLR is monotonicity of the odds $p(\theta)/(1 - p(\theta))$, increasing because p is increasing and $x \mapsto x/(1 - x)$ is increasing on $(0, 1)$. Fix m and let $\tilde{f}_m$ be the posterior after m alone; Bayes' rule gives $\bar{\mu}_\iota(m, 1) - \mathrm{E}_{\tilde{f}_m}[\theta] = \mathrm{Cov}_{\tilde{f}_m}(\theta, p(\theta))/\mathrm{E}_{\tilde{f}_m}[p(\theta)] \ge 0$ and $\bar{\mu}_\iota(m, 0) - \mathrm{E}_{\tilde{f}_m}[\theta] = -\mathrm{Cov}_{\tilde{f}_m}(\theta, p(\theta))/(1 - \mathrm{E}_{\tilde{f}_m}[p(\theta)]) \le 0$, so $\bar{\mu}_\iota(m, 1) \ge \bar{\mu}_\iota(m, 0)$.

Part (b). Set $\rho := \iota_1/\iota_2 \in [0,1)$ and garble the ι_2 recommendation Y_2 by $K(1 \mid 1) = 1$, $K(0 \mid 0) = \rho$, $K(1 \mid 0) = 1-\rho$. Then $\Pr(Y_1 = 1 \mid \theta) = (1-\rho) + \rho\, p_{\iota_2}(\theta) = 1 - \iota_1 G_\varepsilon(s_0 - \theta) = p_{\iota_1}(\theta)$, so the ι_1 experiment is a garbling of the ι_2 experiment; hence ι_2 Blackwell-dominates ι_1.

Part (c). By part (a), $\bar{\mu}_\iota(m,1) \geq \bar{\mu}_\iota(m,0)$ for all m; since $\mu_\iota(m,y)$ is affine in $\bar{\mu}_\iota(m,y)$ with weight $\tau(m)$ independent of the recommendation y, also $\mu_\iota(m,1) \geq \mu_\iota(m,0)$ for all m. Let $m_0 := m_\iota^{\mathrm{req}}(u,0)$; then $\mu_\iota(m_0,0) \geq u$ gives $\mu_\iota(m_0,1) \geq u$, so m_0 is feasible for $m_\iota^{\mathrm{req}}(u,1)$ and $m_\iota^{\mathrm{req}}(u,1) \leq m_0 = m_\iota^{\mathrm{req}}(u,0)$.

Part (d). Since $\partial p_\iota(\theta)/\partial\iota = -G_\varepsilon(s_0-\theta) \leq 0$, differentiating under the integral gives $\partial\mathcal{E}/\partial\iota = -\int_{\underline{\theta}_0}^{\overline{\theta}_0} L((m-\theta)_+, r, \sigma)\, G_\varepsilon(s_0-\theta)\, f(\theta)\, d\theta \leq 0$, as $L \geq 0$, $G_\varepsilon \geq 0$, $f \geq 0$. □

S10.4 Proofs of main-text committee complementarity results

Lemma 9 (Gate false-negative cost is increasing in independence). *The gate false-negative cost* $\mathrm{FN}^{\mathrm{gate}}(\iota) := \int_0^{\overline{\theta}_0} \big(1 - q_\iota(\theta)\big)\, \theta\, f(\theta)\, d\theta$ *is strictly increasing and linear in* ι *on* $[0,1]$, *with* $\mathrm{FN}^{\mathrm{gate}}(0) = 0$, *and is independent of* (u, r, σ).

Proof of Lemma 9. Since $q_\iota(\theta) = (1-\iota) + \iota Q(\theta)$, we have $1 - q_\iota(\theta) = \iota\big(1 - Q(\theta)\big)$. Hence

$$\mathrm{FN}^{\mathrm{gate}}(\iota) = \iota \int_0^{\overline{\theta}_0} \big(1 - Q(\theta)\big)\, \theta\, f(\theta)\, d\theta.$$

The integral is a positive constant (since $Q(\theta) < 1$ for $\theta < \overline{\theta}_0$ and $\theta > 0$ on the domain of integration), so $\mathrm{FN}^{\mathrm{gate}}$ is strictly increasing and linear in ι with $\mathrm{FN}^{\mathrm{gate}}(0) = 0$. No term depends on (u, r, σ). □

Proof of Proposition 6. Differentiating under the integral:

$$\frac{\partial\, \mathrm{EC}}{\partial \iota} = \int_{\theta^*}^{u} \big(Q(\theta) - 1\big)\, L\big(m^{\mathrm{req}} - \theta, r, \sigma\big)\, f(\theta)\, d\theta \;\leq\; 0,$$

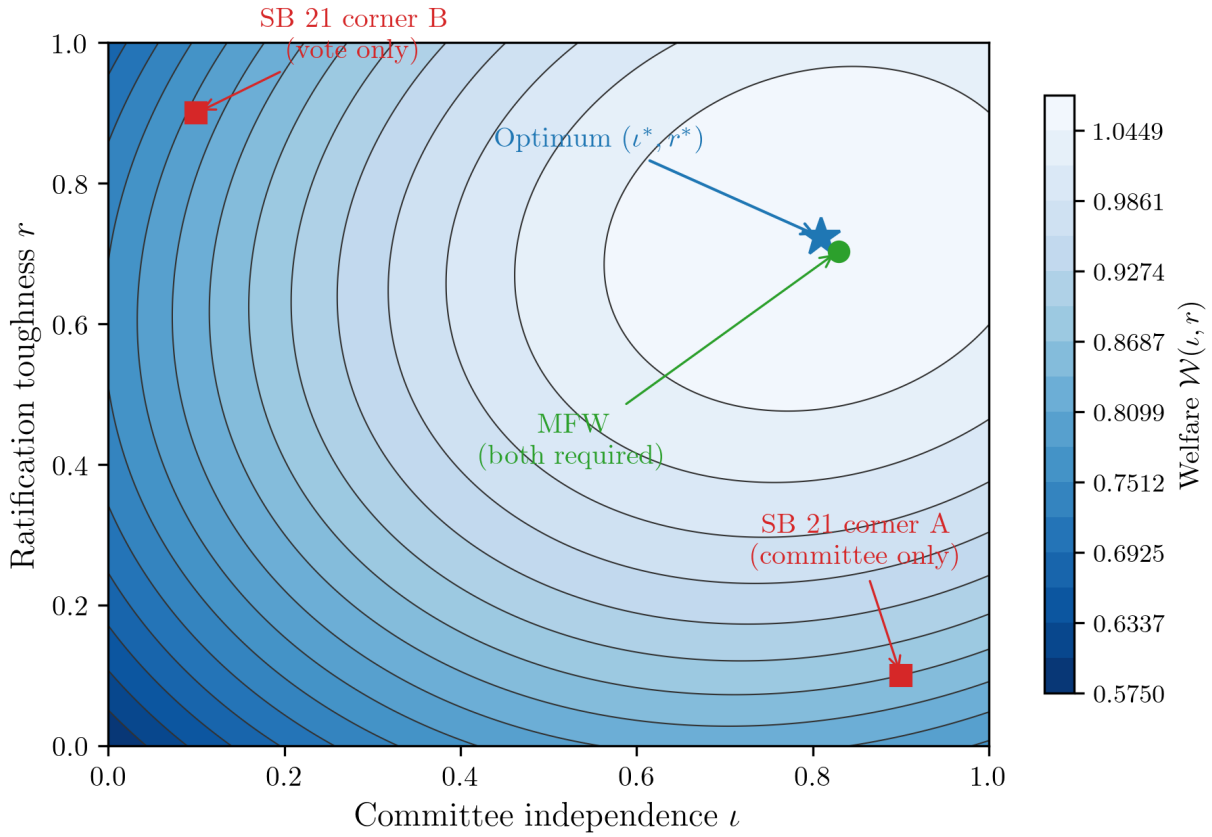
since $Q(\theta) \leq 1$ and $L \geq 0$. Differentiating again with respect to r:

$$\frac{\partial^2\, \mathrm{EC}}{\partial \iota\, \partial r} = \int_{\theta^*}^{u} \big(Q(\theta) - 1\big)\, L_r\big(m^{\mathrm{req}} - \theta, r, \sigma\big)\, f(\theta)\, d\theta.$$

Each factor in the integrand satisfies $Q(\theta) - 1 \leq 0$ and $L_r \geq 0$, so the integrand is weakly negative pointwise. Under the stated non-degeneracy condition, the integral is strictly negative. □

Proof of Corollary 4. The gate false-negative cost $\mathrm{FN}^{\mathrm{gate}}(\iota)$ is independent of r (Lemma 9), so its cross-partial vanishes. The approved-value integral at fixed θ^* depends on ι through $q_\iota(\theta)$ but not on r, so its cross-partial also vanishes. The only surviving term is $-\omega\, \partial^2 \mathrm{EC}/\partial\iota\, \partial r \geq 0$ by Proposition 6. □

Figure 7: Welfare supermodularity over committee independence and ratification toughness



Lighter shading indicates higher welfare $\mathcal{W}(\iota, r)$. The optimum (blue star) is interior, reflecting the complementarity between the two instruments. The MFW point (green circle), which requires both an independent committee and a majority-of-the-minority vote, is near the optimum. In this *fixed-allocation* welfare surface the two SB 21 corners (red squares)—committee only (ι high, r low) and vote only (ι low, r high)—are welfare-inferior to the interior optimum. The contour lines bend toward the (high, high) corner, reflecting the fixed-allocation supermodularity: the marginal return to committee independence is increasing in ratification toughness (Corollary 4). Stylized welfare function, drawn at a fixed allocation (Remark 6.6.2).